MATH3204

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Contents

1 Vector **3**
1.1 Vector Spaces . . . 3
1.2 Vector Norms . . . 7

2 Matrix **10**
2.1 Matrices . . . 10
2.1.1 Rectangular Array of Scalars . . . 10
2.1.2 Linear Maps Between Two Vector Spaces . . . 13
2.2 Matrix Subspaces . . . 14
2.3 Pseudo-inverse . . . 16
2.4 Matrix Norms . . . 18
2.4.1 Entry-wise Matrix Norms . . . 18
2.4.2 Induced by Vector Norms . . . 18
2.4.3 Condition number . . . 19
2.5 Matrix Spectral Properties . . . 20
2.5.1 Eigenvalues and Eigenvectors . . . 20
2.5.2 Singular Value Decomposition . . . 26
2.6 Special Matrix Types . . . 31
2.6.1 Diagonal Matrix . . . 31
2.6.2 Triangular Matrix . . . 32
2.6.3 Hessenberg Matrix . . . 33
2.6.4 Permutation Matrix . . . 34
2.6.5 Projection Matrix . . . 34
2.6.6 Positive (semi-)definite . . . 36
2.7 Matrix Factorizations . . . 38
2.7.1 LU factorisation . . . 38
2.7.2 QR Factorization . . . 41

3 Direct Methods **43**
3.1 Direct Methods . . . 43
3.2 Least Squares . . . 46
3.2.1 Usual method for solving OLS . . . 48
3.2.2 Computing QR . . . 51

4 Iterative Methods for Linear Systems **54**
4.1 Stationary Methods . . . 54
4.1.1 Basic Relaxation Methods . . . 55
4.1.2 Convergence of Basic Relaxation Methods . . . 56
4.2 Polynomial Acceleration . . . 57

5 Fundamentals **61**
5.1 Topology/Geometry . . . 61
5.2 Calculus . . . 64

6 Optimality and Convexity **70**
6.1 Optimal Points 70
6.2 Unconstrained Optimization 70
6.3 Constrained Optimization 73
6.4 Convexity 75

7 Unconstrained Optimization **80**
7.1 Descent Method 80
7.1.1 Descent Direction 81
7.1.2 Step-size 82
7.2 Gradient Descent 84

Chapter 1

Vector

1.1 Vector Spaces

Vector spaces are the fundamental objects for linear algebra. Finite dimensional vector spaces are the fundamental settings for numerical linear algebra.

Definition 1.1.1

Formally, a field is a set $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) together with two operations called addition, "+", and multiplication, "×", which satisfy the following field axioms:

- Associativity: $\forall\, a, b, c \in \mathbb{F}, \quad a + (b + c) = (a + b) + c, \quad a \times (b \times c) = (a \times b) \times c$
- Commutativity: $\forall\, a, b \in \mathbb{F}, \quad a + b = b + a, \quad a \times b = b \times a$
- Additive and multiplicative identity: $\exists\, 0 \in \mathbb{F}$ and $\exists\, 1 \in \mathbb{F}, \quad a + 0 = a, \quad a \times 1 = a$
- Additive inverses: $\forall\, a \in \mathbb{F}, \quad \exists\, -a \in \mathbb{F}, \quad a + (-a) = 0$
- Multiplicative inverses: $\forall\, a \neq 0 \in \mathbb{F}, \quad \exists\, 1/a \in \mathbb{F}, \quad a \times (1/a) = 1$
- Distributivity of multiplication over addition: $\forall\, a, b, c \in \mathbb{F}, \quad a \times (b + c) = (a \times b) + (a \times c)$

Example.

Examples are $\mathbb{R}, \mathbb{C}$, and $\mathbb{Q}$.

Notation: Scalars are typically denoted by lowercase letters, e.g., "c".

Definition 1.1.2

A vector space $\mathcal{V}$ over a field $\mathbb{F}$ is a set of objects (called vectors) together with operations of vector addition "+" and scalar multiplication "×", such that the following hold for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and scalars $a, b \in \mathbb{F}$:

1. Closure of vector addition: $\mathbf{u} + \mathbf{v} \in \mathcal{V}$
2. Commutativity of addition: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. Associativity of addition: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
4. Additive identity: $\exists\, \mathbf{0} \in \mathcal{V}$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u} = \mathbf{0} + \mathbf{u}$
5. Additive inverse: $\exists\, -\mathbf{u} \in \mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0} = (-\mathbf{u}) + \mathbf{u}$
6. ...

Notation: Vectors are typically denoted by boldface lowercase letters, e.g., $\mathbf{v}$.

In short, for a vector space $\mathcal{V}$ over a field $\mathbb{F}$, we have

$$a\mathbf{u} + b\mathbf{v} \in \mathcal{V}, \quad \forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V}, \quad \forall\, a, b \in \mathbb{F}$$

Definition 1.1.3

For a given field $\mathbb{F}$ and a given positive integer d, the set $\mathbb{F}^d$ of d-tuples with entries from $\mathbb{F}$ forms a vector space over $\mathbb{F}$ under element-wise addition in $\mathbb{F}^d$:

Example.

$$\alpha \begin{bmatrix} a \\ b \end{bmatrix} + \beta \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} \alpha a + \beta c \\ \alpha b + \beta d \end{bmatrix}, \quad a, b, c, d, \alpha, \beta \in \mathbb{F}$$

Example.

- $\mathbb{R}^d$ is a real vector space, i.e., a vector space over $\mathbb{R}$
- $\mathbb{C}^d$ is a complex vector space, i.e., a vector space over $\mathbb{C}$
  - $\mathbb{C}^d$ is also a real vector space, i.e., a vector space over $\mathbb{R}$

Convention: Elements of $\mathbb{F}^d$ are always presented as column vectors.

Definition 1.1.4

A subspace $\mathcal{W}$ of a vector space $\mathcal{V}$ over a field $\mathbb{F}$ is a subset $\mathcal{W} \subseteq \mathcal{V}$ that is, by itself, a vector space over $\mathbb{F}$, i.e.,

$$a\mathbf{u} + b\mathbf{v} \in \mathcal{W}, \quad \forall\, \mathbf{u}, \mathbf{v} \in \mathcal{W}, \quad \forall\, a, b \in \mathbb{F}$$

Example.

$$\left\{ \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \middle| a, b \in \mathbb{R} \right\} \quad \text{is a subspace of } \mathbb{R}^3.$$

- *An intersection of subspaces is always a subspace*
- *A union of subspaces need not be a subspace*
- *The subsets $\{\mathbf{0}\}$ and $\mathcal{V}$ are always subspaces of $\mathcal{V}$, so they are often called trivial subspaces*
- *A subspace $\mathcal{W}$ of $\mathcal{V}$ is said to be nontrivial if it is different from both $\{\mathbf{0}\}$ and $\mathcal{V}$*
- *A subspace of $\mathcal{V}$ is said to be a proper subspace if it is not equal to $\mathcal{V}$*
- *Since a vector space always contains the zero vector, a subspace cannot be empty*

Definition 1.1.5

Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$ and $S \subseteq \mathcal{V}$ be a subset. $\mathrm{Span}(S)$ is all linear combinations of all finitely many vectors in S, i.e.,

$$\mathrm{Span}(S) = \left\{ \sum_{i=1}^{k} a_i \mathbf{v}_i \;\middle|\; \mathbf{v}_1, \ldots, \mathbf{v}_k \in S,\ a_1, \ldots, a_k \in \mathbb{F},\ k = 1, 2, \ldots \right\}$$

Example.

> Let $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_d\} = \mathcal{V}$. Q: Are there any redundancy among these vectors? i.e., can the span of a proper subset of these vectors also form $\mathcal{V}$?

A: The answer lies in whether these vectors are linearly independent or not.

Definition 1.1.6

A finite set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in a vector space $\mathcal{V}$ over a field $\mathbb{F}$ is linearly dependent if and only if there are scalars $a_1, \ldots, a_k \in \mathbb{F}$, not all zero, such that

$$\mathbf{0} = \sum_{i=1}^{k} a_i \mathbf{v}_i.$$

In words, a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly dependent if and only if, at least, one of the vectors is a linear combination of some of the others. We can actually throw out that redundant vector without changing the span of the original list.

Since some vectors in the list can be expressed as linear combinations of others, we can remove these redundant vectors without altering the span.

Definition 1.1.7

Alternatively, a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if the only way to make $\sum_{i=1}^{k} a_i \mathbf{v}_i = \mathbf{0}$ true is to have all $a_i = 0$.

Take any $\mathbf{v} \in \mathcal{V}$. By definition of span, there are scalars $\alpha_1, \alpha_2, \ldots, \alpha_d$ for which we have

$$\mathbf{v} = \sum_{i=1}^{d} \alpha_i \mathbf{v}_i.$$

Suppose there are also $\beta_1, \beta_2, \ldots, \beta_d$ for which we can write

$$\mathbf{v} = \sum_{i=1}^{d} \beta_i \mathbf{v}_i.$$

So, we have

$$\mathbf{0} = \sum_{i=1}^{d} (\alpha_i - \beta_i) \mathbf{v}_i.$$

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_d$ are linearly independent, then it must be that $\alpha_i = \beta_i$, i.e., the representation of $\mathbf{v}$ is unique.

Definition 1.1.8

A set of vectors that is linearly independent and spans some vector space forms a basis for that vector space. In other words, a (possibly countably infinite) set $\mathcal{B}$ is a basis for the vector space $\mathcal{V}$ if and only if $\text{Span}(\mathcal{B}) = \mathcal{V}$ and $\mathcal{B}$ is linearly independent.

- *Any $\mathbf{v} \in \mathcal{V}$ can be represented uniquely in terms of elements in $\mathcal{B}$, i.e., there is one and only one way to choose $\mathbf{b}_1, \ldots, \mathbf{b}_k \in \mathcal{B}$ and $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that*

$$\mathbf{v} = \sum_{i=1}^{k} \alpha_i \mathbf{b}_i$$

- *A vector space is finite-dimensional if it has a finite basis*

Definition 1.1.9

The dimension of a vector space $\mathcal{V}$ over $\mathbb{F}$ is the cardinality, i.e., the number of vectors, of any basis of $\mathcal{V}$ over $\mathbb{F}$.

Fact 1.1.10

- If $\mathcal{W}$ is a subspace of $\mathcal{V}$, then $\dim(\mathcal{W}) \leq \dim(\mathcal{V})$
- If $\mathcal{W}$ is a subspace of $\mathcal{V}$ and $\dim(\mathcal{W}) = \dim(\mathcal{V})$, then $\mathcal{W} = \mathcal{V}$
- Let $\dim(\mathcal{V}) = d$. Then:
  - every system of linearly independent vectors of $\mathcal{V}$ has at most d elements, and
  - any basis of $\mathcal{V}$ has exactly d elements ("Dimension Theorem").
- The only vector space with dimension 0 is $\{\mathbf{0}\}$

Example.

$\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is the canonical/standard basis for $\mathbb{R}^3$

Example.

$$\left\{ \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \middle| \, a, b \in \mathbb{R} \right\} \quad \text{is a 2-dimensional subspace of } \mathbb{R}^3.$$

Example.

The vector space $\mathbb{C}^d$ has dimension d over the field $\mathbb{C}$ but dimension $2d$ over the field $\mathbb{R}$. For $\mathbb{C}^2$:

- If $\mathbb{F} = \mathbb{R}$, then
$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$
- If $\mathbb{F} = \mathbb{C}$, then
$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} i \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ i \end{pmatrix} \right\}$$

Example.

The infinite list of monomials $\mathcal{B} = \{1, t, t^2, t^3, \dots\}$ is a basis for the real vector space of all polynomials with real coefficients. It is easy to see that each polynomial is a unique linear combination of (finitely many) elements in the basis.

Definition 1.1.11

A list of vectors $\mathbf{v}_1, \dots, \mathbf{v}_m \in \mathbb{C}^n$ is orthogonal if

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \mathbf{v}_i^* \mathbf{v}_j = \mathbf{v}_j^* \mathbf{v}_i = 0, \quad \forall i, j \in \{1, 2, \dots, m\}.$$

If, in addition, $\|\mathbf{v}_i\|^2 = \langle \mathbf{v}_i, \mathbf{v}_i \rangle = 1, \quad \forall i \in \{1, 2, \dots, m\}$, then the list is orthonormal.

Theorem 1.1.12

Every orthonormal list of vectors in $\mathbb{C}^n$ is linearly independent

Proof.

$$\mathbf{0} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$$

$$\Rightarrow 0 = \left\langle \sum_{i=1}^{n} \alpha_i \mathbf{v}_i, \sum_{i=1}^{n} \alpha_i \mathbf{v}_i \right\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} \langle \alpha_i \mathbf{v}_i, \alpha_j \mathbf{v}_j \rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i^* \alpha_j \, \mathbf{v}_i^* \mathbf{v}_j = \sum_{i=1}^{n} \alpha_i^* \alpha_i \, \mathbf{v}_i^* \mathbf{v}_i = \sum_{i=1}^{n} |\alpha_i|^2$$

$$\Rightarrow \alpha_i = 0, \ \forall i.$$

□

Fact 1.1.13

- For a list of m orthonormal vectors in $\mathbb{C}^n$, we must have $m \leq n$.
- Any list of m orthonormal vectors in $\mathbb{C}^n$ forms a basis for their span as a m-dimensional subspace of $\mathbb{C}^n$.

Definition 1.1.14

Let $\mathcal{U}$ and $\mathcal{V}$ be vector spaces over the same field $\mathbb{F}$. The mapping $f : \mathcal{U} \to \mathcal{V}$ is called linear if

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{U}, \quad \forall \alpha, \beta \in \mathbb{F}.$$

Definition 1.1.15

"Differentiation" $f : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ defined as $f(p) = p'$

Definition 1.1.16

"Integration" $f : \mathcal{P}(\mathbb{R}) \to \mathbb{R}$ defined as $f(p) = \int p(x) \, \mathrm{d}x$

We will soon see that, if $\mathcal{U}$ and $\mathcal{V}$ are finite dimensional, such a linear map f can be represented by a more tangible object, which we call a matrix!

1.2 Vector Norms

Norms encapsulate the notions of size and distance in a vector space.

- a vector is thought of as "small" if its norm is small
- vectors $\mathbf{x}$ and $\mathbf{y}$ are "close" if the norm of the difference $\mathbf{x} - \mathbf{y}$ is small

Approximations and convergence throughout numerical linear algebra and optimization is measured using norms

Definition 1.2.1

Given a vector space $\mathcal{V}$ over $\mathbb{F}$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$), a norm is a non-negative real-valued function $\|\cdot\| : \mathcal{V} \to [0, +\infty)$ with the following properties:

- Sub-additivity or Triangle Inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$
- Absolute Homogeneity: $\|\alpha\mathbf{u}\| = |\alpha|\|\mathbf{u}\|$
- Positive Definiteness: $\|\mathbf{u}\| = 0 \iff \mathbf{u} = \mathbf{0}$

A vector space together with a given norm $\|\cdot\|$ is called a normed vector space.

Lemma 1.2.2: Reverse Triangle Inequality

Any norm satisfies

$$|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x} - \mathbf{y}\|.$$

Proof.

$$\mathbf{x} = \mathbf{y} + (\mathbf{x} - \mathbf{y}) \Rightarrow \|\mathbf{x}\| = \|\mathbf{y} + (\mathbf{x} - \mathbf{y})\| \leq \|\mathbf{y}\| + \|\mathbf{x} - \mathbf{y}\|$$

$$\mathbf{y} = \mathbf{x} + (\mathbf{y} - \mathbf{x}) \Rightarrow \|\mathbf{y}\| = \|\mathbf{x} + (\mathbf{y} - \mathbf{x})\| \leq \|\mathbf{x}\| + \|\mathbf{y} - \mathbf{x}\|$$

So $\|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|$ and $\|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|$, which implies

$$|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x} - \mathbf{y}\|.$$

□

The most important class of vector norms for $\mathbb{C}^d$ are the p-norms.

Definition 1.2.3: Vector p-norms

- ℓ_1 or Manhattan norm: $\|\mathbf{x}\|_1 = \sum_{i=1}^{d} |x_i|$
- ℓ_2 or Euclidean norm:
$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^{d} |x_i|^2\right)^{1/2} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$$
- ℓ_∞ or max norm:
$$\|\mathbf{x}\|_\infty = \max_{i=1,\ldots,d} |x_i|$$
- More generally, ℓ_p-norm:
$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^{d} |x_i|^p\right)^{1/p} \quad \text{for } 1 \leq p < \infty$$

Note: The definition still makes sense $0 < p < 1$, but the resulting function does not define a norm because it violates the triangle inequality.

Note: Often the subscript "2" in $\|\mathbf{x}\|_2$ is dropped, i.e., $\|\mathbf{x}\|$ implies Euclidean norm.

Example.

The unit ball $\mathcal{B} = \left\{\mathbf{x} \in \mathbb{R}^2 \,\middle|\, \|\mathbf{x}\|_p = 1\right\}$ for $p = 1, 1.5, 2, 3, 6, \infty$.

Lemma 1.2.4: Cauchy–Schwarz Inequality

We have

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_2 \, \|\mathbf{y}\|_2,$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent, that is, if and only if $\mathbf{x} = \alpha \mathbf{y}$ for some $\alpha \in \mathbb{C}$.

Lemma 1.2.5: Hölder's Inequality for $\mathbb{C}^d$

For any $1 \leq p, q \leq \infty$ such that $\dfrac{1}{p} + \dfrac{1}{q} = 1$, we have

$$\big|\langle \mathbf{x}, \mathbf{y} \rangle\big| \leq \|\mathbf{x}\|_p \, \|\mathbf{y}\|_q,$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent, that is, if and only if $\mathbf{x} = \alpha \, \mathbf{y}$ for some $\alpha \in \mathbb{C}$.

For $p = 2$, this is the Cauchy–Schwarz inequality.

Theorem 1.2.6: Unitary Invariance of Euclidean Norm

Given any matrix $\mathbf{U} \in \mathbb{C}^{m \times d}$ with $m \geq d$ and orthonormal columns, we have

$$\|\mathbf{U}\mathbf{x}\| = \|\mathbf{x}\|.$$

If $m = d$, it is unitary matrix.

Proof.

$$\|\mathbf{U}\mathbf{x}\|^2 = \langle \mathbf{U}\mathbf{x}, \mathbf{U}\mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{U}^*\mathbf{U}\mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2.$$

□

Chapter 2

Matrix

2.1 Matrices

Definition 2.1.1

The fundamental object of study in numerical linear algebra is matrix.

One can think of matrices in two ways:

1. Rectangular array of scalars
2. Linear maps between two vector spaces

Notation: Matrices are typically denoted by boldface uppercase letters, e.g., $\mathbf{A}$.

2.1.1 Rectangular Array of Scalars

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{F}^{m\times n}$$

Sometimes, a_{ij} is denoted by $a_{ij} = [\mathbf{A}]_{ij}$

- Square: $m = n$
- "Tall and Skinny": $m > n$
- "Short and Fat": $m < n$

Definition 2.1.2

Transpose ($\mathbf{A}^{\mathrm{T}}$): for any $\mathbf{A} \in \mathbb{F}^{m\times n}$, $[\mathbf{A}^{\mathrm{T}}]_{ij} = [\mathbf{A}]_{ji}$.

Definition 2.1.3

Conjugate transpose / Adjoint / Hermitian transpose ($\mathbf{A}^*$ or $\mathbf{A}^{\mathrm{H}}$): for any $\mathbf{A} \in \mathbb{C}^{m\times n}$,

$$[\mathbf{A}^*]_{ij} = [\overline{\mathbf{A}}]_{ji}, \quad \text{i.e.,} \quad \mathbf{A}^* = \left(\overline{\mathbf{A}}\right)^{\mathrm{T}}$$

Theorem 2.1.4

$$(\mathbf{AB})^{\mathrm{T}} = \mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}, \quad (\mathbf{AB})^* = \mathbf{B}^*\mathbf{A}^*$$
$$(\mathbf{A}+\mathbf{B})^{\mathrm{T}} = \mathbf{A}^{\mathrm{T}} + \mathbf{B}^{\mathrm{T}}, \quad (\mathbf{A}+\mathbf{B})^* = \mathbf{A}^* + \mathbf{B}^*$$

Definition 2.1.5

- Any $\mathbf{A} \in \mathbb{F}^{n\times n}$ is symmetric if $\mathbf{A}^{\mathrm{T}} = \mathbf{A}$
- Any $\mathbf{A} \in \mathbb{F}^{n\times n}$ is skew-symmetric if $\mathbf{A}^{\mathrm{T}} = -\mathbf{A}$
- Any $\mathbf{A} \in \mathbb{F}^{n\times n}$ is orthogonal if $\mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{I}$
- Any $\mathbf{A} \in \mathbb{C}^{n\times n}$ is Hermitian if $\mathbf{A}^* = \mathbf{A}$
- Any $\mathbf{A} \in \mathbb{C}^{n\times n}$ is skew-Hermitian if $\mathbf{A}^* = -\mathbf{A}$
- Any $\mathbf{A} \in \mathbb{C}^{n\times n}$ is unitary if $\mathbf{A}^*\mathbf{A} = \mathbf{I}$
- Any $\mathbf{A} \in \mathbb{C}^{n\times n}$ is normal if $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$

Theorem 2.1.6

Hermitian matrix has real main diagonal entries

Theorem 2.1.7

Skew-Hermitian matrix has pure imaginary main diagonal entries

Definition 2.1.8

Sum: For any $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{m\times n}$, we have $[\mathbf{A}+\mathbf{B}]_{ij} = [\mathbf{A}]_{ij} + [\mathbf{B}]_{ij}$.

Definition 2.1.9

Scalar multiplication: For any $\mathbf{A} \in \mathbb{F}^{m\times n}$ and scalar λ, we have $[\lambda\mathbf{A}]_{ij} = \lambda[\mathbf{A}]_{ij}$

Definition 2.1.10

Matrix product: For any $\mathbf{A} \in \mathbb{F}^{m\times n}$ and $\mathbf{B} \in \mathbb{F}^{n\times p}$, we have $\mathbf{AB} \in \mathbb{F}^{m\times p}$, where

$$\mathbf{A} = \begin{bmatrix} -\mathbf{a}_1^*- \\ -\mathbf{a}_2^*- \\ \vdots \\ -\mathbf{a}_n^*- \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{bmatrix} \quad \Rightarrow \quad [\mathbf{AB}]_{ij} = \langle \mathbf{a}_i, \mathbf{b}_j \rangle$$

That is, $[\mathbf{AB}]_{ij}$ is the inner product of the ith row of $\mathbf{A}$ and the jth column of $\mathbf{B}$.
Alternatively, let

$$\mathbf{A} = \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{b}_1^*- \\ -\mathbf{b}_2^*- \\ \vdots \\ -\mathbf{b}_n^*- \end{bmatrix} \quad \Rightarrow \quad \mathbf{AB} = \sum_{i=1}^{n} \mathbf{a}_i \mathbf{b}_i^*$$

That is, $\mathbf{AB}$ is the sum of outer products of columns of $\mathbf{A}$ and the corresponding rows of $\mathbf{B}$.

The matrix product is associative and distributive with respect to matrix sum, but it is not in general commutative.

Multiplication in block forms:

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{bmatrix}$$

Outer-product form:

$$\begin{aligned} \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} &= \begin{bmatrix} \mathbf{A}_{11} \\ \mathbf{A}_{21} \end{bmatrix} \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \end{bmatrix} + \begin{bmatrix} \mathbf{A}_{12} \\ \mathbf{A}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{A}_{11}\mathbf{B}_{11} & \mathbf{A}_{11}\mathbf{B}_{12} \\ \mathbf{A}_{21}\mathbf{B}_{11} & \mathbf{A}_{21}\mathbf{B}_{12} \end{bmatrix} + \begin{bmatrix} \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{22}\mathbf{B}_{22} \end{bmatrix} \end{aligned}$$

Definition 2.1.11

$\det : \mathbb{F}^{n\times n} \to \mathbb{F}$, $\det(\mathbf{A}) = \sum_{\pi\in\mathcal{P}} \operatorname{sign}(\pi)\, a_{1\pi_1} a_{2\pi_2} \cdots a_{n\pi_n}$.

Theorem 2.1.12

- $\det(\mathbf{A}^{\mathrm{T}}) = \det(\mathbf{A})$
- $\det(\mathbf{A}^*) = \det(\overline{\mathbf{A}}) = \overline{\det(\mathbf{A})}$
- $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$
- $\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$
- $\det(\alpha\mathbf{A}) = \alpha^n \det(\mathbf{A}), \quad \forall \alpha \in \mathbb{F}$

Proposition 2.1.13

For any unitary matrix $\mathbf{U} \in \mathbb{C}^{n\times n}$, show that $|\det(\mathbf{U})| = 1$.

Proof. Since $\mathbf{U}$ is unitary, we have $\mathbf{U}^*\mathbf{U} = \mathbf{I}$. So we have

$$1 = \det(\mathbf{I}) = \det(\mathbf{U}^*\mathbf{U}) = \det(\mathbf{U}^*)\det(\mathbf{U}) = \overline{\det(\mathbf{U})}\det(\mathbf{U}) = |\det(\mathbf{U})|^2.$$

□

Definition 2.1.14

$\mathrm{Trace} : \mathbb{F}^{n\times n} \to \mathbb{F}$, $\mathrm{Trace}(\mathbf{A}) = \sum_i a_{ii}$.

Theorem 2.1.15

- $\mathrm{Trace}(\mathbf{A}^{\mathrm{T}}) = \mathrm{Trace}(\mathbf{A})$
- $\mathrm{Trace}(\mathbf{A}^*) = \mathrm{Trace}(\overline{\mathbf{A}})$
- $\mathrm{Trace}(\mathbf{AB}) \neq \mathrm{Trace}(\mathbf{A})\mathrm{Trace}(\mathbf{B})$
- $\mathrm{Trace}(\mathbf{ABC}) = \mathrm{Trace}(\mathbf{CAB}) = \mathrm{Trace}(\mathbf{BCA})$
- $\mathrm{Trace}(\mathbf{A}+\mathbf{B}) = \mathrm{Trace}(\mathbf{A}) + \mathrm{Trace}(\mathbf{B})$
- $\mathrm{Trace}(\alpha\mathbf{A}) = \alpha\,\mathrm{Trace}(\mathbf{A}), \quad \forall \alpha \in \mathbb{F}$

2.1.2 Linear Maps Between Two Vector Spaces

Let $\mathcal{U}$ be a vector space of $\dim(\mathcal{U}) = n$ over $\mathbb{F}$ with a basis $\mathcal{B}_\mathcal{U}$

$$\mathcal{B}_\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\} \quad \Longrightarrow \quad \mathbf{U} \triangleq [\mathbf{u}_1 \mid \mathbf{u}_2 \mid \cdots \mid \mathbf{u}_n]$$

Let $\mathcal{V}$ be a vector space of $\dim(\mathcal{V}) = m$ over $\mathbb{F}$ with a basis $\mathcal{B}_\mathcal{V}$

$$\mathcal{B}_\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\} \quad \Longrightarrow \quad \mathbf{V} \triangleq [\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_m]$$

Consider the linear transformation $f : \mathcal{U} \to \mathcal{V}$

$$f(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) = \alpha f(\mathbf{u}_1) + \beta f(\mathbf{u}_2), \quad \forall\, \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}, \quad \alpha, \beta \in \mathbb{F}.$$

For any $x \in \mathcal{U}$, we have

$$\mathbf{x} = \sum_{i=1}^{n} \alpha_i u_i = \mathbf{U}\,[\mathbf{x}]_{B_u} \quad \text{where} \quad \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_d \end{pmatrix} \triangleq [\mathbf{x}]_{B_u}.$$

and

$$f(\mathbf{x}) = \sum_{j=1}^{m} \beta_j v_j = \mathbf{V}\,[f(\mathbf{x})]_{B_v} \quad \text{where} \quad \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_d \end{pmatrix} \triangleq [f(\mathbf{x})]_{B_v}.$$

We also have

$$f(\mathbf{x}) = f\left(\sum_{i=1}^{n} \alpha_i \mathbf{u}_i\right) = \sum_{i=1}^{n} \alpha_i f(\mathbf{u}_i) = \sum_{i=1}^{n} \alpha_i \sum_{j=1}^{m} a_{ji}\mathbf{v}_j = \sum_{i=1}^{n}\sum_{j=1}^{m} a_{ji}\alpha_i \mathbf{v}_j = \sum_{j=1}^{m} \mathbf{v}_j \left(\sum_{i=1}^{n} a_{ji}\alpha_i\right)$$

$$\Rightarrow f(\mathbf{x}) = \underbrace{(\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_m)}_{\mathbf{V}} \cdot \underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_{\mathbf{A}} \cdot \underbrace{\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}}_{[\mathbf{x}]_{\mathcal{B}_\mathcal{U}}}$$

So we ended up having

$$f(\mathbf{x}) = \mathbf{V} \cdot [f(\mathbf{x})]_{\mathcal{B}_\mathcal{V}} = \mathbf{V} \cdot \mathbf{A} \cdot [\mathbf{x}]_{\mathcal{B}_\mathcal{U}}$$

But $\mathbf{V}$ is a basis, so the representation of $f(\mathbf{x})$ with respect to $\mathbf{V}$ must be unique, which means

$$[f(\mathbf{x})]_{\mathcal{B}_\mathcal{V}} = \mathbf{A} \cdot [\mathbf{x}]_{\mathcal{B}_\mathcal{U}}$$

The $m \times n$ matrix $\mathbf{A}$ defined by the scalars a_{ij} is called the matrix representation of f in the ordered bases $\mathcal{B}_\mathcal{U}$ and $\mathcal{B}_\mathcal{V}$.

Theorem 2.1.16: Matrices and Linear Maps

Let $\mathcal{U}$ and $\mathcal{V}$ be, respectively, n and m dimensional vector spaces, and consider their respective basis $B_\mathcal{U}$ and $B_\mathcal{V}$. Suppose $f : \mathcal{U} \to \mathcal{V}$ is a linear map. Then, there exists a unique matrix $\mathbf{A} \in \mathbb{F}^{m\times n}$ such that

$$[f(\mathbf{x})]_{B_\mathcal{V}} = \mathbf{A} \cdot [\mathbf{x}]_{B_\mathcal{U}}, \quad \forall\, \mathbf{x} \in \mathcal{U}.$$

Recipe: The ith column of $\mathbf{A}$ is just $[f(\mathbf{u}_i)]_{\mathcal{B}_\mathcal{V}}$, i.e., find $f(\mathbf{u}_i)$ and write it in the basis $B_\mathcal{V}$:

$$[f(\mathbf{u}_i)]_{\mathcal{B}_\mathcal{V}} = \mathbf{A} \cdot [\mathbf{u}_i]_{\mathcal{B}_\mathcal{U}} = \mathbf{A} \cdot \mathbf{e}_i = \mathbf{a}_i$$

Example.

> Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ be defined as $f(x, y) = (x + y,\ 2x - y,\ x - y)$. Let the bases be:
>
> $$\mathcal{B}_{\mathbb{R}^2} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \quad \mathcal{B}_{\mathbb{R}^3} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$
>
> What is the corresponding matrix?

$$f(1, 0) = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}_{\mathcal{B}_{\mathbb{R}^3}}, \qquad f(0, 1) = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}_{\mathcal{B}_{\mathbb{R}^3}}$$

So,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 1 & -1 \end{bmatrix}$$

That is, for any $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$, its matrix representation is:

$$f(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + y \\ 2x - y \\ x - y \end{bmatrix}$$

Example.

> Let $f(x, y) = (x + y,\ 2x - y,\ x - y)$ be as before, but let's change the basis to
>
> $$\tilde{\mathcal{B}}_{\mathbb{R}^2} = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}, \quad \tilde{\mathcal{B}}_{\mathbb{R}^3} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} \right\}$$
>
> What is the corresponding matrix now?

$$f(2, 1) = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} \Rightarrow \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 1 & -2 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}$$

$$f(1, 1) = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} \Rightarrow \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 1 & -2 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

So,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 1 & -2 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 1 & 0 \end{bmatrix} = \begin{pmatrix} 7 & 7 \\ -2 & -3 \\ 2 & 2 \end{pmatrix}$$

2.2 Matrix Subspaces

So, any linear map on a finite dimensional vector space can be represented uniquely by a matrix. So, let's consider the matrix $\mathbf{A} \in \mathbb{F}^{m \times n}$, i.e.,

$$\mathbf{A} : \mathbb{F}^n \longrightarrow \mathbb{F}^m$$

Definition 2.2.1

$\mathrm{dom}(\mathbf{A}) = \mathbb{F}^n$

Definition 2.2.2

$\text{Range}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{F}^m \mid \mathbf{y} = \mathbf{A}\mathbf{x}, \text{ for some } \mathbf{x} \in \mathbb{F}^n\}$

Fact 2.2.3

- $\text{Range}(\mathbf{A})$ is a subspace of $\mathbb{F}^m$
- $\dim(\text{Range}(\mathbf{A})) = \text{Rank}(\mathbf{A})$

Definition 2.2.4

$\text{Null}(\mathbf{A}) = \text{Kernel}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{F}^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\}$

Fact 2.2.5

- $\text{Null}(\mathbf{A})$ is a subspace of $\mathbb{F}^n$
- $\dim(\text{Null}(\mathbf{A})) = \text{Nullity}(\mathbf{A})$

Theorem 2.2.6: Rank–Nullity Theorem

For any $\mathbf{A} \in \mathbb{F}^{m\times n}$, we have

$$\dim(\text{Range}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A})) = n$$

Definition 2.2.7: Orthogonal Complement

$S^\perp = \{\mathbf{v} \mid \langle \mathbf{v}, \mathbf{w}\rangle = 0, \ \forall \mathbf{w} \in S\}$

Theorem 2.2.8: Four Fundamental Subspaces

If $\mathbf{A} \in \mathbb{C}^{m\times n}$, then

$$\text{Null}(\mathbf{A}) = \text{Range}(\mathbf{A}^*)^\perp \quad \text{and} \quad \text{Null}(\mathbf{A}^*) = \text{Range}(\mathbf{A})^\perp$$

Proof. Let $\mathbf{x} \in \text{Null}(\mathbf{A})$. Take any $\mathbf{y} \in \text{Range}(\mathbf{A}^*)$, we have $\mathbf{y} = \mathbf{A}^*\mathbf{z}$ for some $\mathbf{z}$. So we have

$$\langle \mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{A}^*\mathbf{z}\rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{z}\rangle = 0,$$

which implies $\mathbf{x} \in \text{Range}(\mathbf{A}^*)^\perp$, and hence

$$\text{Null}(\mathbf{A}) \subseteq \text{Range}(\mathbf{A}^*)^\perp.$$

Conversely, let $\mathbf{x} \in \text{Range}(\mathbf{A}^*)^\perp$, which means for any $\mathbf{y} \in \text{Range}(\mathbf{A}^*)$, we have $\langle \mathbf{x}, \mathbf{y}\rangle = 0$. In particular, choosing $\mathbf{y} = \mathbf{A}^*\mathbf{A}\mathbf{x}$ implies

$$0 = \langle \mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{A}^*\mathbf{A}\mathbf{x}\rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x}\rangle = \|\mathbf{A}\mathbf{x}\|^2,$$

which gives $\mathbf{A}\mathbf{x} = 0$, which means

$$\mathbf{x} \in \text{Null}(\mathbf{A}), \quad \text{i.e.,} \quad \text{Range}(\mathbf{A}^*)^\perp \subseteq \text{Null}(\mathbf{A}).$$

The other statement is also proved similarly. □

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$

Definition 2.2.9

$\operatorname{colsp}(\mathbf{A}) = \operatorname{Range}(\mathbf{A}) = \{\mathbf{A}\mathbf{x} \mid \mathbf{x} \in \mathbb{C}^n\}$

Definition 2.2.10

$\operatorname{rowsp}(\mathbf{A}) = \operatorname{Range}(\mathbf{A}^{\mathrm{T}}) = \left\{\mathbf{A}^{\mathrm{T}}\mathbf{x} \mid \mathbf{x} \in \mathbb{C}^m\right\}$

Theorem 2.2.11

$\dim(\operatorname{colsp}(\mathbf{A})) = \dim(\operatorname{rowsp}(\mathbf{A})) = \operatorname{Rank}(\mathbf{A}) \leq \min\{m, n\}$

Definition 2.2.12

$\mathbf{A}$ is full rank if $\operatorname{Rank}(\mathbf{A}) = \min\{m, n\}$

Theorem 2.2.13

$\operatorname{Rank}(\mathbf{A}) = \operatorname{Rank}(\mathbf{A}^{\mathrm{T}}) = \operatorname{Rank}(\mathbf{A}^*)$.
$\operatorname{Rank}(\mathbf{A}^*\mathbf{A}) = \operatorname{Rank}(\mathbf{A})$

Theorem 2.2.14

Let $\mathbf{A} \in \mathbb{C}^{m\times m}$, $\mathbf{B} \in \mathbb{C}^{m\times p}$, and $\mathbf{C} \in \mathbb{C}^{p\times p}$, with $\mathbf{A}$ and $\mathbf{C}$ non-singular, then

$$\operatorname{Rank}(\mathbf{AB}) = \operatorname{Rank}(\mathbf{B}) = \operatorname{Rank}(\mathbf{BC}) = \operatorname{Rank}(\mathbf{ABC}),$$

In other words, pre- or post-multiplication by a non-singular matrix leaves rank unchanged.

2.3 Pseudo-inverse

Definition 2.3.1

Let $\mathbf{A} \in \mathbb{F}^{n\times n}$ be said to be non-singular if

$$\mathbf{A}\mathbf{x} = 0 \iff \mathbf{x} = 0$$

Theorem 2.3.2

If $\mathbf{A} \in \mathbb{F}^{m\times n}$ with $m < n$, i.e., a "short and fat" matrix, then it is necessarily singular.

Theorem 2.3.3

If $\mathbf{A} \in \mathbb{F}^{n\times n}$ is non-singular, then

$$(\mathbf{A}^{-1})^{\mathrm{T}} = (\mathbf{A}^{\mathrm{T}})^{-1} \triangleq \mathbf{A}^{-\mathrm{T}}$$

Theorem 2.3.4

If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is non-singular, then

$$(\mathbf{A}^{-1})^* = (\mathbf{A}^*)^{-1} \triangleq \mathbf{A}^{-*}$$

Theorem 2.3.5

A matrix $\mathbf{A} \in \mathbb{F}^{n\times n}$ being non-singular is equivalent to:

- $\text{Rank}(\mathbf{A}) = n$
- $\exists!\, \mathbf{A}^{-1} \in \mathbb{F}^{n\times n}$ such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$
- $\det(\mathbf{A}) \neq 0$
- $\dim(\text{Range}(\mathbf{A})) = n$ and $\dim(\text{Null}(\mathbf{A})) = 0$
- $\text{Null}(\mathbf{A}) = \{\mathbf{0}\}$
- $\mathbf{A}$ has linearly independent rows and columns
- The linear system $\mathbf{Ax} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{F}^n$

Theorem 2.3.6: Pseudo-inverse

For any $\mathbf{A} \in \mathbb{F}^{m\times n}$, there exists a unique matrix $\mathbf{A}^\dagger \in \mathbb{F}^{n\times m}$ that satisfies the following four properties:

- $\mathbf{AA}^\dagger\mathbf{A} = \mathbf{A}$
- $\mathbf{A}^\dagger\mathbf{AA}^\dagger = \mathbf{A}^\dagger$
- $(\mathbf{AA}^\dagger)^* = \mathbf{AA}^\dagger$
- $(\mathbf{A}^\dagger\mathbf{A})^* = \mathbf{A}^\dagger\mathbf{A}$

Theorem 2.3.7

When $\mathbf{A}$ has full column rank, we have a left inverse:

$$\mathbf{A}^\dagger = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*, \quad \text{so } \mathbf{A}^\dagger\mathbf{A} = \mathbf{I}$$

Theorem 2.3.8

When $\mathbf{A}$ has full row rank, we have a right inverse:

$$\mathbf{A}^\dagger = \mathbf{A}^*(\mathbf{AA}^*)^{-1}, \quad \text{so } \mathbf{AA}^\dagger = \mathbf{I}$$

Theorem 2.3.9

If $\mathbf{A}$ is invertible, then its pseudo-inverse is its inverse.

Theorem 2.3.10

- $(\mathbf{A}^\dagger)^\dagger = \mathbf{A}$
- $(\mathbf{A}^\dagger)^{\mathrm{T}} = (\mathbf{A}^{\mathrm{T}})^\dagger, \quad (\mathbf{A}^\dagger)^* = (\mathbf{A}^*)^\dagger$
- In general, $(\mathbf{AB})^\dagger \neq \mathbf{B}^\dagger\mathbf{A}^\dagger$

2.4 Matrix Norms

2.4.1 Entry-wise Matrix Norms

Recall: An $m \times n$ matrix can be viewed as a vector in an mn-dimensional space.

Therefore, any mn-dimensional p-norm can therefore be used for measuring the "size" of such a matrix, i.e.,

$$\|\mathbf{A}\|_p = \|\mathrm{vec}(\mathbf{A})\|_p = \left(\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^p\right)^{1/p}.$$

This is sometimes called "entry-wise matrix norms".

The most famous of these vector norms on matrices is the Frobenius norm.

Definition 2.4.1: Frobenius norm

Given any $\mathbf{A} \in \mathbb{C}^{m\times n}$, the ℓ_2 norm of the associated mn-dimensional vector is the Frobenius norm of the matrix, i.e.,

$$\|\mathbf{A}\|_F \triangleq \left(\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2\right)^{1/2}.$$

It is easy to see that $\|\mathbf{A}\|_F = \sqrt{\mathrm{Trace}(\mathbf{A}^*\mathbf{A})}$ (We'll see later why we necessarily have $\mathrm{Trace}(\mathbf{A}^*\mathbf{A}) \geq 0$).

Theorem 2.4.2: Unitary Invariance of Frobenius Norm

Given any matrix $\mathbf{U} \in \mathbb{C}^{p\times m}$ with $p \geq m$ and orthonormal columns, we have

$$\|\mathbf{UA}\|_F = \|\mathbf{A}\|_F.$$

Proof.

$$\|\mathbf{UA}\|_F^2 = \mathrm{Trace}(\mathbf{A}^*\mathbf{U}^*\mathbf{UA}) = \mathrm{Trace}(\mathbf{A}^*\mathbf{A}) = \|\mathbf{A}\|_F^2.$$

□

2.4.2 Induced by Vector Norms

Definition 2.4.3

Consider a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$. Given any vector norm $\|\cdot\|_p$ for some $p \geq 1$, the corresponding induced matrix norm is defined as

$$\|\mathbf{A}\|_p \triangleq \max_{\substack{\mathbf{x}\in\mathbb{C}^n \\ \|\mathbf{x}\|_p=1}} \|\mathbf{Ax}\|_p = \max_{\substack{\mathbf{x}\in\mathbb{C}^n \\ \mathbf{x}\neq\mathbf{0}}} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p}.$$

For any $\mathbf{x}$, we always have $\|\mathbf{Ax}\|_p \leq \|\mathbf{A}\|_p\|\mathbf{x}\|_p$ for any $p \geq 1$. Induced norms give an indication of the maximum factor by which $\mathbf{A}$ can "stretch" a vector.

Theorem 2.4.4: Sub-multiplicative property

For any two matrices $\mathbf{A} \in \mathbb{C}^{m\times n}$ and $\mathbf{B} \in \mathbb{C}^{n\times p}$, we have $\|\mathbf{AB}\|_p \leq \|\mathbf{A}\|_p \|\mathbf{B}\|_p$

Often the subscript "2" in $\|\mathbf{A}\|_2$ is dropped, i.e., $\|\mathbf{A}\|$ implies the induced 2-Norm, which is at times called the spectral norm.

Example.

Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$, as a mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Theorem 2.4.5: Unitary Invariance of Induced 2-Norm

Given any matrix $\mathbf{U} \in \mathbb{C}^{p\times m}$ with orthonormal columns, we have

$$\|\mathbf{UA}\| = \|\mathbf{A}\|,$$

where the norm here is induced 2-norm.

Proof. The proof is immediate by noticing that for any $\mathbf{x} \in \mathbb{C}^m$, we have

$$\|\mathbf{UAx}\| = \|\mathbf{Ax}\|.$$

□

2.4.3 Condition number

Having vector-induced matrix norms allows us to measure the sensitivity of linear systems to perturbations in the problem data.

Consider $\mathbf{Ax} = \mathbf{b}$. An important question we will run into later is to ask "how accurate we can expect the solution of a linear system to be if the right-hand side $\mathbf{b}$ is perturbed by $\delta\mathbf{b}$?"

More specifically, suppose $\mathbf{A} \in \mathbb{C}^{n\times n}$ is invertible and $\mathbf{b} \in \mathbb{C}^n$. Let $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. Now suppose, $\hat{\mathbf{b}} = \mathbf{b} + \delta\mathbf{b}$ and $\hat{\mathbf{x}} = \mathbf{A}^{-1}\hat{\mathbf{b}}$. How close is $\hat{\mathbf{x}}$ to $\mathbf{x}$?

This can be encapsulated in the notion of the matrix condition number.

$$\left.\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ \mathbf{A}\hat{\mathbf{x}} &= \hat{\mathbf{b}} \end{aligned}\right\} \Longrightarrow \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}) = \delta\mathbf{b} \Longrightarrow (\hat{\mathbf{x}} - \mathbf{x}) = \mathbf{A}^{-1}\delta\mathbf{b}.$$

Now using a vector p-norm and its induced matrix norm, we have

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_p \leq \|\mathbf{A}^{-1}\|_p \, \|\delta\mathbf{b}\|_p,$$

and

$$\|\mathbf{b}\|_p = \|\mathbf{Ax}\|_p \leq \|\mathbf{A}\|_p \, \|\mathbf{x}\|_p,$$

which imply

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|_p}{\|\mathbf{x}\|_p} \leq \underbrace{\|\mathbf{A}\|_p \, \|\mathbf{A}^{-1}\|_p}_{\kappa_p(\mathbf{A})} \frac{\|\delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p}.$$

Definition 2.4.6

Condition number: measures the worst-case change in output relative to the input

$$\kappa_p(\mathbf{A}) = \frac{\lambda_{\max}}{\lambda_{\min}} = \|\mathbf{A}\|_p \, \|\mathbf{A}^{-1}\|_p \geq 1$$

Intuitively, the condition number is the worst-case scenario in term of the relative change in the output as compared with the relative change in the input, i.e.,

$$\text{Condition number} = \max_{\text{all possible changes in the input}} \left(\frac{\text{Relative change in the output}}{\text{Relative change in the input}} \right)$$

The input is $\mathbf{b}$ *and output is* $\mathbf{x}$.

- A well-conditioned problem is one in which all small perturbations of input lead to only small changes in the output.
- An ill-conditioned problem is one in which some small perturbation in the input leads to a large change in the output.

Lemma 2.4.7

We always have $\kappa_p(\mathbf{A}) \geq 1$.

Proof.

$$1 = \|\mathbf{I}\|_p = \|\mathbf{A}\mathbf{A}^{-1}\|_p \leq \|\mathbf{A}\|_p \, \|\mathbf{A}^{-1}\|_p,$$

where we used sub-multiplicative property for matrix induced p-norm.

□

Note: Even though one can define the condition number for any $p \geq 1$, from here on out we will only consider $p = 2$, as it overwhelmingly appears in the literature.

The bound is not pessimistic, i.e., for a given $\mathbf{A}$, there are $\mathbf{b}$ and $\delta\mathbf{b}$ for which we have

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} = \kappa(\mathbf{A}) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}$$

Even though, we motivated condition number by perturbing $\mathbf{b}$, we could have also perturbed $\mathbf{x}$ or $\mathbf{A}$. However, all these perturbations result in the same notion of $\kappa(\mathbf{A})$.

Definition 2.4.8

We can generalize the notion of condition number for tall and skinny, but full rank, matrices as

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \, \|\mathbf{A}^{\dagger}\|.$$

Theorem 2.4.9

A singular matrix has the condition number that is infinite.

Generally speaking, in solving $\mathbf{A}\mathbf{x} = \mathbf{b}$, one can expect to "lose $\log_{10} \kappa(\mathbf{A})$ digits" in computing the solution, except under very special circumstances.

2.5 Matrix Spectral Properties

2.5.1 Eigenvalues and Eigenvectors

We saw that viewing matrices as linear maps is a more informative perspective than treating them merely as rectangular arrays of numbers. From this viewpoint, matrices map vectors to other vectors. A vector has both direction and scale; however, it can be argued that the direction is by far the more important of the two.

Given a matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$, are there nonzero vectors that are mapped to a possibly scaled version of themselves, i.e., their direction is invariant under $\mathbf{A}$?

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Definition 2.5.1

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. If we have

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \in \mathbb{C}^n,\ \mathbf{v} \neq \mathbf{0},\ \lambda \in \mathbb{C},$$

then λ is called an eigenvalue of $\mathbf{A}$ and $\mathbf{v}$ is called an eigenvector of $\mathbf{A}$ associated with λ.

- *Eigenpairs are only defined for square matrices* $\mathbf{A}$
- *Eigenvector can never be the zero vector... otherwise, the definition is vacuous*

Definition 2.5.2

The pair $(\lambda, \mathbf{v})$ is an eigenpair for $\mathbf{A}$. $(\lambda, \mathbf{v})$ is an eigenpair $\iff$ $(\lambda\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$.

If λ is an eigenvalue of $\mathbf{A}$, then $\lambda\mathbf{I} - \mathbf{A}$ is singular. So, eigenvalues are the roots of the characteristic polynomial of $\mathbf{A}$, i.e., $\det(\lambda\mathbf{I} - \mathbf{A}) = 0$, where $\det(\lambda\mathbf{I} - \mathbf{A})$ is a polynomial of degree exactly n in λ, i.e.,

$$\det(\lambda\mathbf{I} - \mathbf{A}) = p_n(\lambda) = \sum_{k=0}^{n} c_k\lambda^k, \quad c_n \neq 0$$

Q: Does any matrix have an eigenpair?
A: Over $\mathbb{C}$ yes, but over $\mathbb{R}$, not necessarily

Example.

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \det\left(\begin{bmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix}\right) = (1-\lambda)^2 - (-1) = \lambda^2 - 2\lambda + 2 = 0 \Longrightarrow \lambda = 1 \pm i$$

Thus, finding eigenvalues $\equiv$ Solving polynomial equations.

By the Fundamental Theorem of Algebra, any matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ has exactly $\underline{n}$ eigenvalues, not necessarily all distinct, among the complex numbers.

Definition 2.5.3: Spectrum

The spectrum of $\mathbf{A} \in \mathbb{C}^{n\times n}$, denoted by $\operatorname{spec}(\mathbf{A})$, is the set of all eigenvalues of $\mathbf{A}$.

$$\operatorname{spec}(\mathbf{A}) = \{\lambda \in \mathbb{C} \mid \exists \mathbf{v} \neq \mathbf{0},\ \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\}$$

Definition 2.5.4: Spectral Radius

$$\rho(\mathbf{A}) \triangleq \max_{\lambda \in \operatorname{spec}(\mathbf{A})} |\lambda|$$

Theorem 2.5.5

$(\lambda, c\mathbf{v})$ is an eigenpair for $\mathbf{A}$ for any $c \in \mathbb{C}$, if $(\lambda, \mathbf{v})$ is an eigenpair of $\mathbf{A}$.

Theorem 2.5.6

- If $\mathbf{A} \in \mathbb{C}^{n\times n}$ and $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, then $\overline{\mathbf{A}}\,\overline{\mathbf{v}} = \overline{\lambda}\,\overline{\mathbf{v}}$
- If $\mathbf{A} \in \mathbb{R}^{n\times n}$ and $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, then $\mathbf{A}\,\overline{\mathbf{v}} = \overline{\lambda}\,\overline{\mathbf{v}}$, i.e., for real matrices, if $(\lambda, \mathbf{v})$ is an eigenpair, then so is $(\overline{\lambda}, \overline{\mathbf{v}})$. In other words, for real matrices, complex eigenvalues come in conjugate pairs.

Theorem 2.5.7

If $\mathbf{A} \in \mathbb{C}^{n\times n}$ and λ is an eigenvalue of $\mathbf{A}$, then $\overline{\lambda}$ is an eigenvalue of $\mathbf{A}^*$, but eigenvectors might not be related.

Theorem 2.5.8

- $\det(\mathbf{A}) = \prod_{i=1}^{n} \lambda_i$
- $\mathrm{Trace}(\mathbf{A}) = \sum_{i=1}^{n} \lambda_i$

Theorem 2.5.9

- $\mathrm{spec}(\mathbf{A}) = \mathrm{spec}(\mathbf{A}^{\mathrm{T}})$ and $\mathrm{spec}(\overline{\mathbf{A}}) = \mathrm{spec}(\mathbf{A}^*)$
- In general, $\mathrm{spec}(\mathbf{A}) \neq \mathrm{spec}(\mathbf{A}^*)$, but always $\rho(\mathbf{A}) = \rho(\mathbf{A}^*)$
- $\forall \alpha \in \mathbb{C}$, $\rho(\alpha\mathbf{A}) = |\alpha|\rho(\mathbf{A})$ and $\rho(\mathbf{A}^k) = [\rho(\mathbf{A})]^k$

Let $p(t)$ be a given polynomial of degree k, i.e.,

$$p(t) = \sum_{i=0}^{k} a_i t^i.$$

We can define matrix polynomial in an analogous way:

Definition 2.5.10

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. Then a matrix polynomial of degree k is defined as

$$p(\mathbf{A}) = \sum_{i=0}^{k} a_i \mathbf{A}^i$$

for $a_i \in \mathbb{C}, i = 1, 2, \ldots, k$.

- The polynomial is monic if $a_k = 1$.
- Polynomial factorization carries over to matrices as well, i.e.,

$$p(\mathbf{A}) = \prod_{i=1}^{k} (\mathbf{A} - \beta_i \mathbf{I}), \quad \beta_i \in \mathbb{C}, \quad i = 1, \ldots, k.$$

Theorem 2.5.11: Spectral Mapping Theorem for Matrix Polynomials

- If $(\lambda, \mathbf{v})$ is an eigenpair for $\mathbf{A} \in \mathbb{C}^{n\times n}$, then $(p(\lambda), \mathbf{v})$ is an eigenpair of $p(\mathbf{A})$
- Conversely, if $k \geq 1$ and μ is an eigenvalue of $p(\mathbf{A})$, then there is some eigenvalue λ of $\mathbf{A}$ such that $\mu = p(\lambda)$

Proof.

- Note that $\mathbf{A}^i\mathbf{v} = \mathbf{A}^{i-1}\mathbf{A}\mathbf{v} = \lambda\mathbf{A}^{i-1}\mathbf{v} = \cdots = \lambda^i\mathbf{v}$. So,

$$p(\mathbf{A})\mathbf{v} = \left(\sum_{i=0}^{k} a_i\mathbf{A}^i\right)\mathbf{v} = \sum_{i=0}^{k} a_i\lambda^i\mathbf{v} = \left(\sum_{i=0}^{k} a_i\lambda^i\right)\mathbf{v} = p(\lambda)\mathbf{v}$$

- Proof of converse is omitted...

□

Theorem 2.5.12

$\mathbf{A}$ is singular if and only if $0 \in \operatorname{spec}(\mathbf{A})$

Proof.

$$\begin{aligned}\mathbf{A} \text{ is singular} &\iff \exists \mathbf{v} \neq \mathbf{0} \text{ s.t. } \mathbf{A}\mathbf{v} = \mathbf{0}\\ &\iff \exists \mathbf{v} \neq \mathbf{0} \text{ s.t. } \mathbf{A}\mathbf{v} = 0\mathbf{v}\\ &\iff 0 \in \operatorname{spec}(\mathbf{A})\end{aligned}$$

□

Theorem 2.5.13

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$, and $\lambda, \mu \in \mathbb{C}$. Then

$$\lambda \in \operatorname{spec}(\mathbf{A}) \iff \lambda + \mu \in \operatorname{spec}(\mathbf{A} + \mu\mathbf{I}).$$

Proof.

$$\begin{aligned}\lambda \in \operatorname{spec}(\mathbf{A}) &\iff \exists \mathbf{v} \neq \mathbf{0} \text{ s.t. } \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\\ &\iff \mathbf{A}\mathbf{v} + \mu\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}\\ &\iff (\mathbf{A} + \mu\mathbf{I})\mathbf{v} = (\lambda + \mu)\mathbf{v}\\ &\iff \lambda + \mu \in \operatorname{spec}(\mathbf{A} + \mu\mathbf{I})\end{aligned}$$

□

Theorem 2.5.14

If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is Hermitian, then all its eigenvalues are real.

Proof. Suppose $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$. We have,

$$\lambda\mathbf{v}^*\mathbf{v} = \mathbf{v}^*\mathbf{A}\mathbf{v} = \mathbf{v}^*\mathbf{A}^*\mathbf{v}.$$

On the other hand,

$$\begin{aligned}\lambda\mathbf{v}^*\mathbf{v} = \mathbf{v}^*\mathbf{A}\mathbf{v} &\iff (\lambda\mathbf{v}^*\mathbf{v})^* = (\mathbf{v}^*\mathbf{A}\mathbf{v})^*\\ \iff \overline{\lambda}(\mathbf{v}^*\mathbf{v}) = (\mathbf{A}\mathbf{v})^*(\mathbf{v}^*)^* &\iff \overline{\lambda}\mathbf{v}^*\mathbf{v} = \mathbf{v}^*\mathbf{A}^*\mathbf{v}.\end{aligned}$$

So $\lambda = \overline{\lambda}$, which implies $\lambda \in \mathbb{R}$. □

For certain types of matrices, computing eigenvalues is immediate.

Example.

(Lower or Upper) Triangular Matrix:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ & a_{22} & a_{23} & \cdots & a_{2n} \\ & & a_{33} & \cdots & a_{3n} \\ & & & \ddots & \vdots \\ 0 & & & & a_{nn} \end{pmatrix} \in \mathbb{C}^{n\times n}$$

$$p_{\mathbf{A}}(\lambda) = \prod_{i=1}^{n}(\lambda - a_{ii}) \Longrightarrow \operatorname{spec}(\mathbf{A}) = \{a_{11}, a_{22}, \ldots, a_{nn}\}$$

Example.

More generally for block (lower or upper) Triangular Matrix:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} & \cdots & \mathbf{A}_{1k} \\ & \mathbf{A}_{22} & \mathbf{A}_{23} & \cdots & \mathbf{A}_{2k} \\ & & \mathbf{A}_{33} & \cdots & \mathbf{A}_{3k} \\ & & & \ddots & \vdots \\ 0 & & & & \mathbf{A}_{kk} \end{pmatrix} \in \mathbb{C}^{n\times n}$$

$$p_{\mathbf{A}}(\lambda) = \prod_{i=1}^{k} p_{\mathbf{A}_{ii}}(\lambda) \Longrightarrow \operatorname{spec}(\mathbf{A}) = \bigcup_{i=1}^{k} \operatorname{spec}(\mathbf{A}_{ii})$$

Since diagonal/triangular matrices are nice, is there a way to "transform" all matrices to look diagonal/triangular without affecting their spectrum?

Definition 2.5.15

Let $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n\times n}$. We say that $\mathbf{B}$ is similar to $\mathbf{A}$ if there exists a non-singular matrix $\mathbf{S} \in \mathbb{C}^{n\times n}$ such that

$$\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}.$$

Two similar matrices share the same spectrum and the same characteristic polynomial.

Theorem 2.5.16

If $\mathbf{A}$ and $\mathbf{B}$ are similar, then they have the same characteristic polynomial.

Proof.

$$\begin{aligned} p_{\mathbf{B}}(\lambda) &= \det(\lambda\mathbf{I} - \mathbf{B}) = \det(\lambda\mathbf{I} - \mathbf{S}^{-1}\mathbf{A}\mathbf{S}) \\ &= \det(\lambda\mathbf{S}^{-1}\mathbf{S} - \mathbf{S}^{-1}\mathbf{A}\mathbf{S}) \\ &= \det(\mathbf{S}^{-1}(\lambda\mathbf{I} - \mathbf{A})\mathbf{S}) \\ &= \det(\mathbf{S}^{-1})\det(\lambda\mathbf{I} - \mathbf{A})\det(\mathbf{S}) \\ &= \det(\lambda\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}(\lambda) \end{aligned}$$

□

Since diagonal matrices are the simplest, we might be interested in knowing which matrices are similar to diagonal matrices?

Definition 2.5.17: Diagonalizable Matrix

If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is similar to a diagonal matrix, then $\mathbf{A}$ is said to be diagonalizable.

Theorem 2.5.18

The matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is diagonalizable if and only if it has n linearly independent eigenvectors.

Proof. We have $\mathbf{A}\mathbf{v}_i = \lambda_i \mathbf{v}_i$, for $i = 1, \dots, n$. So, we can write

$$\mathbf{A} \underbrace{\begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n\end{bmatrix}}_{\mathbf{V}} = \underbrace{\begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n\end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix}\lambda_1 & 0 & \dots & 0\\ 0 & \lambda_2 & \dots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \dots & \lambda_n\end{bmatrix}}_{\mathbf{\Lambda}}.$$

If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ are linearly independent, the matrix $\mathbf{V} \in \mathbb{C}^{n\times n}$ is invertible, and hence

$$\mathbf{A} = \mathbf{V}^{-1}\,\mathbf{\Lambda}\,\mathbf{V}$$

(diagonalisation of $\mathbf{A}$). □

Simple Criterion: If all eigenvalues of $\mathbf{A} \in \mathbb{C}^{n\times n}$ are simple, i.e., if $\mathbf{A}$ has n unique eigenvalues, then $\mathbf{A}$ is diagonalizable.

Theorem 2.5.19

Let $\mathbf{A}$ be diagonalizable, and define $\mathbf{V} = [\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_n] \in \mathbb{C}^{n\times n}$, to be the set of linearly independent eigenvectors of $\mathbf{A}$. Then

$$\mathbf{V}^{-1}\mathbf{A}\mathbf{V} = \begin{pmatrix}\lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n\end{pmatrix} \triangleq \mathbf{\Lambda}$$

Here, λ_i's are eigenvalues of $\mathbf{A}$.

The matrix $\mathbf{V}$ is not necessarily unique, e.g., $\alpha\mathbf{V}$ for any $\alpha \neq 0$ works too

As a rule of thumb, almost every matrix in $\mathbb{C}^{n\times n}$ is diagonalizable.

More precisely: the set of matrices in $\mathbb{C}^{n\times n}$ that are not diagonalizable has Lebesgue measure zero.

For a unitary matrix $\mathbf{U} \in \mathbb{C}^{n\times n}$, we have $\mathbf{U}^{-1} = \mathbf{U}^*$. So when is a matrix $\mathbf{A}$ unitarily similar to a diagonal matrix, i.e.,

$$\mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{\Lambda}?$$

Definition 2.5.20: Unitarily Diagonalizable Matrix

We say that $\mathbf{A}$ is unitarily diagonalizable if it is unitarily similar to a diagonal matrix.

Recall: $\mathbf{A} \in \mathbb{C}^{n\times n}$ is normal if and only if $\mathbf{A}\mathbf{A}^* = \mathbf{A}^*\mathbf{A}$.

Theorem 2.5.21

A matrix is unitarily diagonalizable if and only if it is normal.

- *Among complex matrices, all unitary, Hermitian, and skew-Hermitian matrices are normal.*
- *Among real matrices, all orthogonal, symmetric, and skew-symmetric matrices are normal.*

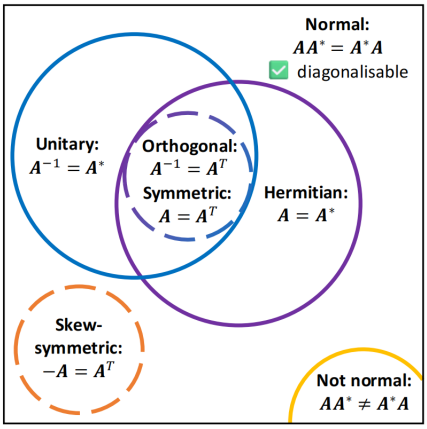


Figure 2.1: Matrix Family Diagram

It is not the case that all normal matrices are either unitary or (skew-)Hermitian,

Example.

> $\forall a, b \in \mathbb{C}$,
> $$\begin{bmatrix} a & b \\ -b & a \end{bmatrix},$$
> is normal and has eigenvalues $a \pm ib$.

2.5.2 Singular Value Decomposition

Singular Value Decomposition

We saw when a square matrix is unitarily diagonalizable, i.e., it is similar to a diagonal matrix by unitary similarity transformation, i.e., $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{\Lambda}$. But a non-normal matrix is not unitarily diagonalizable.

Question: If we relax "similarity" and allow for unitary matrices to be selected independently, can we still "diagonalize" a matrix? i.e., instead of unitary similarity, we seek unitary equivalence.

Singular Value Decomposition (SVD): Any matrix of any size can be "diagonalized" by a suitable pre- and post-multiplication by unitary matrices.

Theorem 2.5.22: Singular Value Decomposition (SVD) - General Matrix

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$, $q = \min\{m, n\}$, and denote $\text{Rank}(\mathbf{A}) = r \leq q$. There are two unitary matrices $\mathbf{U} \in \mathbb{C}^{m\times m}$ and $\mathbf{V} \in \mathbb{C}^{n\times n}$, and diagonal matrix

$$\mathbf{\Sigma}_q = \begin{pmatrix} \sigma_1 & & & & & & \\ & \sigma_2 & & & & & \\ & & \ddots & & & & \\ & & & \sigma_r & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix} \in \mathbb{R}^{q\times q},$$

such that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$, and $\mathbf{U}^*\mathbf{A}\mathbf{V} = \mathbf{\Sigma} \in \mathbb{R}^{m\times n}$, where

$$\mathbf{\Sigma} = \begin{cases} \mathbf{\Sigma}_m & m = n \\ \begin{bmatrix} \mathbf{\Sigma}_m & \mathbf{0}_{m\times(n-m)} \end{bmatrix} & m \leq n \\ \begin{bmatrix} \mathbf{\Sigma}_n \\ \mathbf{0}_{(m-n)\times n} \end{bmatrix} & m \geq n \end{cases}$$

Definition 2.5.23

Diagonal entries of $\mathbf{\Sigma}_q$ are called the singular values of $\mathbf{A}$

Definition 2.5.24

Columns of $\mathbf{U}$ and $\mathbf{V}$ are called, respectively, left and right singular vectors of $\mathbf{A}$

$$\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i, \quad i = 1, \ldots, \min\{m, n\}.$$

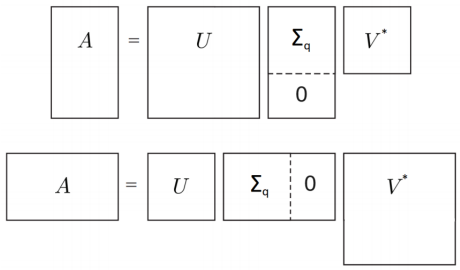


Figure 2.2: Singular Value Decomposition

- *The unitary factors in SVD are never unique, e.g., we may replace* $\mathbf{U}$ *with* $-\mathbf{U}$ *and* $\mathbf{V}$ *with* $-\mathbf{V}$.
- *The matrix* $\mathbf{\Sigma}$ *is determined up to permutation of its diagonal entries.*

Convention: To make $\mathbf{\Sigma}$ unique, one often requires that the singular values be arranged in non-increasing order (as in our definition here), but other choices are possible.

Geometric Interpretation of SVD

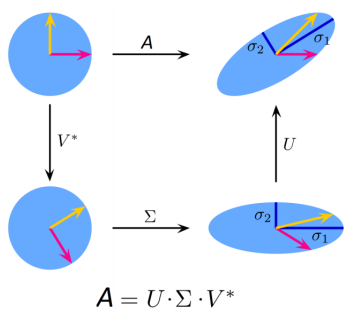


Figure 2.3: Geometric Interpretation of SVD

Theorem 2.5.25

Let $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| = 1\}$ and consider $\mathcal{E} \triangleq \{\mathbf{A}\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \in \mathcal{S}\}$. Then

$$\mathcal{E} = \mathbf{U}\mathcal{E}_0, \quad \text{where} \quad \mathcal{E}_0 = \left\{ \mathbf{y} \in \mathbb{R}^n \mid \sum_{i=1}^{n} \frac{y_i^2}{\sigma_i^2} = 1 \right\}.$$

Recall: For a given matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$, we saw that the condition number allows us to measure the sensitivity of problems with respect to perturbations. When spectral norm is used, condition number

can also be equivalently written (in below, we assume $\mathbf{A}$ has full-column rank):

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \, \|\mathbf{A}^\dagger\| = \frac{\sigma_1}{\sigma_n}$$

This way, we also obtain a nice geometric interpretation of $\kappa(\mathbf{A})$ as the amount of distortion, caused by $\mathbf{A}$, to a unit sphere, i.e., it measures how badly $\mathbf{A}$ distorts the geometry of the space.

Properties of SVD

Theorem 2.5.26

Any real matrix has a singular value decomposition in which all three factors are real.

Theorem 2.5.27

For any $\mathbf{A} \in \mathbb{C}^{m\times n}$, we have

$$\sigma_i = \sqrt{\lambda_i(\mathbf{A}^*\mathbf{A})} = \sqrt{\lambda_i(\mathbf{A}\mathbf{A}^*)}, \quad i = 1, 2, \ldots, \mathrm{Rank}(\mathbf{A}).$$

Proof. Assume, without loss of generality, that $m \geq n$. Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$. We have

$$\mathbf{A}^*\mathbf{A} = \mathbf{V}\mathbf{\Sigma}^{\mathrm{T}}\mathbf{U}^*\mathbf{U}\mathbf{\Sigma}\mathbf{V}^* = \mathbf{V}\left(\mathbf{\Sigma}^{\mathrm{T}}\mathbf{\Sigma}\right)\mathbf{V}^* = \mathbf{V}\mathbf{\Sigma}_n^2\mathbf{V}^* \Longrightarrow \ldots$$

Similarly,

$$\mathbf{A}\mathbf{A}^* = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*\mathbf{V}\mathbf{\Sigma}^{\mathrm{T}}\mathbf{U}^* = \mathbf{U}\begin{bmatrix} \mathbf{\Sigma}_n^2 & \mathbf{0}_{n\times(m-n)} \\ \mathbf{0}_{(m-n)\times n} & \mathbf{0}_{(m-n)\times(m-n)} \end{bmatrix}\mathbf{U}^* \Longrightarrow \ldots$$

□

Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ be SVD of $\mathbf{A} \in \mathbb{C}^{m\times n}$ and for some $1 \leq r \leq \min\{m, n\}$,

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_{\min\{m,n\}} = 0.$$

Theorem 2.5.28

$\mathrm{Rank}(\mathbf{A}) = r$, and $\|\mathbf{A}\| = \sigma_1$

Theorem 2.5.29

Compact SVD: Discard zero entries on $\mathbf{\Sigma}$ to get $\mathbf{A} = \mathbf{U}_r\mathbf{\Sigma}_r\mathbf{V}_r^*$ where $\mathbf{U}_r \in \mathbb{C}^{m\times r}$, $\mathbf{\Sigma}_r \in \mathbb{C}^{r\times r}$, $\mathbf{V}_r \in \mathbb{C}^{n\times r}$ as

$$\mathbf{A} = \underbrace{[\mathbf{u}_1\,\mathbf{u}_2\,\ldots\,\mathbf{u}_r]}_{\mathbf{U}_r}\underbrace{\mathrm{diag}(\sigma_1, \sigma_2, \ldots, \sigma_r)}_{\mathbf{\Sigma}_r}\underbrace{\begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_r\end{bmatrix}^*}_{\mathbf{V}_r^*}$$

$$= \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^* \quad \text{(outer product form)}$$

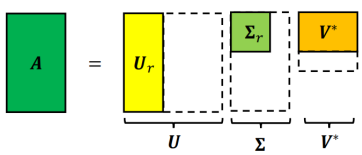


Figure 2.4: Compact SVD

Theorem 2.5.30

- $\text{Null}(\mathbf{A}) = \text{Span}\{\mathbf{v}_{r+1}, \mathbf{v}_{r+2}, \ldots, \mathbf{v}_n\}$
- $\text{Range}(\mathbf{A}) = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r\}$

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V} \qquad (m > n)$$

$$\mathbf{A} = \underbrace{\begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \ldots & \mathbf{u}_r\end{bmatrix}}_{\text{Range}(\mathbf{A})} \underbrace{\begin{bmatrix}\mathbf{u}_{r+1} & \ldots & \mathbf{u}_m\end{bmatrix}}_{\text{Null}(\mathbf{A}^*)} \left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \ddots & & \mathbf{0}_{r\times(n-r)} \\ & & \sigma_r & \\ \hline & \mathbf{0}_{(m-r)\times r} & & \mathbf{0}_{(m-r)\times(n-r)} \end{array}\right] \left[\begin{array}{c} \mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \vdots \\ \mathbf{v}_r^* \\ \hline \mathbf{v}_{r+1}^* \\ \vdots \\ \mathbf{v}_n^* \end{array}\right] \begin{array}{l} \left.\vphantom{\begin{array}{c}\mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \vdots \\ \mathbf{v}_r^*\end{array}}\right\} \text{Range}(\mathbf{A}^*) \\ \left.\vphantom{\begin{array}{c}\mathbf{v}_{r+1}^* \\ \vdots \\ \mathbf{v}_n^*\end{array}}\right\} \text{Null}(\mathbf{A}) \end{array}$$

Theorem 2.5.31

$\mathbf{A}^\dagger = \mathbf{V}\boldsymbol{\Sigma}^\dagger\mathbf{U}^* = \mathbf{V}_r\boldsymbol{\Sigma}_r^{-1}\mathbf{U}_r^*, \quad \|\mathbf{A}^\dagger\| = 1/\sigma_r$

Theorem 2.5.32

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be a normal matrix whose (not necessarily distinct) eigenvalues are $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then the singular values of $\mathbf{A}$ are $|\lambda_1|, |\lambda_2|, \ldots, |\lambda_n|$.

Proof. Since $\mathbf{A}$ is a normal matrix, it can be unitarily diagonalized as $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^*$, where $\boldsymbol{\Lambda}$ is the diagonal matrix of eigenvalues of $\mathbf{A}$. Now, we have

$$\sigma_i = \sqrt{i^{\text{th}} \text{ eigenvalue of } \mathbf{A}^*\mathbf{A}} = \sqrt{i^{\text{th}} \text{ eigenvalue of } \overline{\mathbf{U}}\mathbf{U}^*\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^*}$$

$$= \sqrt{i^{\text{th}} \text{ eigenvalue of } \overline{\mathbf{U}}\boldsymbol{\Lambda}\boldsymbol{\Lambda}\mathbf{U}^*} = \sqrt{\overline{\lambda_i(\mathbf{A})}\lambda_i(\mathbf{A})} = \sqrt{|\lambda_i(\mathbf{A})|^2} = |\lambda_i(\mathbf{A})|$$

□

Schatten p-norm

Recall: We discussed two notions of norms (i) entry-wise, and (ii) induced. It turns out, there is also a third notion based on singular values, and is referred to as Schatten norm.

Definition 2.5.33

Schatten p-norm of a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ is defined by applying vector p-norm to the vector of singular values, i.e.,

$$\|\mathbf{A}\|_p = \left(\sum_{i=1}^{\text{Rank}(\mathbf{A})} \sigma_i^p\right)^{1/p}.$$

Theorem 2.5.34

All Schatten norms are sub-multiplicative: $\|\mathbf{AB}\|_p \leq \|\mathbf{A}\|_p\|\mathbf{B}\|_p$

Theorem 2.5.35

They are also unitarily invariant: $\|\mathbf{UA}\|_p = \|\mathbf{A}\|_p$

The most famous cases are $p = 1, 2, \infty$

- $p = 1$: It is called the nuclear norm or the trace norm
- $p = 2$: It coincides with Frobenius norm
- $p = \infty$: It coincides with spectral norm, i.e., ℓ_2-induced norm.
-

Matrix Low-Rank Approximation

Consider a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ with rank $r = \operatorname{rank}(\mathbf{A})$, and let its singular-value decomposition (SVD) be

$$\mathbf{A} \;=\; \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^* \;=\; \mathbf{U}_r\,\boldsymbol{\Sigma}_r\,\mathbf{V}_r^*,$$

where $\mathbf{U}_r \in \mathbb{C}^{m\times r}$, $\boldsymbol{\Sigma}_r \in \mathbb{R}^{r\times r}$, and $\mathbf{V}_r \in \mathbb{C}^{n\times r}$ correspond to the non-zero singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.
Using the outer-product view of a matrix–matrix product, this can be written as a sum of rank-one matrices:

$$\mathbf{A} \;=\; \sum_{i=1}^{r} \sigma_i\,\mathbf{u}_i\mathbf{v}_i^*,$$

where $\mathbf{u}_i$ and $\mathbf{v}_i$ are the i-th columns of $\mathbf{U}$ and $\mathbf{V}$, respectively.
For any $1 \leq k \leq r$, the truncated sum

$$\mathbf{A}_k \;=\; \sum_{i=1}^{k} \sigma_i\,\mathbf{u}_i\mathbf{v}_i^*$$

defines a rank-k matrix that captures part of the "energy" of $\mathbf{A}$. This motivates the notion of *low-rank matrix approximation.*

Theorem 2.5.36: Eckart–Young–Mirsky Theorem

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$ with singular-value decomposition $\mathbf{A} = \sum_{i=1}^{r} \sigma_i\,\mathbf{u}_i\mathbf{v}_i^*$, where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ are the non-zero singular values of $\mathbf{A}$. For any $1 \leq k < r$, define

$$\mathbf{A}_k \;:=\; \sum_{i=1}^{k} \sigma_i\,\mathbf{u}_i\mathbf{v}_i^*$$

Then $\mathbf{A}_k$ is the best rank-k approximation to $\mathbf{A}$ in both the spectral norm and the Frobenius norm, i.e.

$$\min_{\operatorname{rank}(\mathbf{B})\leq k} \|\mathbf{A}-\mathbf{B}\|_2 \;=\; \|\mathbf{A}-\mathbf{A}_k\|_2 \;=\; \sigma_{k+1},$$

$$\min_{\operatorname{rank}(\mathbf{B})\leq k} \|\mathbf{A}-\mathbf{B}\|_F \;=\; \|\mathbf{A}-\mathbf{A}_k\|_F \;=\; \left(\sum_{i=k+1}^{r} \sigma_i^2\right)^{1/2}.$$

2.6 Special Matrix Types

2.6.1 Diagonal Matrix

Definition 2.6.1: Diagonal

$$\mathbf{D} = \begin{bmatrix} d_{11} & & & \\ & d_{22} & & \\ & & \ddots & \\ & & & d_{nn} \end{bmatrix}$$

Theorem 2.6.2

- $\operatorname{spec}(\mathbf{D}) = \{d_{11}, d_{22}, \ldots, d_{nn}\}$
- $\det(\mathbf{D}) = \prod_{i=1}^{n} d_{ii}$

Theorem 2.6.3

$\mathbf{D}$ is non-singular iff $d_{ii} \neq 0,\ \forall i$

$$\begin{bmatrix} d_{11} & & & \\ & d_{22} & & \\ & & \ddots & \\ & & & d_{nn} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} d_{11}a_{11} & d_{11}a_{12} & \cdots & d_{11}a_{1n} \\ d_{22}a_{21} & d_{22}a_{22} & \cdots & d_{22}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{nn}a_{n1} & d_{nn}a_{n2} & \cdots & d_{nn}a_{nn} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} d_{11} & & & \\ & d_{22} & & \\ & & \ddots & \\ & & & d_{nn} \end{bmatrix} = \begin{bmatrix} d_{11}a_{11} & d_{22}a_{12} & \cdots & d_{nn}a_{1n} \\ d_{11}a_{21} & d_{22}a_{22} & \cdots & d_{nn}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{11}a_{n1} & d_{22}a_{n2} & \cdots & d_{nn}a_{nn} \end{bmatrix}$$

Definition 2.6.4: Block-diagonal

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_{11} & & & \\ & \mathbf{D}_{22} & & \\ & & \ddots & \\ & & & \mathbf{D}_{kk} \end{bmatrix}$$

Theorem 2.6.5

- $\operatorname{spec}(\mathbf{D}) = \bigcup_{i=1}^{k} \operatorname{spec}(\mathbf{D}_{ii})$
- $\det(\mathbf{D}) = \prod_{i=1}^{k} \det(\mathbf{D}_{ii})$

Theorem 2.6.6

$\mathbf{D}$ is non-singular iff each $\mathbf{D}_{ii}$ is non-singular

$$\begin{bmatrix} \mathbf{D}_{11} & & & \\ & \mathbf{D}_{22} & & \\ & & \ddots & \\ & & & \mathbf{D}_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \cdots & \mathbf{A}_{1k} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \cdots & \mathbf{A}_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{k1} & \mathbf{A}_{k2} & \cdots & \mathbf{A}_{kk} \end{bmatrix} = \begin{bmatrix} \mathbf{D}_{11}\mathbf{A}_{11} & \mathbf{D}_{11}\mathbf{A}_{12} & \cdots & \mathbf{D}_{11}\mathbf{A}_{1k} \\ \mathbf{D}_{22}\mathbf{A}_{21} & \mathbf{D}_{22}\mathbf{A}_{22} & \cdots & \mathbf{D}_{22}\mathbf{A}_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{D}_{kk}\mathbf{A}_{k1} & \mathbf{D}_{kk}\mathbf{A}_{k2} & \cdots & \mathbf{D}_{kk}\mathbf{A}_{kk} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \cdots & \mathbf{A}_{1k} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \cdots & \mathbf{A}_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{k1} & \mathbf{A}_{k2} & \cdots & \mathbf{A}_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{D}_{11} & & & \\ & \mathbf{D}_{22} & & \\ & & \ddots & \\ & & & \mathbf{D}_{kk} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{D}_{11} & \mathbf{A}_{12}\mathbf{D}_{22} & \cdots & \mathbf{A}_{1k}\mathbf{D}_{kk} \\ \mathbf{A}_{21}\mathbf{D}_{11} & \mathbf{A}_{22}\mathbf{D}_{22} & \cdots & \mathbf{A}_{2k}\mathbf{D}_{kk} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{k1}\mathbf{D}_{11} & \mathbf{A}_{k2}\mathbf{D}_{22} & \cdots & \mathbf{A}_{kk}\mathbf{D}_{kk} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{D}_{11} & & & \\ & \mathbf{D}_{22} & & \\ & & \ddots & \\ & & & \mathbf{D}_{kk} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{D}_{11}^{-1} & & & \\ & \mathbf{D}_{22}^{-1} & & \\ & & \ddots & \\ & & & \mathbf{D}_{kk}^{-1} \end{bmatrix}$$

2.6.2 Triangular Matrix

Definition 2.6.7: Triangular

$$\mathbf{T} = \begin{bmatrix} t_{11} & t_{12} & \cdots & t_{1n} \\ & t_{22} & \cdots & t_{2n} \\ & & \ddots & \vdots \\ & & & t_{nn} \end{bmatrix}$$

Theorem 2.6.8

- $\operatorname{spec}(\mathbf{T}) = \{t_{11}, t_{22}, \ldots, t_{nn}\}$
- $\det(\mathbf{T}) = \prod_{i=1}^{k} t_{ii}$

Theorem 2.6.9

$\mathbf{T}$ is non-singular iff all $t_{ii} \neq 0$

Theorem 2.6.10

$\operatorname{Rank}(\mathbf{T}) \geq \#$ of nonzero t_{ii}

Example.

The singular values of the strictly upper triangular matrix are $0, |t_{12}|, \ldots, |t_{n-1,n}|$.

$$\begin{bmatrix} 0 & t_{12} & & & \\ & 0 & t_{23} & & \\ & & \ddots & \ddots & \\ & & & & t_{n-1,n} \\ & & & & 0 \end{bmatrix}$$

Sparsity patterns

$$\begin{bmatrix} \star & \star & \cdots & \star \\ & \star & \cdots & \star \\ & & \ddots & \vdots \\ & & & \star \end{bmatrix}^{-1} = \begin{bmatrix} \star & \star & \cdots & \star \\ & \star & \cdots & \star \\ & & \ddots & \vdots \\ & & & \star \end{bmatrix}$$

$$\begin{bmatrix} \star & \star & \cdots & \star \\ & \star & \cdots & \star \\ & & \ddots & \vdots \\ & & & \star \end{bmatrix} \times \begin{bmatrix} \star & \star & \cdots & \star \\ & \star & \cdots & \star \\ & & \ddots & \vdots \\ & & & \star \end{bmatrix} = \begin{bmatrix} \star & \star & \cdots & \star \\ & \star & \cdots & \star \\ & & \ddots & \vdots \\ & & & \star \end{bmatrix}$$

Similar patterns hold for lower triangular matrices

Definition 2.6.11: Block-triangular

$$\mathbf{T} = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \cdots & \mathbf{T}_{1k} \\ & \mathbf{T}_{22} & \cdots & \mathbf{T}_{2k} \\ & & \ddots & \vdots \\ & & & \mathbf{T}_{kk} \end{bmatrix}$$

Theorem 2.6.12

- $\operatorname{spec}(\mathbf{T}) = \bigcup_{i=1}^{k} \operatorname{spec}(\mathbf{T}_{ii})$
- $\det(\mathbf{T}) = \prod_{i=1}^{k} \det(\mathbf{T}_{ii})$

Theorem 2.6.13

$\mathbf{T}$ is non-singular iff all $\mathbf{T}_{ii}$'s are non-singular

Theorem 2.6.14

$\operatorname{Rank}(\mathbf{T}) \geq \sum_{i=1}^{k} \operatorname{Rank}(\mathbf{T}_{ii})$

Sparsity pattern is similar to triangular case, but with respect to blocks

2.6.3 Hessenberg Matrix

Definition 2.6.15: Upper/Lower Hessenberg

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ & a_{32} & a_{33} & \cdots & a_{3n} \\ & & \ddots & \ddots & \vdots \\ & & & a_{n,n-1} & a_{nn} \end{bmatrix}$$

A Hessenberg matrix is "almost" triangular.

Definition 2.6.16

An upper Hessenberg matrix $\mathbf{A}$ is said to be *unreduced* if all its sub-diagonal entries are nonzero

Theorem 2.6.17

$\text{Rank}(\mathbf{A})$ of the unreduced matrix is at least $n-1$ since its first $n-1$ columns are independent.

2.6.4 Permutation Matrix

Definition 2.6.18: Permutation Matrices

Permutation Matrices: exactly one entry in each row and column is equal to 1 and all other entries are 0.

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 3 \\ 1 \end{bmatrix}$$

- Left multiplication of a matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ by $n \times n$ permutation matrix $\mathbf{P}$, i.e., $\mathbf{PA}$, permutes the rows of $\mathbf{A}$
- Right multiplication of a matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ by $n \times n$ permutation matrix $\mathbf{P}$, i.e., $\mathbf{AP}$, permutes the columns of $\mathbf{A}$

Theorem 2.6.19

$\mathbf{P}^\top\mathbf{P} = \mathbf{P}\mathbf{P}^\top = \mathbf{I}$, i.e., $\mathbf{P}$ is orthogonal

Theorem 2.6.20

$\det(\mathbf{P}) = \pm 1$, i.e., permutation matrices are non-singular

Theorem 2.6.21

If $\mathbf{P}$ and $\mathbf{Q}$ are permutation matrices, then so is $\mathbf{PQ}$ and $\mathbf{QP}$ (generally $\mathbf{PQ} \neq \mathbf{QP}$)

2.6.5 Projection Matrix

Definition 2.6.22: Projection Matrices

$\mathbf{P} \in \mathbb{C}^{n\times n}$ is a projection/idempotent if $\mathbf{P}^2 = \mathbf{P}$

Recall. Any $\mathbf{P} \in \mathbb{C}^{n\times n} : \mathbf{v} \in \text{Range}(\mathbf{P}) \implies \mathbf{v} = \mathbf{P}\mathbf{w}$, for some $\mathbf{w} \in \mathbb{C}^n$.

Theorem 2.6.23

Projection $\mathbf{P} \in \mathbb{C}^{n\times n} : \mathbf{v} \in \text{Range}(\mathbf{P}) \implies \mathbf{v} = \mathbf{P}\mathbf{v}$.

Proof.

$$\mathbf{v} \in \text{Range}(\mathbf{P}) \implies \exists \mathbf{w} \in \mathbb{C}^n \text{ s.t. } \mathbf{v} = \mathbf{P}\mathbf{w} \implies \mathbf{P}\mathbf{v} = \mathbf{P}^2\mathbf{w} = \mathbf{P}\mathbf{w} = \mathbf{v}$$

□

Theorem 2.6.24

If $\mathbf{P}$ is a projection, then so is $\mathbf{I} - \mathbf{P}$

Proof.

$$(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$$

□

Theorem 2.6.25

$\lambda \in \{0, 1\}$

Proof. $\mathbf{P}\mathbf{v} = \lambda\mathbf{v} \Longrightarrow \mathbf{P}^2\mathbf{v} = \lambda\mathbf{P}\mathbf{v} \Longrightarrow \mathbf{P}\mathbf{v} = \lambda^2\mathbf{v} \Longrightarrow \lambda = \lambda^2 \Longrightarrow \lambda \in \{0, 1\}$ □

Definition 2.6.26

$\mathbf{P} \in \mathbb{C}^{n\times n}$ is an orthogonal projection if $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{P}^* = \mathbf{P}$.

NB: Other than the identity matrix, orthogonal projectors are not orthogonal matrices!

Theorem 2.6.27

For orthogonal projections, we have

- $\text{Range}(\mathbf{P}) \perp \text{Range}(\mathbf{I} - \mathbf{P})$,
- Pythagorean Theorem: $\|\mathbf{v}\|^2 = \|\mathbf{P}\mathbf{v}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$

Proof.

$$\mathbf{v} \in \text{Range}(\mathbf{P}) \Rightarrow \mathbf{v} = \mathbf{P}\mathbf{v}$$

$$\mathbf{v}' \in \text{Range}(\mathbf{I} - \mathbf{P}) \Rightarrow \mathbf{v}' = (\mathbf{I} - \mathbf{P})\mathbf{v}'$$

Then, $(\mathbf{v}, \mathbf{v}') = (\mathbf{P}\mathbf{v}, (\mathbf{I} - \mathbf{P})\mathbf{v}') = (\mathbf{v}, \mathbf{P}^*(\mathbf{I} - \mathbf{P})\mathbf{v}') = (\mathbf{v}, \mathbf{P}(\mathbf{I} - \mathbf{P})\mathbf{v}') = (\mathbf{v}, (\mathbf{P} - \mathbf{P}^2)\mathbf{v}) = (\mathbf{v}, \mathbf{0}) = 0$. Thus, $\mathbf{z} = \underbrace{\mathbf{P}\mathbf{z}}_{\text{Range}(\mathbf{P})} + \underbrace{(\mathbf{I} - \mathbf{P})\mathbf{z}}_{\text{Range}(\mathbf{I}-\mathbf{P})} \Rightarrow \|\mathbf{v}\|^2 = \|\mathbf{P}\mathbf{v}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$. Indeed

$$\|\mathbf{v}\|^2 = \|\mathbf{P}\mathbf{v} + (\mathbf{I} - \mathbf{P})\mathbf{v}\|^2 = \|\mathbf{P}\mathbf{v}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2 + 2(\mathbf{P}\mathbf{v}, (\mathbf{I} - \mathbf{P})\mathbf{v}) = \|\mathbf{P}\mathbf{v}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$$

□

Theorem 2.6.28

Given $\mathbf{Q} \in \mathbb{C}^{m\times n}$ with orthonormal columns (tall), $\mathbf{P} = \mathbf{Q}\mathbf{Q}^*$ is the unique orthogonal projection onto the $\text{Range}(\mathbf{Q})$

Proof.

$$\mathbf{Q}\mathbf{Q}^*\mathbf{Q}\mathbf{Q}^* = \mathbf{Q}\mathbf{I}\mathbf{Q}^* = \mathbf{Q}\mathbf{Q}^*$$

For any $\mathbf{x}$, $\mathbf{P}\mathbf{x} = \mathbf{Q}(\mathbf{Q}^*\mathbf{x}) \in \text{Range}(\mathbf{Q})$; and for any $\mathbf{z} \in \text{Range}(\mathbf{Q})$ (write $\mathbf{z} = \mathbf{Q}\mathbf{y}$), $\mathbf{P}\mathbf{z} = \mathbf{Q}\mathbf{Q}^*\mathbf{Q}\mathbf{y} = \mathbf{Q}\mathbf{y} = \mathbf{z}$, showing that on this subspace $\mathbf{P}$ equals the identity map; this is precisely the defining property of a projection.

□

Example.

> Let $\mathbf{v} \in \mathbb{C}^n$. Then, $\mathbf{P} = \mathbf{v}\mathbf{v}^*/\|\mathbf{v}\|^2$ is the orthogonal projection along $\mathbf{v}$.

Let $\mathbf{w}$ be an arbitrary vector. Then the projection onto $\mathbf{v}$ is $\dfrac{(\mathbf{v}, \mathbf{w})}{\|\mathbf{v}\|}\dfrac{\mathbf{v}}{\|\mathbf{v}\|} = \left(\dfrac{\mathbf{v}}{\|\mathbf{v}\|}\dfrac{\mathbf{v}^*}{\|\mathbf{v}\|}\right)\mathbf{w}$.

Theorem 2.6.29

Given $\mathbf{A} \in \mathbb{C}^{m\times n}$, $\mathbf{P} = \mathbf{A}\mathbf{A}^\dagger$ is the unique orthogonal projection onto the Range($\mathbf{A}$)

Proof.

$$\mathbf{A}\mathbf{A}^\dagger\mathbf{A}\mathbf{A}^\dagger = \mathbf{A}\mathbf{A}^\dagger.$$

□

2.6.6 Positive (semi-)definite

Definition 2.6.30: Positive (semi-)definite

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$:

- PD: $\mathbf{A} \succ 0 \iff \langle \mathbf{x}, \mathbf{A}\mathbf{x}\rangle > 0, \quad \forall \mathbf{x} \in \mathbb{C}^n \setminus \{0\}$
- PSD: $\mathbf{A} \succeq 0 \iff \langle \mathbf{x}, \mathbf{A}\mathbf{x}\rangle \geq 0, \quad \forall \mathbf{x} \in \mathbb{C}^n$
- $\{\mathbf{A} \in \mathbb{C}^{n\times n} \mid \mathbf{A} \succ 0\} \subsetneq \{\mathbf{A} \in \mathbb{C}^{n\times n} \mid \mathbf{A} \succeq 0\}$

Negative (semi-)definite is defined similarly

Theorem 2.6.31

If $\mathbf{A} \succeq 0$, then $\mathbf{A}$ must be Hermitian.

Proof. Define $\mathbf{B} \triangleq \frac{1}{2}(\mathbf{A} + \mathbf{A}^*)$, $\mathbf{C} \triangleq \frac{1}{2i}(\mathbf{A} - \mathbf{A}^*)$. Clearly, $\mathbf{B}$ and $\mathbf{C}$ are Hermitian and

$$\mathbf{A} = \mathbf{B} + i\mathbf{C}.$$

By assumption, we have $\mathbf{x}^*\mathbf{A}\mathbf{x} \in \mathbb{R}$, and by Hermitian property, we have $\mathbf{x}^*\mathbf{B}\mathbf{x} \in \mathbb{R}$, and $\mathbf{x}^*\mathbf{C}\mathbf{x} \in \mathbb{R}$. So,

$$\mathbf{x}^*\mathbf{A}\mathbf{x} = \mathbf{x}^*\mathbf{B}\mathbf{x} + i\mathbf{x}^*\mathbf{C}\mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{C}^n,$$

$$\text{which implies } \mathbf{C} = \mathbf{0}, \text{ and hence } \mathbf{A} = \mathbf{B}, \text{ i.e., } \mathbf{A} = \mathbf{A}^*.$$

□

Theorem 2.6.32

For any matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$, we have $\mathbf{A}^*\mathbf{A} \succeq 0$ and $\mathbf{A}\mathbf{A}^* \succeq 0$.

Proof. Take any $\mathbf{x} \in \mathbb{C}^n$. We have

$$\langle \mathbf{x}, \mathbf{A}^*\mathbf{A}\mathbf{x}\rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x}\rangle = \|\mathbf{A}\mathbf{x}\|^2 \geq 0.$$

The case of $\mathbf{A}\mathbf{A}^*$ is similar.

□

Theorem 2.6.33

- If $\text{Rank}(\mathbf{A}) = n \Rightarrow \mathbf{A}^*\mathbf{A} \succ 0$
- If $\text{Rank}(\mathbf{A}) = m \Rightarrow \mathbf{A}\mathbf{A}^* \succ 0$

Theorem 2.6.34

- $\mathbf{A} \succ 0 \Rightarrow \lambda_i(\mathbf{A}) > 0, \quad i = 1, \ldots, n$
- $\mathbf{A} \succeq 0 \Rightarrow \lambda_i(\mathbf{A}) \geq 0, \quad i = 1, \ldots, n$
- $\mathbf{A} \prec 0 \Rightarrow \lambda_i(\mathbf{A}) < 0, \quad i = 1, \ldots, n$
- $\mathbf{A} \preceq 0 \Rightarrow \lambda_i(\mathbf{A}) \leq 0, \quad i = 1, \ldots, n$
- For $\mathbf{A} \succ 0$, the eigen decomposition and SVD coincide
- Every PD matrix is invertible and its inverse is also PD
- For $\mathbf{A}, \mathbf{B} \succ 0$ and $\alpha > 0$, we have $\alpha\mathbf{A} \succ 0$ and $\mathbf{A} + \mathbf{B} \succ 0$
- If $\mathbf{A} \succeq 0$, then $\mathbf{B}^*\mathbf{A}\mathbf{B} \succeq 0, \quad \forall \mathbf{B} \in \mathbb{C}^{n \times m}$
- If $\mathbf{A} \succ 0$ and $\mathbf{B}$ has full column rank, then $\mathbf{B}^*\mathbf{A}\mathbf{B} \succ 0$
- *Square Root:* $\mathbf{A} \succeq 0 \Leftrightarrow \exists \mathbf{B} \succeq 0$ such that $\mathbf{B}^2 = \mathbf{A}$

  (This should not be confused with the Cholesky factor, which is sometimes called the square root: the factor here is symmetric while the factor there is triangular.)

Definition 2.6.35

$\mathbf{A}$ is indefinite if and only if it is Hermitian and it has at least one positive and one negative eigenvalue.

NB: $\mathbf{A} \not\succeq 0$ does not imply $\mathbf{A} \prec 0$ (unlike ordering in real numbers...hence "$\succeq$" is called partial order)

Definition 2.6.36: Loewner Partial-Order For Hermitian Matrices

$$\mathbf{A} \succ \mathbf{B} \iff \mathbf{A} - \mathbf{B} \succ 0$$
$$\mathbf{A} \succeq \mathbf{B} \iff \mathbf{A} - \mathbf{B} \succeq 0$$

Theorem 2.6.37

A few interesting properties (after ordering eigenvalues):

- $\mathbf{A} \succeq \mathbf{B} \Rightarrow \lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}), \quad i = 1, 2, \ldots, n$
- $\mathbf{A} \succeq \mathbf{B}$ and $\mathbf{A} \neq \mathbf{B} \Rightarrow \exists i \in \{1, 2, \ldots, n\},\ \lambda_i(\mathbf{A}) > \lambda_i(\mathbf{B})$
- $\mathbf{A} \succ \mathbf{B} \Rightarrow \lambda_i(\mathbf{A}) > \lambda_i(\mathbf{B}), \quad i = 1, 2, \ldots, n$

Definition 2.6.38: Diagonally Dominant

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

- A matrix is DD if $|a_{ii}| \geq \sum_{i \neq j} |a_{ij}|, \quad \forall i$
- A matrix is strictly DD if $|a_{ii}| > \sum_{i \neq j} |a_{ij}|, \quad \forall i$

Theorem 2.6.39: Levy–Desplanques theorem

A SDD matrix is non-singular

Theorem 2.6.40

- If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is Hermitian, diagonally dominant with non-negative diagonals $a_{ii} \geq 0\ \forall i$, then $\mathbf{A}$ is positive semi-definite.
- If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is Hermitian, strictly diagonally dominant with positive diagonals $a_{ii} > 0\ \forall i$, then $\mathbf{A}$ is positive definite.

Converse is not necessarily true,

Example.

$$\mathbf{A} = \begin{bmatrix} 1 & -3/2 \\ -3/2 & 4 \end{bmatrix} \quad \text{is PD but not diagonally dominant.}$$

2.7 Matrix Factorizations

We have already seen Eigen-decomposition, and SVD (there are a few other decompositions related to the spectrum of the matrix such as Schur decomposition and Jordan normal form). Now, we see a few more important matrix factorizations that are used in various applications

2.7.1 LU factorisation

Definition 2.7.1: LU factorisation

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \underbrace{\begin{bmatrix} \ell_{11} & & & \\ \ell_{21} & \ell_{22} & & \\ \vdots & \vdots & \ddots & \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{nn} \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ & u_{22} & \cdots & u_{2n} \\ & & \ddots & \vdots \\ & & & u_{nn} \end{bmatrix}}_{\mathbf{U}} = \mathbf{LU}$$

where $\mathbf{L}$ is Lower triangular and $\mathbf{U}$ is Upper triangular

- *Applicable to square matrices*
- *These factors result from the Gaussian elimination!*

LU factorization is an example of triangular factorizations.

Theorem 2.7.2

LU factorization of a given matrix may or may not exist, and if it exists, it need not be unique

Example.

$\mathbf{A} = \mathbf{LU} = (\mathbf{L}/2)(2\mathbf{U})$

Suppose $\mathbf{A}$ is non-singular and admits an $\mathbf{LU}$ decomposition. Note that

$$\begin{bmatrix} \ell_{11} & & & \\ \ell_{21} & \ell_{22} & & \\ \vdots & \vdots & \ddots & \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{nn} \end{bmatrix} = \begin{bmatrix} 1 & & & \\ \frac{\ell_{21}}{\ell_{11}} & 1 & & \\ \vdots & \vdots & \ddots & \\ \frac{\ell_{n1}}{\ell_{11}} & \frac{\ell_{n2}}{\ell_{22}} & \cdots & 1 \end{bmatrix} \begin{bmatrix} \ell_{11} & & & \\ & \ell_{22} & & \\ & & \ddots & \\ & & & \ell_{nn} \end{bmatrix}$$

and similarly for $\mathbf{U}$.

Theorem 2.7.3: LDU factorisation

Suppose $\mathbf{A}$ is non-singular and admits an $\mathbf{LU}$ decomposition. $\mathbf{A} = \mathbf{LDU}$ ("LDU Factorization"), where $\mathbf{D}$ is a diagonal matrix with non-zero diagonal elements, and $\mathbf{L}$ and $\mathbf{U}$ are unit triangular matrices, i.e., $\ell_{ii} = u_{ii} = 1,\ \forall i$.

$$\underbrace{\begin{bmatrix} 1 & & & \\ \frac{\ell_{21}}{\ell_{11}} & 1 & & \\ \vdots & \vdots & \ddots & \\ \frac{\ell_{n1}}{\ell_{11}} & \frac{\ell_{n2}}{\ell_{22}} & \cdots & 1 \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} \ell_{11}u_{11} & & & \\ & \ell_{22}u_{22} & & \\ & & \ddots & \\ & & & \ell_{nn}u_{nn} \end{bmatrix}}_{\mathbf{D}} \underbrace{\begin{bmatrix} 1 & \frac{u_{12}}{u_{11}} & \cdots & \frac{u_{1n}}{u_{11}} \\ & 1 & \cdots & \frac{u_{2n}}{u_{22}} \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}}_{\mathbf{U}}$$

The factors $\mathbf{L}$, $\mathbf{D}$, and $\mathbf{U}$ are uniquely determined in the LDU factorization.

Singular matrices can have LU factorization

Example.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

For singular matrices, LU need not be unique even if we require $\mathbf{L}$ (or $\mathbf{U}$) to be unit lower (upper) triangular (you cannot have them both be unit triangular (why?))

Example.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 2-a \end{bmatrix}, \quad \forall a.$$

As mentioned before, LU factorization may not exist even for non-singular matrices. Even if it exists, computing it can be numerically unstable.

Example.

LU factorization may not exist:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix} \quad \Longrightarrow \quad \ell_{11}u_{11} = 0$$

So, either $\mathbf{L}$ or $\mathbf{U}$ must be singular, but $\mathbf{A}$ is not singular.

Example.

LU factorization can be numerically unstable:

$$\begin{bmatrix} 10^{-20} & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 10^{20} & 1 \end{bmatrix} \begin{bmatrix} 10^{-20} & 1 \\ 0 & 1-10^{20} \end{bmatrix}$$

The round-off error may result in $1 - 10^{20} \to -10^{20}$. But then,

$$\begin{bmatrix} 1 & 0 \\ 10^{20} & 1 \end{bmatrix} \begin{bmatrix} 10^{-20} & 1 \\ 0 & -10^{20} \end{bmatrix} = \begin{bmatrix} 10^{-20} & 1 \\ 1 & 0 \end{bmatrix}$$

It turns out that a proper permutation in rows (or columns) is sufficient for LU factorization to exist and to be numerically stable in practice.

Theorem 2.7.4

For any $\mathbf{A} \in \mathbb{C}^{n\times n}$, there is a permutation matrix $\mathbf{P} \in \mathbb{R}^{n\times n}$ such that $\mathbf{PA} = \mathbf{LU}$

Computing LU factorization requires $2n^3/3 + \mathcal{O}(n^2)$ flops (e.g., Doolittle algorithm, Crout algorithm, etc). Adding row (or column) permutation, i.e., partial pivoting, adds only a quadratic term. Partial pivoting is explosively unstable for certain matrices, yet stable in practice for almost all real applications.

Theorem 2.7.5: LDL Factorization

If $\mathbf{A} \succ 0$, then $\mathbf{A} = \mathbf{LDL}^*$ ("LDL Factorization") always exists and $\mathbf{D}$ is real positive diagonal.

So for PD matrices, this factorization can be done at half the memory/flops (this is related to Cholesky Decomposition).
For positive (semi-) definite matrices, the LU factorization is referred to as the Cholesky factorization.

Theorem 2.7.6: Cholesky Factorization

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be Hermitian. Then

- $\mathbf{A}$ is positive semidefinite (respectively, positive definite) if and only if there is a lower triangular matrix $\mathbf{L} \in \mathbb{C}^{n\times n}$ with nonnegative (respectively, positive) diagonal entries such that $\mathbf{A} = \mathbf{LL}^*$.
- Furthermore, if $\mathbf{A}$ is positive definite, $\mathbf{L}$ is unique, i.e., there is only one lower triangular matrix $\mathbf{L}$ with strictly positive diagonal entries such that $\mathbf{A} = \mathbf{LL}^*$.
- Finally, if $\mathbf{A}$ is real, $\mathbf{L}$ may be taken to be real.

It is similar to LU factorization in that it gives triangular factors, i.e., $\mathbf{A} = \mathbf{LL}^*$ but it is only for positive semidefinite matrices.

Theorem 2.7.7

- For a PSD matrix, Cholesky factorization always exists.
- For a PD matrix, Cholesky factorization is unique.

By exploiting symmetry, Cholesky factorization requires $n^3/3 + \mathcal{O}(n^2)$ flops, i.e., half as much as LU decompositions for general matrices. It can be done by imposing the quality $\mathbf{A} = \mathbf{LL}^*$, element-wise! All of the issues with stability Gaussian elimination vanish for Cholesky factorization, i.e., this algorithm is always stable!
Relation with LDL: When $\mathbf{A} \succ 0$, from LDL decomposition $\mathbf{A} = \mathbf{LDL}^*$, we can get the Cholesky factorization as $\mathbf{A} = (\mathbf{L}\sqrt{\mathbf{D}})(\mathbf{L}\sqrt{\mathbf{D}})^*$. Conversely, we can also get the factors in $\mathbf{LDL}^*$ from the Cholesky factor.

2.7.2 QR Factorization

Theorem 2.7.8: QR Factorization

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$ be given. Then, there is always a unitary $\mathbf{Q} \in \mathbb{C}^{m\times m}$ and an upper triangular $\mathbf{R} \in \mathbb{C}^{m\times n}$ with non-negative diagonal entries such that $\mathbf{A} = \mathbf{QR}$. If $\mathbf{A}$ is of full rank, then there exists a unique $\mathbf{R}$ that has all positive diagonal elements. If $\mathbf{A}$ is square, $\mathbf{Q}$ is also unique. Furthermore, if $\mathbf{A}$ is real, then the factors $\mathbf{Q}$ and $\mathbf{R}$ may be taken to be real.

Just like **LU** is a reduction to upper-triangular form by Gaussian elimination, **QR** is a reduction to upper-triangular form by Gram-Schmidt.

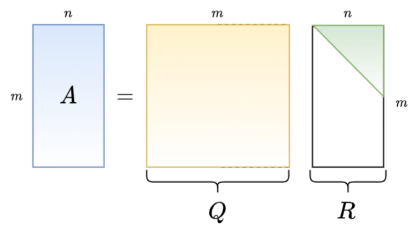


Figure 2.5: QR Factorization

Theorem 2.7.9: "Thin" or "Reduced" QR Factorization

Let $m \geq n$ and $\mathbf{A} \in \mathbb{C}^{m\times n}$ be given. Then, there is a $\mathbf{Q} \in \mathbb{C}^{m\times n}$ with orthonormal columns and an upper triangular $\mathbf{R} \in \mathbb{C}^{n\times n}$ with nonnegative main diagonal entries such that $\mathbf{A} = \mathbf{QR}$. If $\mathrm{Rank}(\mathbf{A}) = n$, then the factors $\mathbf{Q}$ and $\mathbf{R}$ are uniquely determined and the diagonal entries of $\mathbf{R}$ are all positive.

$$\mathbf{A} = \underbrace{\mathbf{Q}}_{m\times m}\underbrace{\mathbf{R}}_{m\times n} = \begin{bmatrix}\mathbf{Q}_1 & \mathbf{Q}_2\end{bmatrix}\begin{bmatrix}\mathbf{R}_1\\ 0\end{bmatrix} = \underbrace{\mathbf{Q}_1}_{m\times n}\underbrace{\mathbf{R}_1}_{n\times n}$$

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$ with $\mathrm{Rank}(\mathbf{A}) = n \leq m$. Then we have:

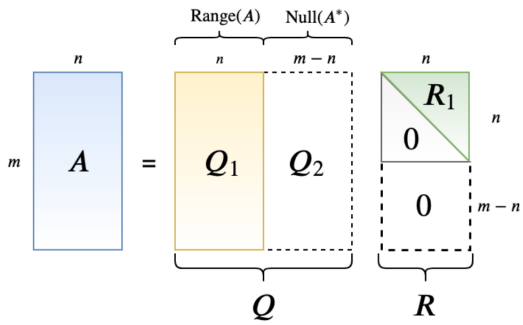


Figure 2.6: $\mathrm{rank}(\mathbf{A}) = n \leq m$

Now, let $\mathbf{A} \in \mathbb{C}^{m\times n}$ with $\mathrm{rank}(\mathbf{A}) = r < n \leq m$. Then, some diagonal element of $\mathbf{R}$ must be zero. Even though QR factorization still exists, the columns of $\mathbf{Q}$ might not give any information about $\mathrm{Range}(\mathbf{A})$, e.g., for any s and c such that $s^2 + c^2 = 1$ we have

$$\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}c & -s\\ s & c\end{bmatrix}\begin{bmatrix}0 & s\\ 0 & c\end{bmatrix}.$$

However, one can permute the columns of $\mathbf{A}$ to obtain "more informative" QR factorization as

$$\mathbf{AP} = \mathbf{Q}\begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2 \\ 0 & 0\end{bmatrix} = \mathbf{Q}_r\begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2\end{bmatrix},$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{R}_1 \in \mathbb{C}^{r\times r}$ is non-singular and upper triangular and $\mathbf{Q}_r \in \mathbb{C}^{m\times r}$ has orthonormal columns.

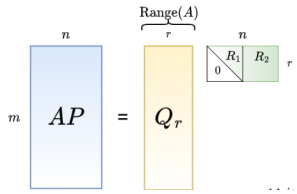


Figure 2.7: $\text{rank}(\mathbf{A}) = r < n \leq m$

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$ with $\text{rank}(\mathbf{A}) = m \leq n$. Then, we have

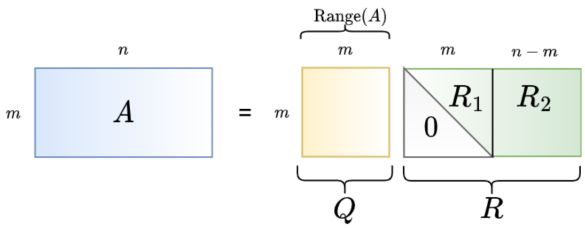


Figure 2.8: $\text{rank}(\mathbf{A}) = m \leq n$

Now, if $\mathbf{A} \in \mathbb{C}^{m\times n}$ with $\text{rank}(\mathbf{A}) = r < m \leq n$, Again, some diagonal element of $\mathbf{R}$ must be zero. So, again as before, one can permute the columns of $\mathbf{A}$ to obtain

$$\mathbf{AP} = \mathbf{Q}\begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2 \\ 0 & 0\end{bmatrix} = \mathbf{Q}_r\begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2\end{bmatrix},$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{R}_1 \in \mathbb{C}^{r\times r}$ is non-singular and upper triangular and $\mathbf{Q}_r \in \mathbb{C}^{m\times r}$ has orthonormal columns.

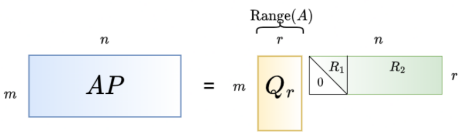


Figure 2.9: $\text{rank}(\mathbf{A}) = r < m \leq n$

Chapter 3

Direct Methods

Generally speaking, numerical linear algebra algorithms can be categorized into two main groups:

- Direct Methods:
  - The procedure takes finite number of steps.
  - On completion, one has the exact solution (up to round-off errors).
  - No useful approximation if one stops early.
  - Not possible to "warm-start".
- Iterative Methods:
  - The approximate solution, called the iterate, is updated iteratively.
  - Some iterative methods take infinite number of steps.
  - Some others give the exact solution in finite number of steps.
  - Most often the solution is approximated well if one stops early.
  - Possible to "warm-start".

We study these in two main contexts: (1) Linear Systems, and (2) Least Squares.
We now consider numerical methods for solving a linear system of equations:

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{C}^{n\times n}, \quad \mathbf{b} \in \mathbb{C}^n$$

It is difficult to overstate the importance of linear systems. Just like calculus, it is implicitly or explicitly everywhere! The most fundamental problem in linear algebra is solving linear systems.
The linear system may have no solution, exactly one solution, or infinitely many solutions:

- Consistent: the linear system has a solution
  - $\mathbf{A}$ is non-singular: unique solution
  - $\mathbf{A}$ is singular but $\mathbf{b} \in \text{Range}(\mathbf{A})$: infinitely many solutions
- Inconsistent: the linear system has no solution
  - $\mathbf{A}$ must be singular and $\mathbf{b} \notin \text{Range}(\mathbf{A})$

Here, we only consider the non-singular case.

3.1 Direct Methods

Definition 3.1.1: Direct Methods

Procedures to solve a linear system of equations that take a finite number of steps to arrive at the exact solution

Gaussian elimination and its variants. These methods often only involve the matrix $\mathbf{A}$, i.e., $\mathbf{b}$ is often not needed! **The essence of these methods amounts decomposition of A into a product of matrices that have a simpler form**. We already saw $\mathbf{A} = \mathbf{LU}$, $\mathbf{PA} = \mathbf{LU}$, and Cholesky $\mathbf{A} = \mathbf{LL}^*$.

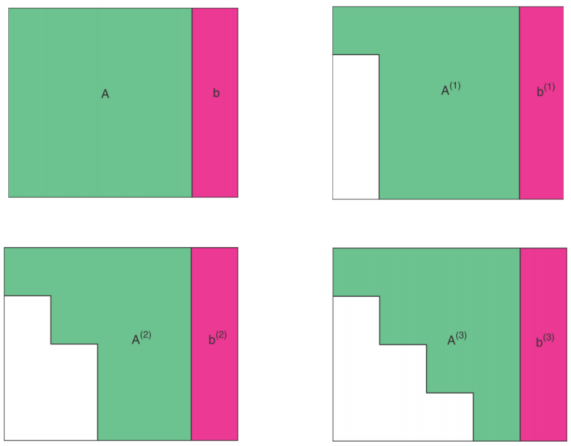


Figure 3.1: Gaussian Elimination

Which is the same as:

$$\mathbf{Ax} = \mathbf{b} \Rightarrow \mathbf{LUx} = \mathbf{b} \Rightarrow \underbrace{\mathbf{Ux}}_{\mathbf{A}^{(3)}} = \underbrace{\mathbf{L}^{-1}\mathbf{b}}_{\mathbf{b}^{(3)}}$$

$$\underbrace{\begin{pmatrix} 1 & & & & \\ -\frac{a_{21}}{a_{11}} & 1 & & & \\ -\frac{a_{31}}{a_{11}} & & 1 & & \\ \vdots & & & \ddots & \\ -\frac{a_{n1}}{a_{11}} & & & & \ddots \\ & & & & & 1 \end{pmatrix}}_{\tilde{\mathbf{L}}^{(1)}} \overbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1,n-1} & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2,n-1} & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3,n-1} & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{n,n-1} & a_{nn} \end{pmatrix}}^{\mathbf{A}} = \overbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1,n-1} & a_{1n} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \cdots & a_{2,n-1}^{(1)} & a_{2n}^{(1)} \\ 0 & a_{32}^{(1)} & a_{33}^{(1)} & \cdots & a_{3,n-1}^{(1)} & a_{3n}^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_{n2}^{(1)} & a_{n3}^{(1)} & \cdots & a_{n,n-1}^{(1)} & a_{nn}^{(1)} \end{pmatrix}}^{\mathbf{A}^{(1)}}$$

$$\underbrace{\begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & -\frac{a_{32}^{(1)}}{a_{22}^{(1)}} & 1 & & \\ & \vdots & & \ddots & \\ & -\frac{a_{n2}^{(1)}}{a_{22}^{(1)}} & & & \ddots \\ & & & & & 1 \end{pmatrix}}_{\tilde{\mathbf{L}}^{(2)}} \overbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1,n-1} & a_{1n} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \cdots & a_{2,n-1}^{(1)} & a_{2n}^{(1)} \\ 0 & a_{32}^{(1)} & a_{33}^{(1)} & \cdots & a_{3,n-1}^{(1)} & a_{3n}^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_{n2}^{(1)} & a_{n3}^{(1)} & \cdots & a_{n,n-1}^{(1)} & a_{nn}^{(1)} \end{pmatrix}}^{\mathbf{A}^{(1)}} = \overbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1,n-1} & a_{1n} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \cdots & a_{2,n-1}^{(1)} & a_{2n}^{(1)} \\ 0 & 0 & a_{33}^{(2)} & \cdots & a_{3,n-1}^{(2)} & a_{3n}^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & a_{n3}^{(2)} & \cdots & a_{n,n-1}^{(2)} & a_{nn}^{(2)} \end{pmatrix}}^{\mathbf{A}^{(2)}}$$

Keep doing so, we get

$$\underbrace{\tilde{\mathbf{L}}^{(n-1)} \cdots \tilde{\mathbf{L}}^{(2)} \tilde{\mathbf{L}}^{(1)}}_{\tilde{\mathbf{L}}} \mathbf{A} = \mathbf{U} \implies \mathbf{A} = \underbrace{\tilde{\mathbf{L}}^{-1}}_{\mathbf{L}} \mathbf{U}$$

$$\downarrow$$

$$\tilde{\mathbf{L}}^{-1} = \left(\tilde{\mathbf{L}}^{(n-1)} \cdots \tilde{\mathbf{L}}^{(2)} \tilde{\mathbf{L}}^{(1)}\right)^{-1} = \left[\tilde{\mathbf{L}}^{(1)}\right]^{-1} \left[\tilde{\mathbf{L}}^{(2)}\right]^{-1} \cdots \left[\tilde{\mathbf{L}}^{(n-1)}\right]^{-1}$$

First,

$$\tilde{\mathbf{L}}^{(2)} = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & -\frac{a_{32}^{(1)}}{a_{22}^{(1)}} & 1 & & \\ & \vdots & & \ddots & \\ & -\frac{a_{n2}^{(1)}}{a_{22}^{(1)}} & & & 1 \end{pmatrix} \Rightarrow [\tilde{\mathbf{L}}^{(2)}]^{-1} = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & \frac{a_{32}^{(1)}}{a_{22}^{(1)}} & 1 & & \\ & \vdots & & \ddots & \\ & \frac{a_{n2}^{(1)}}{a_{22}^{(1)}} & & & 1 \end{pmatrix}$$

Second, let's define

$$\boldsymbol{\ell}_1 = \begin{pmatrix} 0 \\ \frac{a_{21}}{a_{11}} \\ \frac{a_{31}}{a_{11}} \\ \vdots \\ \frac{a_{n1}}{a_{11}} \end{pmatrix}, \qquad \boldsymbol{\ell}_2 = \begin{pmatrix} 0 \\ 0 \\ \frac{a_{32}^{(1)}}{a_{22}^{(1)}} \\ \vdots \\ \frac{a_{n2}^{(1)}}{a_{22}^{(1)}} \end{pmatrix}$$

We note that

$$\overbrace{\begin{pmatrix} 1 & & & & \\ \frac{a_{21}}{a_{11}} & 1 & & & \\ \frac{a_{31}}{a_{11}} & & 1 & & \\ \vdots & & & \ddots & \\ \frac{a_{n1}}{a_{11}} & & & & 1 \end{pmatrix}}^{[\tilde{\mathbf{L}}^{(1)}]^{-1}} = \mathbf{I} + \begin{pmatrix} 0 \\ \frac{a_{21}}{a_{11}} \\ \frac{a_{31}}{a_{11}} \\ \vdots \\ \frac{a_{n1}}{a_{11}} \end{pmatrix} \begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix} = \mathbf{I} + \boldsymbol{\ell}_1 \mathbf{e}_1^{\mathrm{T}},$$

and similarly we can show $[\tilde{\mathbf{L}}^{(2)}]^{-1} = \mathbf{I} + \boldsymbol{\ell}_2 \mathbf{e}_2^{\mathrm{T}}$. Hence,

$$\left[\tilde{\mathbf{L}}^{(1)}\right]^{-1}\left[\tilde{\mathbf{L}}^{(2)}\right]^{-1} = \left(\mathbf{I} + \boldsymbol{\ell}_1 \mathbf{e}_1^{\mathrm{T}}\right)\left(\mathbf{I} + \boldsymbol{\ell}_2 \mathbf{e}_2^{\mathrm{T}}\right) = \mathbf{I} + \boldsymbol{\ell}_1 \mathbf{e}_1^{\mathrm{T}} + \boldsymbol{\ell}_2 \mathbf{e}_2^{\mathrm{T}} + \boldsymbol{\ell}_1 \underbrace{\langle \mathbf{e}_1, \boldsymbol{\ell}_2 \rangle}_{=0} \mathbf{e}_2^{\mathrm{T}} = \mathbf{I} + \boldsymbol{\ell}_1 \mathbf{e}_1^{\mathrm{T}} + \boldsymbol{\ell}_2 \mathbf{e}_2^{\mathrm{T}}.$$

and so,

$$\left[\tilde{\mathbf{L}}^{(1)}\right]^{-1}\left[\tilde{\mathbf{L}}^{(2)}\right]^{-1} = \begin{pmatrix} 1 & & & & \\ \frac{a_{21}}{a_{11}} & 1 & & & \\ \frac{a_{31}}{a_{11}} & \frac{a_{32}^{(1)}}{a_{22}^{(1)}} & 1 & & \\ \vdots & \vdots & & \ddots & \\ \frac{a_{n1}}{a_{11}} & \frac{a_{n2}^{(1)}}{a_{22}^{(1)}} & & & \ddots \\ & & & & 1 \end{pmatrix}$$

Definition 3.1.2

Algorithm solving $\mathbf{Ax} = \mathbf{b}$ using PLU factorization ($\mathbf{PA} = \mathbf{LU}$)

1. $\tilde{\mathbf{b}} = \mathbf{Pb}$
2. Forward Solve: $\mathbf{Ly} = \tilde{\mathbf{b}}$
3. Backward Solve: $\mathbf{Ux} = \mathbf{y}$

PLU factorization takes $2n^3/3 + \mathcal{O}(n^2)$ FLOPS. Forward/Backward substitutions, i.e., triangular system solves, cost only $\mathcal{O}(n^2)$ FLOPS.
In practice, $\mathbf{P}$ is not stored as a matrix. Instead, the permutations are represented in a vector.
For diagonally dominant and PD matrices, partial pivoting is not required. We may perform the LU/PLU decomposition once and then solve different linear systems for different $\mathbf{b}$ at an $\mathcal{O}(n^2)$ cost for each. However, we have to store the factors.

Example.

> Never form $\mathbf{A}^{-1}$ explicitly!

If we had $\mathbf{A}^{-1}$, then we could have found $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ in $\mathcal{O}(n^2)$ FLOPS. So suppose, we would like to

explicitly find $\mathbf{A}^{-1}$. One way, as good as any other way, is to form LU decomposition of $\mathbf{A}$ and then apply it to columns of identity $\mathbf{I} = [\mathbf{e}_1 \mid \mathbf{e}_2 \mid \cdots \mid \mathbf{e}_n]$, i.e.,

$$\mathbf{A}\mathbf{x}_i = \mathbf{e}_i \Rightarrow \mathbf{L}\mathbf{U}\mathbf{x}_i = \mathbf{e}_i \Rightarrow [\mathbf{x}_1 \mid \mathbf{x}_2 \mid \cdots \mid \mathbf{x}_n] = [\mathbf{A}^{-1}\mathbf{e}_1 \mid \mathbf{A}^{-1}\mathbf{e}_2 \mid \cdots \mid \mathbf{A}^{-1}\mathbf{e}_n] = \mathbf{A}^{-1}\mathbf{I} = \mathbf{A}^{-1}$$

The initial LU takes $2n^3/3$ FLOPS, each forward and backward will take $2n^2$ FLOPS, so the total is

$$2n^3/3 + 2n^2 \cdot n = 8n^3/3 \text{ FLOPS.}$$

Forming $\mathbf{A}^{-1}$ explicitly and then multiplying by $\mathbf{b}$ is generally not recommended:

- It is more computationally expensive than by simply using LU decomposition
- $\mathbf{A}$ can be sparse, whereas $\mathbf{A}^{-1}$ can become very dense
- Round-off errors can be more pronounced and magnified with increasing matrix size.
- Finally, it simply has no advantage over any other method to offset the disadvantage!

For many sparse matrices, direct methods can be done very fast and effective. One such class of sparse matrices is banded matrices:

Definition 3.1.3

The matrix $\mathbf{A}$ is banded if, other than inside a band of diagonals, all other elements are zero.

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1q} & & & & \\ \vdots & \ddots & & \ddots & & & \\ a_{p1} & & \ddots & & \ddots & & \\ & \ddots & & \ddots & & \ddots & \\ & & \ddots & & \ddots & & \ddots \\ & & & \ddots & & \ddots & & a_{n-q+1,n} \\ & & & & \ddots & & \ddots & \vdots \\ & & & & & \cdots & a_{n,n-p+1} & a_{nn} \end{pmatrix}$$

$$\mathbf{L} = \begin{pmatrix} 1 & & & & & \\ \vdots & \ddots & & & & \\ l_{p1} & \ddots & 1 & & & \\ & \ddots & \ddots & \ddots & & \\ & & \ddots & \ddots & \ddots & \\ & & & \ddots & l_{n-p+1} & 1 \end{pmatrix}, \qquad \mathbf{U} = \begin{pmatrix} u_{11} & \cdots & u_{1q} & & & \\ & \ddots & \ddots & \ddots & & \\ & & \ddots & \ddots & \ddots & \\ & & & \ddots & \ddots & u_{n-q+1,n} \\ & & & & \ddots & \vdots \\ & & & & & u_{nn} \end{pmatrix}$$

If $p, q \ll n$, then LU decomposition can be carried out in merely $\mathcal{O}(n)$ time and storage! With pivoting, however, the sparsity may be altered and the cost may increase.

3.2 Least Squares

We saw that linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a solution when $\mathbf{b} \in \text{Range}(\mathbf{A})$. But what about when $\mathbf{b} \notin \text{Range}(\mathbf{A})$. We obviously will not get $\mathbf{A}\mathbf{x} = \mathbf{b}$ for any $\mathbf{x}$, but we might be able to make the difference "$\mathbf{b} - \mathbf{A}\mathbf{x}$" small for some $\mathbf{x}$.

Definition 3.2.1

Ordinary least-squares:

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2, \quad \mathbf{A} \in \mathbb{R}^{m \times n}, \quad \mathbf{b} \in \mathbb{R}^m$$

For simplicity, we keep things real, but everything we say here applies to complex settings as well.

Example.

Polynomial data fitting

Suppose we have m data points $\{(t_i, b_i)\}_{i=1}^m$. Suppose we'd like to fit a polynomial of degree at most $n-1$.
If $m > n$, we cannot generally interpolate all data points, i.e., we cannot generally fit the data exactly (e.g., interpolating 3 points with a line is generally impossible). But, we can approximate b_i's...
Consider a polynomial of degree at most $n-1$ as

$$p_{n-1}(t) = \sum_{j=0}^{n-1} x_j t^j$$

For each b_i, we get an approximation for b_i as

$$\hat{b}_i = p_{n-1}(t_i) = \sum_{j=0}^{n-1} x_j t_i^j = \langle \mathbf{x}, \mathbf{t}_i \rangle, \quad i = 1, \ldots, m,$$

where

$$\mathbf{x} \triangleq \begin{bmatrix} x_0 & x_1 & x_2 & \ldots & x_{n-1} \end{bmatrix}^\top, \quad \mathbf{t}_i \triangleq \begin{bmatrix} 1 & t_i & t_i^2 & \ldots & t_i^{n-1} \end{bmatrix}^\top, \quad i = 1, \ldots, m.$$

Putting it all together, we get

$$\underbrace{\begin{bmatrix} 1 & t_1 & t_1^2 & \ldots & t_1^{n-1} \\ 1 & t_2 & t_2^2 & \ldots & t_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_m & t_m^2 & \ldots & t_m^{n-1} \end{bmatrix}}_{\mathbf{A}\text{:Vandermonde Matrix}} \underbrace{\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{n-1} \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \vdots \\ \hat{b}_m \end{bmatrix}}_{\hat{\mathbf{b}}}$$

We would like to find $\mathbf{x}$ that makes $\hat{\mathbf{b}}$ close to $\mathbf{b}$, hence we do

$$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

$\mathbf{A}$ depends on the choice of basis for polynomials. Here, we used monomial basis (only good for really low degree polynomials).
Data fitting in other norms:

Definition 3.2.2

$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_1$: least absolute deviation is more robust to outliers

Definition 3.2.3

$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_\infty$: a min-max problem to keep the worst-case error at check

These two formulation can be solved using linear programming.

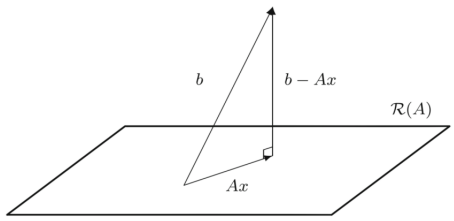


Figure 3.2: $\min_{\mathbf{x}\in\mathbb{R}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2, \quad \mathbf{A} \in \mathbb{R}^{m\times n}, \quad \mathbf{b} \in \mathbb{R}^m$

3.2.1 Usual method for solving OLS

Typical direct methods for solving OLS:

- Normal Equation:
  - Working with Normal equations is simple and intuitive.
  - Solving the normal equation takes $mn^2 + n^3/3$ flops.
  - Solving it naively is generally strongly advised against.
  - There are clever techniques to do this implicitly and more stably using iterative methods.
- QR Factorization:
  - The "standard" direct method is using QR factorization.
  - QR factorization approach takes $2mn^2 - 2n^3/3$ flops. So, QR approach is roughly twice expensive than normal equation, but significantly more robust.
- SVD Factorization:
  - SVD approach takes $2mn^2 + 11n^3$ flops. So, it is roughly the same as QR for $m \gg n$, but more expensive when $m \approx n$.
  - SVD is more robust than QR for (nearly) rank deficient cases.

We need to find a minimizer of

$$f(\mathbf{x}) \triangleq \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2. \quad \text{(We added "1/2" for convenience)}$$

We will see later that $f(\mathbf{x})$ is convex. So, all we need is to find a $\hat{\mathbf{x}}$ such that $\nabla f(\hat{\mathbf{x}}) = 0$.

$$\begin{aligned}
f(\mathbf{x}) &= \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \frac{1}{2}(\mathbf{A}\mathbf{x} - \mathbf{b})^\top(\mathbf{A}\mathbf{x} - \mathbf{b}) \\
&= \frac{1}{2}(\mathbf{x}^\top\mathbf{A}^\top - \mathbf{b}^\top)(\mathbf{A}\mathbf{x} - \mathbf{b}) \\
&= \frac{1}{2}\left(\mathbf{x}^\top\mathbf{A}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{A}^\top\mathbf{b} - \mathbf{b}^\top\mathbf{A}\mathbf{x} + \mathbf{b}^\top\mathbf{b}\right) \\
&= \frac{1}{2}\left(\mathbf{x}^\top\mathbf{A}^\top\mathbf{A}\mathbf{x} - 2\mathbf{b}^\top\mathbf{A}\mathbf{x} + \mathbf{b}^\top\mathbf{b}\right) \\
&= \frac{1}{2}\mathbf{x}^\top\mathbf{A}^\top\mathbf{A}\mathbf{x} - \mathbf{b}^\top\mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{b}^\top\mathbf{b}
\end{aligned}$$

Using Normal Equations

Fact 3.2.4

For all $\mathbf{b} \in \mathbb{R}^p$ and $\mathbf{B} \in \mathbb{R}^{p\times p}$, we have

$$f(\mathbf{x}) = \langle \mathbf{b}, \mathbf{x}\rangle \Rightarrow \nabla f(\mathbf{x}) = \mathbf{b}$$

$$f(\mathbf{x}) = \langle \mathbf{x}, \mathbf{B}\mathbf{x}\rangle \Rightarrow \nabla f(\mathbf{x}) = (\mathbf{B} + \mathbf{B}^\top)\mathbf{x}$$

We obtained

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x} - \mathbf{b}^\top \mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{b}^\top \mathbf{b}.$$

So

$$\nabla f(\mathbf{x}) = \mathbf{A}^\top \mathbf{A}\mathbf{x} - \mathbf{A}^\top \mathbf{b}.$$

Hence, we need to find a $\hat{\mathbf{x}}$ such that $\nabla f(\hat{\mathbf{x}}) = 0$, i.e.,

$$\mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b} \quad \text{(Normal Equations)}.$$

Theorem 3.2.5

A vector $\mathbf{x}$ solves the OLS problem if and only if it can be written as

$$\mathbf{x} = \mathbf{A}^\dagger \mathbf{b} + (\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}, \quad \text{where } \mathbf{y} \in \mathbb{R}^n.$$

- $\mathbf{A}^\dagger \mathbf{b}$: Least-norm solution
- $(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})$: Orthogonal Projection onto $\text{Null}(\mathbf{A})$ $(\mathbf{A}(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A}) = 0)$

Proof. Let's first check that $\mathbf{x}^\dagger \triangleq \mathbf{A}^\dagger \mathbf{b}$ is a particular solution:

$$\mathbf{A}^\top \mathbf{A}\mathbf{x}^\dagger = \mathbf{A}^\top \mathbf{A}\left(\mathbf{A}^\dagger \mathbf{b}\right) = \left(\mathbf{A}^\top \mathbf{A}\mathbf{A}^\dagger\right)\mathbf{b} = \mathbf{A}^\top \mathbf{b}. \quad \text{(Exercise: show that } \mathbf{A}^\top \mathbf{A}\mathbf{A}^\dagger = \mathbf{A}^\top\text{)}$$

Now suppose $\mathbf{x}$ is another solution, i.e., $\mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{A}^\top \mathbf{b}$. We have

$$\begin{cases} \mathbf{A}^\top \mathbf{A}\mathbf{x}^\dagger = \mathbf{A}^\top \mathbf{b} \\ \mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{A}^\top \mathbf{b} \end{cases} \implies \mathbf{A}^\top \mathbf{A}\left(\mathbf{x} - \mathbf{x}^\dagger\right) = 0.$$

Denoting $\mathbf{z} = \mathbf{x} - \mathbf{x}^\dagger$, it follows that $\mathbf{z} \in \text{Null}(\mathbf{A}^\top \mathbf{A})$. Now, it can be shown that $\text{Null}(\mathbf{A}^\top \mathbf{A}) = \text{Null}(\mathbf{A})$ (Exercise: show this), hence $\mathbf{z} \in \text{Null}(\mathbf{A})$. So, we can write it as $\mathbf{z} = \left(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A}\right)\mathbf{y}$ for some $\mathbf{y} \in \mathbb{R}^n$. As a result,

$$\mathbf{x} = \mathbf{x}^\dagger + \left(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A}\right)\mathbf{y}.$$

□

So solutions to the least squares problem or equivalently the normal equation, are of the form

$$\mathbf{x} = \mathbf{A}^\dagger \mathbf{b} + (\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}, \quad \text{where } \mathbf{y} \in \mathbb{R}^n.$$

Proposition 3.2.6

1. Let $\mathbf{x}^\dagger \triangleq \mathbf{A}^\dagger \mathbf{b}$ and consider any other OLS solution $\mathbf{x}$. We have
$$\|\mathbf{x}^\dagger\|_2 \leq \|\mathbf{x}\|_2.$$
2. For any solution $\mathbf{x}$ to OLS,
$$\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x} \perp \text{Range}(\mathbf{A}).$$
3. When $\mathbf{A}$ is full column-rank, the solution is unique and given by
$$\mathbf{x} = (\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{b}.$$

Proof. 1. Take $\mathbf{x}$ to be any OLS solution. We have

$$\|\mathbf{x}\|^2 = \|\mathbf{A}^\dagger \mathbf{b} + (\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\|^2$$

$$= \|\mathbf{A}^\dagger \mathbf{b}\|^2 + \|(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\|^2 + 2\langle \mathbf{A}^\dagger \mathbf{b}, (\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\rangle$$

$$= \|\mathbf{A}^\dagger \mathbf{b}\|^2 + \|(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\|^2 + 2\langle (\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{A}^\dagger \mathbf{b}, \mathbf{y}\rangle$$

$$
\begin{aligned}
&= \|\mathbf{A}^\dagger \mathbf{b}\|^2 + \|(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\|^2 + 2\langle (\mathbf{A}^\dagger - \mathbf{A}^\dagger \mathbf{A}\mathbf{A}^\dagger)\mathbf{b}, \mathbf{y}\rangle \\
&= \|\mathbf{A}^\dagger \mathbf{b}\|^2 + \|(\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\|^2 \\
&\geq \|\mathbf{A}^\dagger \mathbf{b}\|^2 = \|\mathbf{x}^\dagger\|^2.
\end{aligned}
$$

2. Take $\mathbf{x}$ to be any OLS solution. We have

$$
\begin{aligned}
\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x} &= \mathbf{b} - \mathbf{A}\big(\mathbf{A}^\dagger \mathbf{b} + (\mathbf{I} - \mathbf{A}^\dagger \mathbf{A})\mathbf{y}\big) \\
&= \mathbf{b} - \mathbf{A}\mathbf{A}^\dagger \mathbf{b} + (\mathbf{A} - \mathbf{A}\mathbf{A}^\dagger \mathbf{A})\mathbf{y} \\
&= \mathbf{b} - \mathbf{A}\mathbf{A}^\dagger \mathbf{b} = (\mathbf{I} - \mathbf{A}\mathbf{A}^\dagger)\mathbf{b}
\end{aligned}
$$

   Thus, by $\mathbf{A}^\top \mathbf{A}\mathbf{A}^\dagger = \mathbf{A}^\top$,

$$
\mathbf{A}^\top (\mathbf{I} - \mathbf{A}\mathbf{A}^\dagger)\mathbf{b} = (\mathbf{A}^\top - \mathbf{A}^\top \mathbf{A}\mathbf{A}^\dagger)\mathbf{b} = 0.
$$

   So, $\mathbf{r} \in \mathrm{Null}(\mathbf{A}^\top)$, and hence $\mathbf{r} \perp \mathrm{Range}(\mathbf{A})$.

3. It follows by noting that $\mathbf{A}^\dagger = (\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top$ and $\mathbf{A}^\dagger \mathbf{A} = \mathbf{I}$.

□

Example.

Consider

$$
\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon \end{bmatrix}
$$

We have

$$
\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1+\epsilon^2 & 1 & 1 \\ 1 & 1+\epsilon^2 & 1 \\ 1 & 1 & 1+\epsilon^2 \end{bmatrix}
$$

Now if $\eta \leq \epsilon < \sqrt{\eta}$, where η is the rounding unit, then $f(1+\epsilon^2) = 1$, and $\mathbf{A}^\top \mathbf{A}$ is numerically singular, even though $\mathbf{A}$ is full rank.
(In single precision, the rounding unit is $\eta \approx 5 \times 10^{-8}$ while in double precision, it is $\eta \approx 10^{-16}$)

Proposition 3.2.7

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be full column rank. We have $\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa^2(\mathbf{A})$

Proof. Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the economy SVD of $\mathbf{A}$. In particular, $\mathbf{U} \in \mathbb{R}^{m \times n}$ with $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$, $\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$ is diagonal containing the singular values of $\mathbf{A}$, and $\mathbf{V} \in \mathbb{R}^{n \times n}$ is a unitary matrix. We have

$$
\mathbf{A}^\top \mathbf{A} = \mathbf{V}\mathbf{\Sigma}\mathbf{U}^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{V}\mathbf{\Sigma}^2 \mathbf{V}^\top.
$$

Since $\mathbf{A}^\top \mathbf{A} \succeq 0$, its eigendecomposition and SVD coincide. So $\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$ is the SVD decomposition of $\mathbf{A}^\top \mathbf{A}$. Hence

$$
\kappa(\mathbf{A}^\top \mathbf{A}) = \frac{\sigma_1(\mathbf{A}^\top \mathbf{A})}{\sigma_n(\mathbf{A}^\top \mathbf{A})} = \frac{\sigma_1^2(\mathbf{A})}{\sigma_n^2(\mathbf{A})} = \kappa^2(\mathbf{A}).
$$

□

Using QR Factorization

Suppose $\mathbf{A} \in \mathbb{R}^{m \times n}$ is full column rank with its economy QR as $\mathbf{A} = \mathbf{Q}\mathbf{R}$. Let $\mathbf{Q}_\perp$ be the vectors not included in the economy QR.

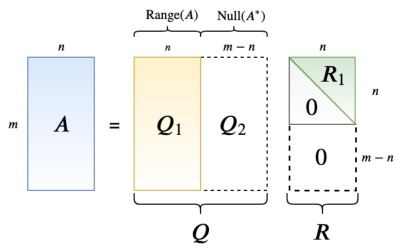


Figure 3.3: $\text{rank}(\mathbf{A}) = n \leq m$

Here $\mathbf{Q}_1 = \mathbf{Q}$ and $\mathbf{Q}_2 = \mathbf{Q}_\perp$.

- Recall that $\mathbf{Q}\mathbf{Q}^\top$ is the orthogonal projection onto $\text{Range}(\mathbf{A})$
- Recall that $\mathbf{Q}_\perp\mathbf{Q}_\perp^\top$ is the orthogonal projection onto $\text{Null}(\mathbf{A}^\top)$
- Recall that $\mathbf{Q}\mathbf{Q}^\top + \mathbf{Q}_\perp\mathbf{Q}_\perp^\top = \mathbf{I}$ (show this) and $\mathbf{Q}_\perp^\top\mathbf{Q} = 0$

We have

$$\begin{aligned}
\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 &= \|\mathbf{Q}\mathbf{R}\mathbf{x} - \mathbf{b}\|^2 = \|\mathbf{Q}\mathbf{R}\mathbf{x} - (\mathbf{Q}\mathbf{Q}^\top + \mathbf{Q}_\perp\mathbf{Q}_\perp^\top)\mathbf{b}\|^2 \\
&= \|\underbrace{(\mathbf{Q}\mathbf{R}\mathbf{x} - \mathbf{Q}\mathbf{Q}^\top\mathbf{b})}_{\text{Range}(\mathbf{Q})} - \underbrace{\mathbf{Q}_\perp\mathbf{Q}_\perp^\top\mathbf{b}}_{\text{Null}(\mathbf{A}^\top)}\| \\
&= \|\mathbf{Q}\mathbf{R}\mathbf{x} - \mathbf{Q}\mathbf{Q}^\top\mathbf{b}\|^2 + \|\mathbf{Q}_\perp\mathbf{Q}_\perp^\top\mathbf{b}\|^2 = \|\mathbf{R}\mathbf{x} - \mathbf{Q}^\top\mathbf{b}\|^2 + \|\mathbf{Q}_\perp\mathbf{Q}_\perp^\top\mathbf{b}\|^2
\end{aligned}$$

So, we have

$$\arg\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \arg\min_{\mathbf{x}} \|\mathbf{R}\mathbf{x} - \mathbf{Q}^\top\mathbf{b}\|^2 = \mathbf{R}^{-1}\mathbf{Q}^\top\mathbf{b}$$

Since $\mathbf{R}$ is square invertible matrix (show this).
We never form $\mathbf{R}^{-1}$ explicitly. Instead, we form $\mathbf{c} = \mathbf{Q}^\top\mathbf{b}$ and then solve $\mathbf{R}\mathbf{x} = \mathbf{c}$ using backward substitution.

Using SVD Factorization

Suppose $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $r \triangleq \text{Rank}(\mathbf{A}) < \min\{m, n\}$. There are infinitely many solutions! We typically choose the least norm solution $\mathbf{A}^\dagger\mathbf{b}$, i.e., $\mathbf{x}^\star = \mathbf{V}_r\mathbf{\Sigma}_r^{-1}\mathbf{U}_r^\top\mathbf{b}$.

3.2.2 Computing QR

There are generally a couple of intuitive ways to compute QR:

- Gram-Schmidt Procedure: Triangular Orthogonalization

$$\mathbf{A}\underbrace{\mathbf{R}_1\mathbf{R}_2\cdots\mathbf{R}_n}_{\mathbf{R}^{-1}} = \mathbf{Q}$$

  We study this here.

- Householder Reflections, Given Rotation: Orthogonal Triangularization

$$\underbrace{\mathbf{Q}_n\cdots\mathbf{Q}_2\mathbf{Q}_1}_{\mathbf{Q}^*}\mathbf{A} = \mathbf{R}$$

  We won't have time to cover these latter techniques during the lecture! :(

Gram-Schmidt Procedure

Consider the economy QR of $\mathbf{A} \in \mathbb{C}^{m\times n}$:

$$\underbrace{[\boldsymbol{a}_1 \mid \boldsymbol{a}_2 \mid \cdots \mid \boldsymbol{a}_n]}_{\mathbf{A}} = \underbrace{[\boldsymbol{q}_1 \mid \boldsymbol{q}_2 \mid \cdots \mid \boldsymbol{q}_n]}_{\mathbf{Q}\in\mathbb{C}^{m\times n}} \underbrace{\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ & r_{22} & \cdots & r_{2n} \\ & & \ddots & \vdots \\ & & & r_{nn} \end{bmatrix}}_{\mathbf{R}\in\mathbb{C}^{n\times n}}$$

We start by setting $r_{11} = \|\boldsymbol{a}_1\|, \quad \boldsymbol{q}_1 = \boldsymbol{a}_1/\|\boldsymbol{a}_1\|$. We then require

$$\boldsymbol{a}_2 = r_{12}\boldsymbol{q}_1 + r_{22}\boldsymbol{q}_2$$

This gives

$$r_{12} = \langle \boldsymbol{q}_1, \boldsymbol{a}_2\rangle, \quad r_{22} = \|\boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1\|, \quad \boldsymbol{q}_2 = \frac{\boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1}{r_{22}}$$

Generalizing the above procedure to the jth column, we get

$$\boldsymbol{a}_j = r_{1j}\boldsymbol{q}_1 + r_{2j}\boldsymbol{q}_2 + \cdots + r_{jj}\boldsymbol{q}_j$$

This gives

$$r_{ij} = \langle \boldsymbol{q}_i, \boldsymbol{a}_j\rangle, \quad i = 1, \ldots, j-1 \quad r_{jj} = \left\| \boldsymbol{a}_j - \sum_{i=1}^{j-1} r_{ij}\boldsymbol{q}_i \right\|, \quad \boldsymbol{q}_j = \frac{\boldsymbol{a}_j - \sum_{i=1}^{j-1} r_{ij}\boldsymbol{q}_i}{r_{jj}}$$

Overall it takes $2mn^2$ flops.

Definition 3.2.8: Algorithm Classical Gram-Schmidt

1. Input: $\mathbf{A}$
2. for $j = 1, 2, \ldots, n$ do
3. $\mathbf{q}_j = \mathbf{a}_j$
4. for $i = 1, \ldots, j-1$ do
5. $r_{ij} = \langle \mathbf{q}_i, \mathbf{a}_j\rangle$
6. $\mathbf{q}_j = \mathbf{q}_j - r_{ij}\mathbf{q}_i$
7. end for
8. $r_{jj} = \|\mathbf{q}_j\|$
9. $\mathbf{q}_j = \mathbf{q}_j/r_{jj}$
10. end for

Issue with the classical Gram-Schmidt

Classical Gram-Schmidt is simple but it is highly numerically unstable if the columns of $\mathbf{A}$ are nearly linearly dependent so that the vectors $\mathbf{q}_i$'s are not quite orthogonal. Recall that

$$r_{jj}\mathbf{q}_j = \mathbf{a}_j - \sum_{i=1}^{j-1} r_{ij}\mathbf{q}_i.$$

What classical Gram-Schmidt is actually doing is

$$r_{jj}\mathbf{q}_j = \mathbf{a}_j - \sum_{i=1}^{j-1}\langle \mathbf{q}_i, \mathbf{a}_j\rangle\mathbf{q}_i = \mathbf{a}_j - \sum_{i=1}^{j-1}\mathbf{q}_i\left(\mathbf{q}_i^*\mathbf{a}_j\right) = \left(\mathbf{I} - \sum_{i=1}^{j-1}\mathbf{q}_i\mathbf{q}_i^*\right)\mathbf{a}_j = \left(\mathbf{I} - \mathbf{Q}_j\mathbf{Q}_j^*\right)\mathbf{a}_j,$$

where $\mathbf{Q}_j = [\,\mathbf{q}_1 \mid \mathbf{q}_2 \mid \cdots \mid \mathbf{q}_{j-1}\,]$.
Remedy:

$$r_{jj}\mathbf{q}_j = \left(\mathbf{I} - \mathbf{Q}_j\mathbf{Q}_j^*\right)\mathbf{a}_j = \left(\mathbf{I} - \sum_{i=1}^{j-1}\mathbf{q}_i\mathbf{q}_i^*\right)\mathbf{a}_j = \left(\prod_{i=1}^{j-1}\left(\mathbf{I} - \mathbf{q}_i\mathbf{q}_i^*\right)\right)\mathbf{a}_j$$
$$= \left(\mathbf{I} - \mathbf{q}_{j-1}\mathbf{q}_{j-1}^*\right)\cdots\left(\mathbf{I} - \mathbf{q}_2\mathbf{q}_2^*\right)\left(\mathbf{I} - \mathbf{q}_1\mathbf{q}_1^*\right)\mathbf{a}_j$$

Definition 3.2.9: Algorithm Modified Gram-Schmidt

1. Input: $\mathbf{A}$
2. for $j = 1, 2, \ldots, n$ do
3. $\mathbf{q}_j = \mathbf{a}_j$
4. for $i = 1, \ldots, j-1$ do
5. $r_{ij} = \langle \mathbf{q}_i, \mathbf{q}_j \rangle$
6. $\mathbf{q}_j = \mathbf{q}_j - r_{ij}\mathbf{q}_i$
7. end for
8. $r_{jj} = \|\mathbf{q}_j\|$
9. $\mathbf{q}_j = \mathbf{q}_j / r_{jj}$
10. end for

Chapter 4

Iterative Methods for Linear Systems

The direct methods have some serious drawbacks:

- They cannot solve inexactly: Sometimes we do not really need to solve the system exactly, e.g., linear systems as sub-problems of nonlinear solvers.
- They require $\mathbf{A}$ to be available explicitly: Sometimes $\mathbf{A}$ is only known as a "blackbox" operator, e.g., we only have access to MVP $\mathbf{A} \cdot \mathbf{v}$ (think of a very large deep learning application where storing Hessian is impossible and we can only do backprop!).

$$\mathbf{x} \longrightarrow \text{BLACK BOX} \longrightarrow \mathbf{A}\mathbf{x}$$

- They cannot use any prior info on the solution: Sometimes, we have a good guess for an approximate solution, and we'd like to use that to "warm start" the method, e.g., the solution to the previous step in a time-dependent problem.
- They can result in fill-ins: When $\mathbf{A}$ is large and sparse, but not tightly banded, LU decomposition can result in dense factors. Even if it is banded, it will result in fill-in within the band.

A note before we delve into iterative methods:
Direct methods such as Gaussian elimination and its variations are the method of choice for many problems. Despite the disadvantages, one should not discount them altogether. For example, if one can "afford" direct methods, they might be preferable since they are less affected by the problem conditioning. Obviously, the correct choice of the algorithm is highly problem dependent.

4.1 Stationary Methods

Stationary methods can be motivated as fixed point iterations.
Suppose we need to solve $f(\mathbf{x}) = 0$ where $f : \mathbb{R}^n \to \mathbb{R}^n$. We might be able to find $g : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$f(\mathbf{x}) = 0 \Longleftrightarrow g(\mathbf{x}) = \mathbf{x}.$$

Example.

For example, $g(\mathbf{x}) = \mathbf{x} + f(\mathbf{x})$ (NB: this might actually be a very bad choice for g in many cases!)

Fixed point iterations: $\mathbf{x}_{k+1} = g(\mathbf{x}_k)$. Here, $f(\mathbf{x}) = \mathbf{A}\mathbf{x} - \mathbf{b}$. So if $g(\mathbf{x}) = \mathbf{x} - f(\mathbf{x}) = (\mathbf{I} - \mathbf{A})\mathbf{x} + \mathbf{b}$, then $\mathbf{x} = (\mathbf{I} - \mathbf{A})\mathbf{x} + \mathbf{b}$. Now we can have very basic fixed-point iterations of the form

$$\mathbf{x}_{k+1} = (\mathbf{I} - \mathbf{A})\mathbf{x}_k + \mathbf{b} = \mathbf{x}_k + (\mathbf{b} - \mathbf{A}\mathbf{x}_k).$$

Splitting: More generally, for $\mathbf{Ax} = \mathbf{b}$, we can split $\mathbf{A} = \mathbf{M} - \mathbf{N}$ in such a way that $\mathbf{M}$ is invertible and hence

$$\mathbf{Ax} = \mathbf{b} \iff \mathbf{x} = \mathbf{M}^{-1}(\mathbf{Nx} + \mathbf{b}) = \underbrace{\mathbf{x} + \mathbf{M}^{-1}(\mathbf{b} - \mathbf{Ax})}_{g(\mathbf{x})}$$

Definition 4.1.1

General Stationary Iterations

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{M}^{-1}(\mathbf{b} - \mathbf{Ax}_k).$$

They are called stationary because

- we are looking for a stationary point, i.e., a fixed point
- $\mathbf{M}$ and $\mathbf{N}$ are constant and do not depend on iterations

Main Idea of Splitting:
If inverting $\mathbf{A}$ is very difficult, it might be worth inverting a much "easier" matrix several times, as opposed to directly inverting $\mathbf{A}$ only once. So we aim to split in such a way that inverting $\mathbf{M}$ can be done easily.
How to choose $\mathbf{M}$? Two Opposing Goals

- Goal 1: Choose $\mathbf{M}$ such that inverting it is easy.
- Goal 2: Choose $\mathbf{M}$ such that $\mathbf{M}^{-1}$ is close to $\mathbf{A}^{-1}$.

Different choices of $\mathbf{M}$ lead to a variety of methods, from the simple iterative methods to very complicated multi-resolution ones.
NB: We do not have to form $\mathbf{M}^{-1}$ explicitly. We only need the solution to $\mathbf{Mp}_k = \mathbf{b} - \mathbf{Ax}_k$, followed by $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{p}_k$.

4.1.1 Basic Relaxation Methods

Basic Relaxation Methods: $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{M}^{-1}(\mathbf{b} - \mathbf{Ax}_k)$ and $\mathbf{A} = \mathbf{D} + \mathbf{E} + \mathbf{F}$, where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

$$\mathbf{E} = \begin{bmatrix} 0 & & & \\ a_{21} & \ddots & & \\ \vdots & \ddots & \ddots & \\ a_{n1} & \cdots & a_{n,n-1} & 0 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 0 & a_{12} & \cdots & a_{1n} \\ & \ddots & \ddots & \vdots \\ & & \ddots & a_{n-1,n} \\ & & & 0 \end{bmatrix}$$

- $\mathbf{M} = \mathbf{D}$: Jacobi method (simultaneous relaxation)
  - All components of $\mathbf{x}_{k+1}$ can be computed simultaneously
- $\mathbf{M} = \mathbf{D} + \mathbf{E}$: Gauss–Seidel method (GS) (It has been noted that this method was never mentioned by Gauss or advocated by Seidel.)
- $\mathbf{M} = \omega^{-1}\mathbf{D} + \mathbf{E}$: Successive over-relaxation (SOR)

There is also block version of these splittings.
NB: When $\mathbf{A}$ is implicit, these methods may not be possible.

4.1.2 Convergence of Basic Relaxation Methods

Let us denote the true solution by $\mathbf{x}^\star$, i.e., $\mathbf{A}\mathbf{x}^\star = \mathbf{b}$. We have

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{M}^{-1}(\mathbf{b} - \mathbf{A}\mathbf{x}_k)$$

$$\mathbf{x}_{k+1} - \mathbf{x}^\star = \mathbf{x}_k - \mathbf{x}^\star + \mathbf{M}^{-1}(\mathbf{A}\mathbf{x}^\star - \mathbf{A}\mathbf{x}_k)$$
$$\mathbf{x}_{k+1} - \mathbf{x}^\star = \mathbf{x}_k - \mathbf{x}^\star - \mathbf{M}^{-1}\mathbf{A}(\mathbf{x}_k - \mathbf{x}^\star)$$
$$\mathbf{x}_{k+1} - \mathbf{x}^\star = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})(\mathbf{x}_k - \mathbf{x}^\star)$$
$$\mathbf{x}_{k+1} - \mathbf{x}^\star = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})^{k+1}(\mathbf{x}_0 - \mathbf{x}^\star)$$
$$\mathbf{e}_{k+1} = \mathbf{T}^{k+1}\mathbf{e}_0$$

where $\mathbf{e}_k \triangleq \mathbf{x}_k - \mathbf{x}^\star$ is the error vector, and $\mathbf{T} \triangleq \mathbf{I} - \mathbf{M}^{-1}\mathbf{A}$ is called the iteration matrix. Thus,

$$\lim_{k\to\infty} \mathbf{e}_k = \mathbf{0},\ \forall \mathbf{x}_0 \iff \lim_{k\to\infty} \mathbf{T}^k = \mathbf{0}$$

Proposition 4.1.2

Sufficient Condition on Convergence:

$$\|\mathbf{T}\| < 1 \implies \lim_{k\to\infty} \mathbf{e}_k = \mathbf{0},$$

where $\|\cdot\|$ is any induced matrix norm.

Proof.

$$\|\mathbf{e}_{k+1}\| = \|\mathbf{T}^{k+1}\mathbf{e}_0\| \le \|\mathbf{T}^{k+1}\|\,\|\mathbf{e}_0\| \le \|\mathbf{T}\|^{k+1}\,\|\mathbf{e}_0\|$$

□

But it turns out that we can completely characterize the convergence of these methods...

Theorem 4.1.3: Necessary and Sufficient Condition on Convergence

$$\rho(\mathbf{T}) < 1 \iff \lim_{k\to\infty} \mathbf{e}_k = \mathbf{0}.$$

Proof. Omitted. □

Non-asymptotic Rate of Convergence:
If $\|\mathbf{T}\| < 1$, from $\|\mathbf{e}_k\| \le \|\mathbf{T}\|^k\,\|\mathbf{e}_0\|$, it follows that after $k \ge \log(\epsilon)/\log(\|\mathbf{T}\|)$, we have $\|\mathbf{e}_k\| \le \epsilon\,\|\mathbf{e}_0\|$ for some $0 < \epsilon < 1$:

$$\|\mathbf{e}_k\| \le \|\mathbf{T}\|^k\|\mathbf{e}_0\| \le \|\mathbf{T}\|^{\log(\varepsilon)/\log(\|\mathbf{T}\|)}\|\mathbf{e}_0\| = \varepsilon\|\mathbf{e}_0\|.$$

Then the factor $\|\mathbf{T}\|$ is called the non-asymptotic rate of convergence.
More generally, we can obtain the asymptotic rate of convergence.

Theorem 4.1.4: Asymptotic Rate of Convergence

$$\limsup_{k\to\infty} \left(\frac{\|\mathbf{e}_k\|}{\|\mathbf{e}_0\|}\right)^{1/k} \le \rho(\mathbf{T}), \quad \forall \mathbf{x}_0.$$

Interpretation of the asymptotic rate of convergence:

$$\forall \delta > 0,\ \exists k_0 \ge 0,\ \text{s.t. } \forall k \ge k_0,\ \|\mathbf{e}_k\| \le (\rho(\mathbf{T}) + \delta)^k\|\mathbf{e}_0\|$$

The asymptotic rate is tighter than the non-asymptotic one since we always have $\rho(\mathbf{T}) \le \|\mathbf{T}\|$ (Exercise: Show this inequality).

Jacobi, GS, and SOR do not necessarily converge for just any non-singular matrix $\mathbf{A}$, e.g., they are not even well-defined if $a_{ii} = 0$ for some i.

Theorem 4.1.5

They converge if $\mathbf{A}$ is strictly diagonally dominant.

For the SOR method:

- $0 < \omega < 2$ is necessary for convergence, i.e., if $\omega \notin (0, 2)$, SOR will diverge for any matrix.
- $0 < \omega < 2$ is sufficient for positive definite systems.
- For $\omega = 1$, SOR is just GS.
- The best results are usually obtained for $1 \leq \omega < 2$.

4.2 Polynomial Acceleration

Polynomial acceleration is a technique to accelerate the convergence of stationary methods for solving $\mathbf{A}\mathbf{x} = \mathbf{b}$.
Throughout this lecture, we assume $\mathbf{A} \succeq 0$ and define $M \triangleq \lambda_{\max}(\mathbf{A})$ and $m \triangleq \lambda_{\min}(\mathbf{A})$.
Recall stationary methods:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{M}^{-1}\big(\mathbf{b} - \mathbf{A}\,\mathbf{x}_k\big).$$

Simple stationary Richardson iteration is when $\mathbf{M} = \alpha^{-1}\mathbf{I}$, i.e.,

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha\big(\mathbf{b} - \mathbf{A}\,\mathbf{x}_k\big).$$

The optimal value of the step-size α is $\alpha = \dfrac{2}{m + M}$.
The convergence is given by

$$\|\mathbf{e}_k\| \leq \left(\frac{\kappa - 1}{\kappa + 1}\right) \|\mathbf{e}_{k-1}\|,$$

where $\kappa \triangleq \dfrac{M}{m}$ is the condition number of $\mathbf{A}$.
Non-stationary Richardson aims to accelerate convergence by

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k\big(\mathbf{b} - \mathbf{A}\,\mathbf{x}_k\big),$$

where $\alpha_k \in \mathbb{R}$.
Let $\mathbf{A}\mathbf{x}^\star = \mathbf{b}$ and $\mathbf{e}_k \triangleq \mathbf{x}_k - \mathbf{x}^\star$. We easily see that

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \alpha_{k-1}\big(\mathbf{b} - \mathbf{A}\mathbf{x}_{k-1}\big)$$

$$\mathbf{x}_k - \mathbf{x}^\star = \mathbf{x}_{k-1} - \mathbf{x}^\star + \alpha_{k-1}\big(\mathbf{A}\mathbf{x}^\star - \mathbf{A}\mathbf{x}_{k-1}\big)$$

$$\mathbf{e}_k = \mathbf{e}_{k-1} - \alpha_{k-1}\mathbf{A}\mathbf{e}_{k-1} = (\mathbf{I} - \alpha_{k-1}\mathbf{A})\,\mathbf{e}_{k-1}$$

$$\mathbf{e}_k = (\mathbf{I} - \alpha_{k-1}\mathbf{A})(\mathbf{I} - \alpha_{k-2}\mathbf{A})\,\mathbf{e}_{k-2}$$

$$\vdots$$

$$\mathbf{e}_k = \prod_{i=0}^{k-1}\big(\mathbf{I} - \alpha_i\mathbf{A}\big)\,\mathbf{e}_0$$

So,

$$\mathbf{e}_k = p_k(\mathbf{A})\,\mathbf{e}_0 \quad \text{where} \quad p_k(\mathbf{A}) \triangleq \prod_{i=0}^{k-1}\big(\mathbf{I} - \alpha_i\mathbf{A}\big).$$

For different choices of $\{\alpha_0, \ldots, \alpha_{k-1}\}$, we get different $p_k(\mathbf{A})$.
Q: What are the best choices of $\{\alpha_0, \ldots, \alpha_{k-1}\}$? Using matrix spectral norm, we obtain

$$\|\mathbf{e}_k\| \leq \|p_k(\mathbf{A})\| \, \|\mathbf{e}_0\|.$$

A: The idea of polynomial acceleration is to find $\{\alpha_0, \ldots, \alpha_{k-1}\}$ that make $\|p_k(\mathbf{A})\|$ small, i.e.,

$$\min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \|p_k(\mathbf{A})\|.$$

We assumed $\mathbf{A} \succ 0$, which in particular implies $\mathbf{A}$ is Hermitian. We can easily see that $p_k(\mathbf{A})$ is also Hermitian (recall that α_i are real).
Recall: If $\mathbf{X}$ is Hermitian, then $\|\mathbf{X}\| = \rho(\mathbf{X}) = \max_{j\in\{1,\ldots,n\}} |\lambda_j(\mathbf{X})|$. So, we have

$$\|p_k(\mathbf{A})\| = \rho\big(p_k(\mathbf{A})\big) = \max_{j\in\{1,\ldots,n\}} \big|\lambda_j\big(p_k(\mathbf{A})\big)\big|.$$

But what is $\lambda(p_k(\mathbf{A}))$?
Recall: If $(\lambda, \mathbf{v})$ is an eigenpair with $\mathbf{A} \in \mathbb{C}^{n\times n}$, then $\big(p(\lambda), \mathbf{v}\big)$ is an eigenpair of $p(\mathbf{A})$.
So, $\lambda(p_k(\mathbf{A})) = p_k(\lambda(\mathbf{A}))$. So,

$$\|p_k(\mathbf{A})\| = \max_{j\in\{1,\ldots,n\}} \big|\lambda_j\big(p_k(\mathbf{A})\big)\big| = \max_{j\in\{1,\ldots,n\}} \big|\, p_k\big(\lambda_j(\mathbf{A})\big)\big| = \max_{j\in\{1,\ldots,n\}} \left|\prod_{i=0}^{k-1} (1 - \alpha_i\, \lambda_j(\mathbf{A}))\right|.$$

Recall that our goal is to find $\{\alpha_0, \ldots, \alpha_{k-1}\}$ that $\|p_k(\mathbf{A})\|$ as small as possible, i.e.,

$$\min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \|p_k(\mathbf{A})\|.$$

this amounts to

$$\min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \max_{j\in\{1,\ldots,n\}} \left|\prod_{i=0}^{k-1} (1 - \alpha_i\, \lambda_j(\mathbf{A}))\right|.$$

Denoting $\lambda_j \triangleq \lambda_j(\mathbf{A})$, $j = 1, \ldots, n$ as the eigenvalues of $\mathbf{A}$, we can equivalently write this as

$$\min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \max_{\lambda\in\{\lambda_1,\lambda_2,\ldots,\lambda_n\}} \left|\prod_{i=0}^{k-1} \big(1 - \alpha_i\lambda\big)\right|.$$

In general, we might not know the eigenvalues of $\mathbf{A}$. But suppose we are able to find a set containing them, i.e.,

$$\{\lambda_1, \lambda_2, \ldots, \lambda_n\} \subseteq S.$$

For example, if the eigenvalues are real, an interval $[a, b]$ with $a \leq \lambda_{\min} \leq \lambda_{\max} \leq b$ is known. Then, since

$$\max_{\lambda\in\{\lambda_1,\lambda_2,\ldots,\lambda_n\}} |p_k(\lambda)| \leq \max_{\lambda\in S} |p_k(\lambda)|,$$

we can consider the looser min–max problem

$$\min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \max_{\lambda\in S} \left|\prod_{i=0}^{k-1} \big(1 - \alpha_i\lambda\big)\right|.$$

So we obtain a "looser" error bound

$$\|\mathbf{e}_k\| \leq \min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \|p_k(\mathbf{A})\| \, \|\mathbf{e}_0\| \leq \min_{\{\alpha_0,\ldots,\alpha_{k-1}\}} \max_{\lambda\in S} |p_k(\lambda)| \, \|\mathbf{e}_0\|,$$

where

$$p_k(\lambda) = \prod_{i=0}^{k-1} (1 - \alpha_i\lambda), \qquad (1)$$

is often called a **residual polynomial**, i.e., $p_k(0) = 1$.
Let Π_k denote the set of all residual polynomials of degree at most k:

$$\Pi_k \triangleq \{\, p(.) \text{ is a polynomial} \mid p(0) = 1,\ \text{degree}(p) \leq k \,\}.$$

Any $p_k \in \Pi_k$ can be uniquely written as (1): $1/\alpha_i$'s are the roots of p_k. So the problem of finding optimal α_i's is equivalent to

$$\min_{p_k \in \Pi_k} \max_{\lambda \in S} |p_k(\lambda)|.$$

Chebyshev polynomials are the key to finding the optimal polynomial. Chebyshev polynomials of the first kind are defined recursively as

$$T_0(x) = 1, \quad T_1(x) = x, \quad \ldots \quad T_{k+1}(x) = 2x\, T_k(x) - T_{k-1}(x), \quad k \geq 1.$$

Alternatively, we also have explicit expressions as

$$T_k(x) = \begin{cases} \cos\big(k \arccos(x)\big), & |x| \leq 1, \\ \dfrac{1}{2}\Big[\big(x + \sqrt{x^2-1}\big)^k + \big(x + \sqrt{x^2-1}\big)^{-k}\Big], & |x| \geq 1. \end{cases}$$

One can show that

$$\min_{p_k \in \Pi_k} \max_{t \in [m,M]} |p_k(t)|$$

is attained **uniquely** by a polynomial of degree exactly k given by

$$p_k^\star(t) = \frac{T_k\left(1 + \dfrac{2(t-M)}{M-m}\right)}{T_k\left(-\dfrac{M+m}{M-m}\right)}.$$

So, now that we have the polynomial, we can

1. find the optimal step-sizes, α_i's, and
2. work out the convergence rate.

The optimal α_i's are given by the reciprocals of the roots of $p_k^\star(t)$ as

$$\alpha_i^{-1} = \frac{1}{2}\left(m + M + (M-m)\cos\left(\frac{\pi\,(2(i+1)-1)}{2k}\right)\right), \quad i = 0, \ldots, k-1.$$

Compare with stationary Richardson iteration $\alpha^{-1} = \dfrac{m+M}{2}$.
We also get

$$\|\mathbf{e}_k\| \leq \min_{p_k \in \Pi_k} \max_{t \in S} |p_k(t)|\, \|\mathbf{e}_0\| = \max_{t \in S} \big|p_k^\star(t)\big|\, \|\mathbf{e}_0\|.$$

It turns out

$$\max_{t \in [m,M]} \big|p_k^\star(t)\big| \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \quad \text{where } \kappa = \frac{M}{m}.$$

So, for the non-stationary Richardson iteration, we get

$$\|\mathbf{e}_k\| \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|\mathbf{e}_0\|.$$

Compare this with the convergence of the stationary Richardson iteration, i.e.,

$$\|\mathbf{e}_k\| \leq \left(\frac{\kappa-1}{\kappa+1}\right)^k \|\mathbf{e}_0\|.$$

Drawbacks of Naïve Chebyshev Acceleration

1. In exact arithmetic, the order in which the parameters α_i's are used does not matter. In practice,

however, this can be very important: the natural ordering may lead to large round-off errors. Hence, without appropriate re-ordering methods, the calculations can be very unstable.

2. One also has to choose the number of iterations k in advance.

3. For Chebyshev acceleration to be effective, a fairly accurate knowledge of the interval $[a, b]$ is required. If this estimate is too crude, the method can be very inefficient.

To alleviate the first two problems, Chebyshev acceleration is stably implemented using the three-term recurrence relation

$$T_{k+1}(x) = 2x\, T_k(x) - T_{k-1}(x).$$

We will see later an alternative method, called conjugate gradient, which also alleviates the last drawback, i.e., it is parameter free.
For $k = 0$

$$\mathbf{x}_1 = \mathbf{x}_0 + \alpha\, \mathbf{r}_0, \quad \text{where } \alpha = \frac{2}{m + M},$$

For $k \geq 1$,

$$\mathbf{x}_{k+1} = \mathbf{x}_{k-1} + \alpha\, \omega_k\, \mathbf{r}_k + \omega_k(\mathbf{x}_k - \mathbf{x}_{k-1}),$$

where $\omega_0 = 2$ and $\omega_k = \left(1 - \frac{\omega_{k-1}}{4\beta^2}\right)^{-1}$. Define $\beta \triangleq 1 - \frac{2M}{M-m}$.
Note: Recall stationary Richardson method: $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha\, \mathbf{r}_k$. Each iteration of accelerated Richardson has nearly the same computational cost as the stationary Richardson method, though it requires slightly more storage.
More generally, such polynomial acceleration techniques can be applied to many stationary methods (under certain conditions).
Chebyshev acceleration typically requires real eigenvalues for $\mathbf{A}$ (in fact, Chebyshev acceleration of Richardson iteration requires real, positive eigenvalues).
Polynomial acceleration, however, can be used in more general settings – even when $\mathbf{A}$ has complex eigenvalues – provided the polynomials are chosen appropriately. It is also possible to derive acceleration schemes adapted to cases where the spectrum lies in two intervals, $[a, b] \cup [c, d]$, with $a < b < 0 < c < d$.

Chapter 5

Fundamentals

5.1 Topology/Geometry

Definition 5.1.1

The line segment between two points $\boldsymbol{x}$ and $\boldsymbol{y}$ in $\mathbb{R}^d$ is the set of points on the straight line joining points, i.e.,

$$\{\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y} \mid \alpha \in [0,1]\} = \{\boldsymbol{y} + \alpha(\boldsymbol{x} - \boldsymbol{y}) \mid \alpha \in [0,1]\}$$

Note: A point $\boldsymbol{z} = \alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}$ where $\alpha \in [0,1]$ is a *convex combination* of $\boldsymbol{x}, \boldsymbol{y}$.

Definition 5.1.2

For a given $\boldsymbol{a} \in \mathbb{R}^d$ and $b \in \mathbb{R}$, a hyperplane is the set of the form

$$\left\{\boldsymbol{x} \in \mathbb{R}^d \mid \langle \boldsymbol{a}, \boldsymbol{x} \rangle = b\right\}$$

- A hyperplane is not a subspace of $\mathbb{R}^d$ unless $b = 0$.
- For $d = 2$, a hyperplane is simply a straight line.
- For $d = 3$, a hyperplane is simply a plane.

Let $\boldsymbol{x}_0$ be any point on this hyperplane, i.e., $b = \langle \boldsymbol{a}, \boldsymbol{x}_0 \rangle$. We can alternatively define the hyperplane as

$$\left\{\boldsymbol{x} \in \mathbb{R}^d \mid \langle \boldsymbol{a}, \boldsymbol{x} \rangle = b\right\} = \left\{\boldsymbol{x} \in \mathbb{R}^d \mid \langle \boldsymbol{a}, \boldsymbol{x} - \boldsymbol{x}_0 \rangle = 0\right\}$$

Definition 5.1.3

Half-spaces: A hyperplane divides $\mathbb{R}^d$ into two halfspaces

$$\left\{\boldsymbol{x} \in \mathbb{R}^d \mid \langle \boldsymbol{a}, \boldsymbol{x} \rangle \leq b\right\}, \quad \left\{\boldsymbol{x} \in \mathbb{R}^d \mid \langle \boldsymbol{a}, \boldsymbol{x} \rangle \geq b\right\}$$

Definition 5.1.4

A polyhedron is defined as the solution set of a finite number of linear equalities and inequalities, i.e.,

$$\mathcal{P} \triangleq \{\boldsymbol{x} \mid \langle \boldsymbol{a}_i, \boldsymbol{x} \rangle \leq b_i,\ i = 1, \ldots, p,\ \langle \boldsymbol{c}_j, \boldsymbol{x} \rangle = d_j,\ j = 1, \ldots, q\} \triangleq \{\boldsymbol{x} \mid A\boldsymbol{x} \leq \boldsymbol{b},\ C\boldsymbol{x} = \boldsymbol{d}\}$$

where

$$A = \begin{bmatrix} \boldsymbol{a}_1 & \ldots & \boldsymbol{a}_p \end{bmatrix}^\top, \quad C = \begin{bmatrix} \boldsymbol{c}_1 & \ldots & \boldsymbol{c}_q \end{bmatrix}^\top$$

A polyhedron is thus the intersection of a finite number of half-spaces and hyper-planes.

Subspaces, lines, rays, half-spaces, hyperplanes are all polyhedra. Bounded polyhedra are called polytopes.

Example.

- The non-negative orthant is a polyhedron defined as

$$\mathbb{R}^d_+ \triangleq \left\{\boldsymbol{x} \in \mathbb{R}^d \mid x_i \geq 0,\ i = 1, \ldots, d\right\}.$$

- The positive orthant is a polyhedron defined as

$$\mathbb{R}^d_{++} \triangleq \left\{\boldsymbol{x} \in \mathbb{R}^d \mid x_i > 0,\ i = 1, \ldots, d\right\}.$$

- The unit-simplex is a polytope defined as

$$\mathcal{E}_d \triangleq \left\{\boldsymbol{x} \in \mathbb{R}^d \mid \boldsymbol{x} \geq 0,\ \langle \mathbf{1}, \boldsymbol{x}\rangle \leq 1\right\}.$$

- The probability simplex is a polytope defined as

$$\Delta_d \triangleq \left\{\boldsymbol{x} \in \mathbb{R}^d \mid \boldsymbol{x} \geq 0,\ \langle \mathbf{1}, \boldsymbol{x}\rangle = 1\right\}.$$

Definition 5.1.5

A set $\mathcal{C}$ is convex if the line segment between any two points in $\mathcal{C}$ lies in $\mathcal{C}$, i.e., if for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{C}$ and any $\alpha \in [0, 1]$, we have $\alpha\boldsymbol{x} + (1 - \alpha)\boldsymbol{y} \in \mathcal{C}$.

Example.

Examples: points, lines, line segments, subspaces, hyperplanes, half-spaces, polyhedra, PSD matrices.

Theorem 5.1.6: Operations Preserving Convexity

- If $\mathcal{C}$ is convex, then so is $\alpha\mathcal{C} \triangleq \{\boldsymbol{y} \mid \boldsymbol{y} = \alpha\boldsymbol{x},\ \boldsymbol{x} \in \mathcal{C}\}$.
- If $\mathcal{C}_1$ and $\mathcal{C}_2$ are convex, then so is $\mathcal{C}_1 + \mathcal{C}_2 \triangleq \{\boldsymbol{y} \mid \boldsymbol{y} = \boldsymbol{x}_1 + \boldsymbol{x}_2,\ \boldsymbol{x}_1 \in \mathcal{C}_1,\ \boldsymbol{x}_2 \in \mathcal{C}_2\}$.
- If $\mathcal{C}$ is convex and f is affine, i.e., $f(\boldsymbol{x}) = A\boldsymbol{x} + \boldsymbol{b}$ where $A \in \mathbb{R}^{n\times d}$ and $\boldsymbol{b} \in \mathbb{R}^n$, then so is $f(\mathcal{C}) \triangleq \{f(\boldsymbol{x}) \mid \boldsymbol{x} \in \mathcal{C}\}$.
- If $\mathcal{C}_1$ and $\mathcal{C}_2$ are convex, then so is $\mathcal{C}_1 \times \mathcal{C}_2 \triangleq \{(\boldsymbol{x}_1, \boldsymbol{x}_2) \mid \boldsymbol{x}_1 \in \mathcal{C}_1,\ \boldsymbol{x}_2 \in \mathcal{C}_2\}$.
- The intersection of any collection of convex sets is convex.

Definition 5.1.7: Local Neighborhood

For any $\varepsilon > 0$, an ε-neighborhood of a point $\mathbf{x} \in \mathbb{R}^d$ is $\mathcal{B}(\mathbf{x}, \varepsilon) \triangleq \{\mathbf{y} \in \mathbb{R}^d \mid \|\mathbf{y} - \mathbf{x}\| \leq \varepsilon\}$. Obviously, different norms define different neighborhoods.

Definition 5.1.8: Interior Point

$\mathbf{x} \in \mathcal{C}$ is an interior point if $\exists\, \varepsilon > 0$ such that $\mathcal{B}(\mathbf{x}, \varepsilon) \subseteq \mathcal{C}$.

Definition 5.1.9: Boundary Point

$\mathbf{x}$ is on the boundary if every neighborhood of $\mathbf{x}$ intersects both $\mathcal{C}$ and its complement.

- Open: contains all its interior points.
- Closed: contains all its boundary points (i.e., its complement is open).
- Bounded: fits inside a finite ball (i.e., $\mathcal{C} \subseteq \mathcal{B}(\mathbf{0}, M)$ for some $M > 0$.)

Theorem 5.1.10: Heine–Borel Theorem

A subset of the Euclidean space $\mathbb{R}^d$ is compact if it is both closed and bounded.

Note: The topological properties of sets in $\mathbb{R}^d$ (or $\mathbb{C}^d$), such as being open, closed, bounded, or compact, do not depend on the norm being used, which is a consequence of the norm equivalence property in finite-dimensional real or complex vector spaces.

Theorem 5.1.11: Level Sets

For $f : \mathcal{C} \to \mathbb{R}$, the sub-level set $\mathcal{S}(f, \alpha) \triangleq \{\mathbf{x} \in \mathcal{C} \mid f(\mathbf{x}) \leq \alpha\}$ is closed if f is continuous and $\mathcal{C}$ is closed.

Theorem 5.1.12: Weierstrass Theorem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be a continuous function, and let $\mathcal{C} \subseteq \mathbb{R}^d$. If either

- $\mathcal{C} \subseteq \mathbb{R}^d$ is compact, or
- the sub-level set $\mathcal{S}(f, \alpha)$ is nonempty and compact for some α, or
- $\mathcal{C} \subseteq \mathbb{R}^d$ is closed and f is coercive, i.e., $\lim_{\|\boldsymbol{x}\| \to \infty} f(\boldsymbol{x}) = \infty$,

then, there exists a point $\boldsymbol{x}^\star \in \mathcal{C}$ such that $f(\boldsymbol{x}^\star) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \mathcal{C}$. In other words, f achieves its minimum in $\mathcal{C}$.

Example.

Examples of functions that do not attain their minimum on $\mathcal{C}$

- $f = e^{-x}$ and $\mathcal{C} = [0, \infty)$
- $f = 1/x$ and $\mathcal{C} = (0, 1)$
- $\mathcal{C} = [-1, 1]$ and

$$f(x) = \begin{cases} 0, & x \in \{-1, 1\} \\ x, & -1 < x < 1 \end{cases}$$

5.2 Calculus

Definition 5.2.1

Let $f : \mathcal{U} \to \mathbb{R}$, $\mathcal{U} \subseteq \mathbb{R}^d$, $\boldsymbol{x} \in \operatorname{int}(\mathcal{U})$ and $\boldsymbol{p} \in \mathbb{R}^d$, $\boldsymbol{p} \neq 0$. If the limit

$$\lim_{t \to 0} \frac{f(\boldsymbol{x} + t\boldsymbol{p}) - f(\boldsymbol{x})}{t},$$

exists, then it is called the directional derivative of f at $\boldsymbol{x}$ along the direction $\boldsymbol{p}$, and at times denoted by $f'(\boldsymbol{x}; \boldsymbol{p})$.

Intuition: This tells us how f changes if we move from $\mathbf{x}$ in the direction $\mathbf{p}$. Think of it as the slope along a chosen path.

Definition 5.2.2

For any $i = 1, \ldots, d$, the directional derivative of f at $\boldsymbol{x}$ along the direction $\boldsymbol{e}_i$ is called the ith partial derivative, i.e.,

$$\frac{\partial f}{\partial x_i}(\boldsymbol{x}) \triangleq \lim_{t \to 0} \frac{f(\boldsymbol{x} + t\boldsymbol{e}_i) - f(\boldsymbol{x})}{t}.$$

Intuition: This measures how f changes if we change only the i-th variable, keeping all others fixed.

Definition 5.2.3

If all the partial derivatives of a function f exist at a point $\boldsymbol{x} \in \mathbb{R}^d$, then the gradient of f at $\boldsymbol{x}$ is defined to be the column vector consisting of all the partial derivatives as

$$\nabla f(\boldsymbol{x}) \triangleq \begin{bmatrix} \frac{\partial f}{\partial x_1}(\boldsymbol{x}) \\ \frac{\partial f}{\partial x_2}(\boldsymbol{x}) \\ \vdots \\ \frac{\partial f}{\partial x_d}(\boldsymbol{x}) \end{bmatrix}$$

Intuition: Think of the gradient as an arrow pointing in the direction where f increases the fastest. Its length tells us how steep the increase is.
Local Linear Approximation: Differential calculus approximates a function near a point by a linear function:

$$f(\mathbf{x} + \mathbf{p}) \approx f(\mathbf{x}) + \text{linear term in } \mathbf{p}.$$

Intuition: Zoom in close enough on a smooth function, and it almost looks like a hyperplane, e.g., a plane (in 2D) or a straight line (in 1D).

Definition 5.2.4

A function f defined on an open set $\mathcal{U} \subseteq \mathbb{R}^d$ is called differentiable at $\boldsymbol{x} \in \mathcal{U}$, if there is a linear map $\ell_{\boldsymbol{x}} : \mathbb{R}^d \to \mathbb{R}$ such that

$$\lim_{\boldsymbol{p} \to 0} \frac{f(\boldsymbol{x} + \boldsymbol{p}) - f(\boldsymbol{x}) - \ell_{\boldsymbol{x}}(\boldsymbol{p})}{\|\boldsymbol{p}\|} = 0.$$

Loosely speaking, f looks linear locally at $\boldsymbol{x}$:

$$f(\boldsymbol{x} + \boldsymbol{p}) \approx f(\boldsymbol{x}) + \ell_{\boldsymbol{x}}(\boldsymbol{p}) + o(\|\boldsymbol{p}\|).$$

Intuition: The differentiable function behaves like a hyperplane very close to $\mathbf{x}$:

$$f(\mathbf{x} + \mathbf{p}) \approx f(\mathbf{x}) + \ell_{\mathbf{x}}(\mathbf{p}) + o(\|\mathbf{p}\|).$$

The linear map $\ell_{\mathbf{x}}$ is essentially the "best linear approximation" of the differentiable function near $\mathbf{x}$.

- $\ell_{\mathbf{x}}$ is unique: $\ell_{\mathbf{x}}(\mathbf{p}) = \langle \nabla f(\mathbf{x}), \mathbf{p} \rangle$.
- All directional derivatives exist: $f'(\mathbf{x}; \mathbf{q}) = \langle \nabla f(\mathbf{x}), \mathbf{q} \rangle$.

Definition 5.2.5

A function is called differentiable over $\mathcal{U}$ if it is differentiable for any $\boldsymbol{x} \in \mathcal{U}$.

Definition 5.2.6

A function f defined on an open set $\mathcal{U} \subseteq \mathbb{R}^d$ is called continuously differentiable over $\mathcal{U}$, and denoted by $f \in \mathcal{C}^1(\mathcal{U})$, if its partial derivatives exist and are continuous on $\mathcal{U}$.

Intuition: The slope (gradient) changes smoothly as you move around, i.e., no sudden jumps.
The definition of continuous differentiability can also be extended to non-open sets by considering an open set $\mathcal{U}$ containing $\mathcal{C}$ on which the function is also defined and continuously differentiable.
Some important things to remember:

- Necessary condition: If a function is discontinuous then it is not differentiable (a jump cannot have a slope).
- Necessary condition: If partial derivatives do not exist then the function can't be differentiable (you need slopes along each axis).
- Sufficient condition: If partial derivatives exist and are continuous, then the function is continuously differentiable (slopes change gradually).
- Inconclusive: If partial derivatives exist but are not continuous, then we need to check differentiability directly using its definition (slope could oscillate wildly).

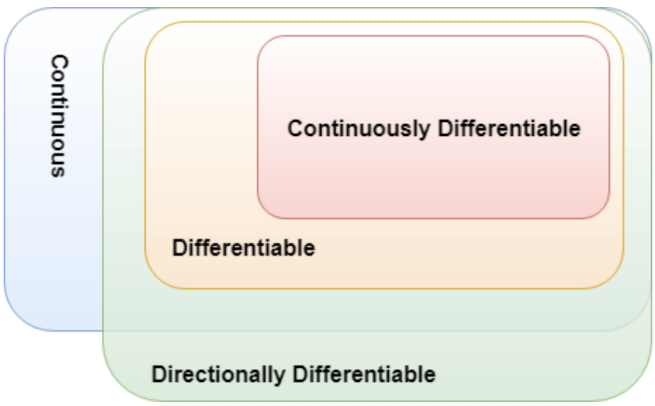


Figure 5.1: Smoothness Hierarchy.

Definition 5.2.7

Let f be defined on an open set $\mathcal{U} \subseteq \mathbb{R}^d$. If all the partial derivatives of f are themselves differentiable on $\mathcal{U}$, the Hessian matrix of f at any $\boldsymbol{x} \in \mathcal{U}$ is defined as

$$\nabla^2 f(\boldsymbol{x}) \triangleq \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\boldsymbol{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\boldsymbol{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_d}(\boldsymbol{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\boldsymbol{x}) & \frac{\partial^2 f}{\partial x_2^2}(\boldsymbol{x}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_d}(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1}(\boldsymbol{x}) & \frac{\partial^2 f}{\partial x_d \partial x_2}(\boldsymbol{x}) & \cdots & \frac{\partial^2 f}{\partial x_d^2}(\boldsymbol{x}) \end{bmatrix}$$

Intuition: The Hessian captures the curvature of f:

- Positive curvature $\to$ curving upward like a bowl
- Negative curvature $\to$ curving downward like a hill
- Zero curvature $\to$ flat

Definition 5.2.8

If all the partial derivatives of f are themselves *continuously* differentiable on $\mathcal{U}$, we say that f is twice continuously differentiable on $\mathcal{U}$, denoted as $f \in \mathcal{C}^2(\mathcal{U})$.

Intuition: Not only is the slope smooth, but the curvature itself changes smoothly.
Note: By Schwarz's theorem (sometimes called Clairaut's theorem), the Hessian is symmetric if $f \in \mathcal{C}^2(\mathcal{U})$. However, it turns out that the differentiability of partial derivatives of f is enough, which is a weaker condition.

Fact 5.2.9

$$f(\mathbf{x}) = \langle \mathbf{b}, \mathbf{x} \rangle \Longrightarrow \begin{cases} \nabla f(\mathbf{x}) = \mathbf{b}, \\ \nabla^2 f(\mathbf{x}) = 0 \end{cases}$$

$$f(\mathbf{x}) = \langle \mathbf{x}, \mathbf{B}\mathbf{x} \rangle \Longrightarrow \begin{cases} \nabla f(\mathbf{x}) = (\mathbf{B} + \mathbf{B}^\mathsf{T})\mathbf{x}, \\ \nabla^2 f(\mathbf{x}) = \mathbf{B} + \mathbf{B}^\mathsf{T} \end{cases}$$

These are "go-to" formulas for linear and quadratic functions, which appear frequently in optimization.

Theorem 5.2.10: Taylor's Theorem

1. If If $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable, then for all $\mathbf{x}$ and $\mathbf{p}$:

$$f(\mathbf{x} + \mathbf{p}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{p} \rangle + \|\mathbf{p}\| \, \delta(\mathbf{p}), \qquad \lim_{\mathbf{p} \to 0} \delta(\mathbf{p}) = 0.$$

   Intuition: f near $\mathbf{x}$ is approximately linear; the error $\delta(\mathbf{p})$ goes to 0 as $\mathbf{p} \to 0$.

2. If f is twice differentiable:

$$f(\mathbf{x} + \mathbf{p}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{p} \rangle + \tfrac{1}{2} \langle \mathbf{p}, \nabla^2 f(\mathbf{x}) \, \mathbf{p} \rangle + \|\mathbf{p}\|^2 \, \delta(\mathbf{p}), \qquad \lim_{\mathbf{p} \to 0} \delta(\mathbf{p}) = 0.$$

   Intuition: Adds curvature information via the Hessian; gives a better local approximation.

Theorem 5.2.11: Taylor's Theorem — Other Formulas

1. If $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable, then for all $\boldsymbol{x}$ and $\boldsymbol{p}$:
$$f(\boldsymbol{x}+\boldsymbol{p}) = f(\boldsymbol{x}) + \langle \nabla f(\boldsymbol{x}+t^*\boldsymbol{p}), \boldsymbol{p}\rangle, \quad t^* \in (0,1).$$
2. If $f : \mathbb{R}^d \to \mathbb{R}$ is continuously differentiable, then for all $\boldsymbol{x}$ and $\boldsymbol{p}$:
$$f(\boldsymbol{x}+\boldsymbol{p}) = f(\boldsymbol{x}) + \int_0^1 \langle \nabla f(\boldsymbol{x}+t\boldsymbol{p}), \boldsymbol{p}\rangle \, dt.$$
3. If $f : \mathbb{R}^d \to \mathbb{R}$ is twice differentiable, then for all $\boldsymbol{x}$ and $\boldsymbol{p}$:
$$f(\boldsymbol{x}+\boldsymbol{p}) = f(\boldsymbol{x}) + \langle \nabla f(\boldsymbol{x}), \boldsymbol{p}\rangle + \frac{1}{2}\langle \boldsymbol{p}, \nabla^2 f(\boldsymbol{x}+t^*\boldsymbol{p})\boldsymbol{p}\rangle, \quad t^* \in (0,1).$$
4. If $f : \mathbb{R}^d \to \mathbb{R}$ is twice continuously differentiable, then for all $\boldsymbol{x}$ and $\boldsymbol{p}$:
$$f(\boldsymbol{x}+\boldsymbol{p}) = f(\boldsymbol{x}) + \langle \nabla f(\boldsymbol{x}), \boldsymbol{p}\rangle + \int_0^1 (1-t)\langle \boldsymbol{p}, \nabla^2 f(\boldsymbol{x}+t\boldsymbol{p})\boldsymbol{p}\rangle \, dt$$
$$\nabla f(\boldsymbol{x}+\boldsymbol{p}) = \nabla f(\boldsymbol{x}) + \int_0^1 \nabla^2 f(\boldsymbol{x}+t\boldsymbol{p})\boldsymbol{p} \, dt.$$

Recall: A function $f : \mathbb{R} \to \mathbb{R}$ is continuous iff for any x, $\lim_{y\to x} f(y) = f(x)$. So as y gets closer to x, $f(y)$ also gets closer to $f(x)$. But if the rate at which $f(y)$ gets closer to $f(x)$ is "controlled" by the rate at which y gets closer to x, then f is Lipschitz continuous, which is smoother than just being continuous.

Definition 5.2.12

A mapping $h : \mathcal{U} \to \mathcal{V}$ is called L-Lipschitz continuous if there is a constant $0 \le L < \infty$ such that
$$\|h(\boldsymbol{x}) - h(\boldsymbol{y})\| \le L\|\boldsymbol{x}-\boldsymbol{y}\|, \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathcal{U}.$$

Intuition: The function cannot change faster than L times the distance between inputs; think of L as a maximum slope.
Note: For us, the norms are always Euclidean (for vectors) and spectral (for matrices).

Definition 5.2.13: Smoothness

- We say f has L_g-Lipschitz gradient if
$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \le L_g\|\boldsymbol{x}-\boldsymbol{y}\|.$$
The class of such functions is often denoted as $\mathcal{C}^{1,1}_{L_g}(\mathbb{R}^d)$.
- We say f has L_H-Lipschitz Hessian if
$$\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \le L_H\|\boldsymbol{x}-\boldsymbol{y}\|.$$
The class of such functions is often denoted as $\mathcal{C}^{2,2}_{L_H}(\mathbb{R}^d)$.

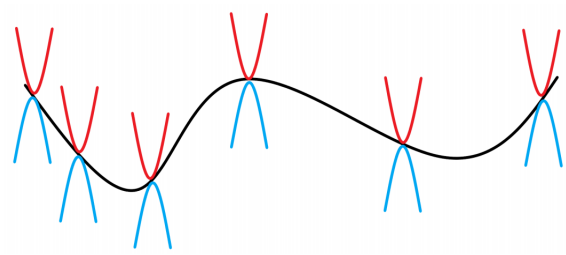
There are also other varieties, e.g., $\mathcal{C}^{2,1}_{L_g}(\mathbb{R}^d)$, i.e., twice continuously differentiable with Lipschitz continuous gradient.

Theorem 5.2.14

If $f \in \mathcal{C}^{1,1}_{L_g}(\mathbb{R}^d)$, then for all $\boldsymbol{x}$ and $\boldsymbol{y}$

$$|f(\boldsymbol{y}) - f(\boldsymbol{x}) - \langle \nabla f(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x}\rangle| \leq \frac{L_g}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2.$$

In words, when $f \in \mathcal{C}^{1,1}_{L_g}(\mathbb{R}^d)$, we can put a quadratic with positive curvature of L_g above and a quadratic with negative curvature of $-L_g$ below f at any point, and f will always lie in between these two over all of $\mathbb{R}^d$. f lies between two quadratic bounds around any point.



Proof. Since f has L_g-Lipschitz gradient, i.e., the gradient is continuous, then f is continuously differentiable. So by Taylor's theorem for all $\boldsymbol{x}$ and $\boldsymbol{y}$, we have

$$f(\boldsymbol{y}) = f(\boldsymbol{x}) + \int_0^1 \langle \nabla f(\boldsymbol{x}+t(\boldsymbol{y}-\boldsymbol{x})), \boldsymbol{y}-\boldsymbol{x}\rangle dt = f(\boldsymbol{x}) + \langle \nabla f(\boldsymbol{x}), \boldsymbol{y}-\boldsymbol{x}\rangle + \int_0^1 \langle \nabla f(\boldsymbol{x}+t(\boldsymbol{y}-\boldsymbol{x})) - \nabla f(\boldsymbol{x}), \boldsymbol{y}-\boldsymbol{x}\rangle dt.$$

Hence,

$$\begin{aligned}
|f(\boldsymbol{y}) - f(\boldsymbol{x}) - \langle \nabla f(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x}\rangle| &= \left| \int_0^1 \langle \nabla f(\boldsymbol{x} + t(\boldsymbol{y} - \boldsymbol{x})) - \nabla f(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x}\rangle dt \right| \\
&\leq \int_0^1 |\langle \nabla f(\boldsymbol{x} + t(\boldsymbol{y} - \boldsymbol{x})) - \nabla f(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x}\rangle| \, dt \\
&\leq \int_0^1 \|\nabla f(\boldsymbol{x} + t(\boldsymbol{y} - \boldsymbol{x})) - \nabla f(\boldsymbol{x})\| \cdot \|\boldsymbol{y} - \boldsymbol{x}\| dt \\
&\leq \int_0^1 L_g \|t(\boldsymbol{y} - \boldsymbol{x})\| \cdot \|\boldsymbol{y} - \boldsymbol{x}\| dt \\
&= \int_0^1 t L_g \|\boldsymbol{y} - \boldsymbol{x}\|^2 dt \\
&= L_g \|\boldsymbol{y} - \boldsymbol{x}\|^2 \int_0^1 t dt = \frac{L_g}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2.
\end{aligned}$$

□

Theorem 5.2.15

Suppose f is twice continuously differentiable. Then

$$f \in \mathcal{C}^{2,1}_{L_g}(\mathbb{R}^d) \iff -L_g I \preceq \nabla^2 f(\boldsymbol{x}) \preceq L_g I, \quad \forall \boldsymbol{x}.$$

Intuition:

Lipschitz gradient $\iff$ rate of change of gradient is bounded $\iff$ Bounded Hessian

Proof. Omitted. □

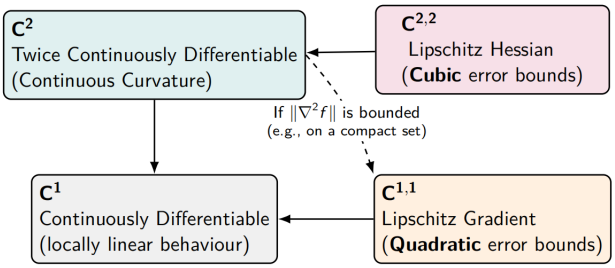
Theorem 5.2.16

If $f \in \mathcal{C}^{2,2}_{L_H}(\mathbb{R}^d)$, then for all $\boldsymbol{x}$ and $\boldsymbol{y}$

$$\left\|\nabla f(\boldsymbol{y}) - \nabla f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{x})(\boldsymbol{y}-\boldsymbol{x})\right\| \leq \frac{L_H}{2}\|\boldsymbol{y}-\boldsymbol{x}\|^2,$$

$$\left| f(\boldsymbol{y}) - f(\boldsymbol{x}) - \langle \nabla f(\boldsymbol{x}), \boldsymbol{y}-\boldsymbol{x}\rangle - \frac{1}{2}\langle \boldsymbol{y}-\boldsymbol{x}, \nabla^2 f(\boldsymbol{x})(\boldsymbol{y}-\boldsymbol{x})\rangle \right| \leq \frac{L_H}{6}\|\boldsymbol{y}-\boldsymbol{x}\|^3.$$

Intuition: f is so smooth that its gradient behaves almost linearly, and f behaves almost quadratically near any point, with a controlled cubic error.



Chapter 6

Optimality and Convexity

6.1 Optimal Points

Definition 6.1.1

$\boldsymbol{x}^\star \in \mathcal{C}$ is a global solution if and only if

$$f(\boldsymbol{x}^\star) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \mathcal{C}.$$

- Intuition: It is the best solution in the entire feasible set $\mathcal{C}$.
- Strict global minimizer: Inequality is strict for all $\mathbf{x} \neq \mathbf{x}^\star$.
- Reality check: Finding a global minimizer is often hard or impossible for complex problems.

Definition 6.1.2

$\boldsymbol{x}^\star \in \mathcal{C}$ is a local solution if and only if there exists an ε-neighborhood $\mathcal{B}(\boldsymbol{x}^\star, \varepsilon) \subseteq \mathcal{C}$ of $\boldsymbol{x}^\star$ such that

$$f(\boldsymbol{x}^\star) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \mathcal{B}(\boldsymbol{x}^\star, \varepsilon).$$

- "Near" means within a small neighborhood (a ball) around $\mathbf{x}^\star$, i.e., $\mathcal{B}(\mathbf{x}^\star, \varepsilon)$ for some $\varepsilon > 0$.
- Intuition: It is the best solution in a neighborhood around $\mathbf{x}^\star$.
- Strict local minimizer: Inequality is strict for all other points in that neighborhood.

6.2 Unconstrained Optimization

Definition 6.2.1: Generic Unconstrained Optimization

$$\min_{\boldsymbol{x} \in \mathbb{R}^d} f(\boldsymbol{x})$$

Theorem 6.2.2: First-Order Necessary Condition

If $\boldsymbol{x}^\star$ is an unconstrained local minimizer and f is differentiable in an open neighborhood of $\boldsymbol{x}^\star$, then $\nabla f(\boldsymbol{x}^\star) = 0$.

- Intuition: At a local minimum, the slope of the function must be flat; otherwise, you could move in the opposite direction of the gradient to decrease f, meaning you were not at a minimum.

- Contrapositive form: If $\nabla f(\mathbf{x}^\star) \neq 0$, then $\mathbf{x}^\star$ cannot be a local minimizer.
- Constrained case:
$$\min_{\mathbf{x}\in\mathcal{C}} f(\mathbf{x}).$$
If $\mathbf{x}^\star \in \operatorname{int}(\mathcal{C})$, the same conclusion holds.

Proof. Omitted. □

Definition 6.2.3

Any point $\boldsymbol{x}^\star \in \mathbb{R}^d$ for which $\nabla f(\boldsymbol{x}^\star) = 0$ is called a stationary point.

Note: Any minimizer must be a stationary point, but the converse is not necessarily true, e.g., $f(x) = x^3$. In fact, maximizers also satisfy the same condition, e.g., $f(x) = -x^2$. So to distinguish between minimizers, maximizers, and saddle points, we need to rely on more than just gradient information.

Theorem 6.2.4: Second-Order Necessary Conditions

If $\boldsymbol{x}^\star$ is an unconstrained local minimizer and f is twice differentiable in an open neighborhood of $\boldsymbol{x}^\star$, then in addition to $\nabla f(\boldsymbol{x}^\star) = 0$, we must have
$$\nabla^2 f(\boldsymbol{x}^\star) \succeq 0.$$

- Intuition: At a local minimum, not only is the slope zero (gradient vanishes), but the surface does not curve downward (the Hessian is positive semi-definite).
- Contrapositive form: If $\nabla f(\mathbf{x}^\star) \neq 0$ or the Hessian is not positive semi-definite, then $\mathbf{x}^\star$ cannot be a local minimizer.
- Constrained case: For constrained problems
$$\min_{\mathbf{x}\in\mathcal{C}} f(\mathbf{x}),$$
if $\mathbf{x}^\star \in \operatorname{int}(\mathcal{C})$, the same conclusion holds.

Proof. Omitted. □

Example.

Using second-order necessary condition we can classify stationary points a bit better. For example, we can definitively say that the stationary point of $f(x) = -x^2$ is not a local minimizer. Another example is $f(\boldsymbol{x}) = x_1^2 - x_2^2$.
$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} 2x_1 \\ -2x_2 \end{bmatrix} = 0 \Rightarrow x_1 = x_2 = 0$$
$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \not\succeq 0 \Rightarrow \boldsymbol{x} \text{ is not a minimizer}$$

Example.

Second-order necessary conditions are not enough/sufficient in some cases. $f(x_1, x_2) = x_1^2 - x_2^4$:
$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ -4x_2^3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x_1 = x_2 = 0.$$
$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & -12x_2^2 \end{bmatrix} \implies \nabla^2 f(\mathbf{0}) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \succeq 0.$$

But $\mathbf{x} = \mathbf{0}$ is not a local min:

$$f(0,t) = -t^4 < 0 \quad (t \neq 0), \qquad f(s,0) = s^2 \geq 0.$$

Hence, arbitrarily close to $(0,0)$, f takes on both negative and positive values. Conclusion: $(0,0)$ is a saddle point even though the Hessian is PSD.

Theorem 6.2.5: Second-Order Sufficient Conditions

Consider unconstrained optimization of f that is twice differentiable in an open neighborhood of $\boldsymbol{x}^\star$. If $\nabla f(\boldsymbol{x}^\star) = 0$ and $\nabla^2 f(\boldsymbol{x}^\star) \succ 0$, then $\boldsymbol{x}^\star$ is a strict local minimum of f.

- Intuition: A positive definite Hessian at a stationary point means the function curves upward in all directions (like the bottom of a perfectly shaped bowl), guaranteeing a strict local minimum.
- These conditions are sufficient but not necessary: for example, $f(x) = x^4$ has a local minimum at $x = 0$, but $\nabla^2 f(0) = 0$.
- Constrained case: For constrained problems

$$\min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x}),$$

  if $\mathbf{x}^\star \in \operatorname{int}(\mathcal{C})$, the same conclusion holds.

Proof. Omitted. □

Definition 6.2.6: Saddle points

Stationary points for which there exist directions $\mathbf{p}$ and $\mathbf{q}$ such that for all $\alpha > 0$ sufficiently small, we have

$$f(\mathbf{x}) < f(\mathbf{x} + \alpha \mathbf{p}),$$
$$f(\mathbf{x}) > f(\mathbf{x} + \alpha \mathbf{q}).$$

In other words, saddle point is neither a local minimum point nor a local maximum point.

Theorem 6.2.7: Sufficient Condition for a Saddle Point

Suppose f is twice differentiable in an open neighborhood of $\mathbf{x}^*$,

$$\nabla f(\mathbf{x}^*) = 0 \quad \text{and} \quad \nabla^2 f(\mathbf{x}^*) \text{ is indefinite.}$$

Then $\mathbf{x}^*$ is a saddle point of f. In other words, when $\nabla f(\mathbf{x}^*) = 0$ and $\nabla^2 f(\mathbf{x}^*)$ has both strictly positive and strictly negative eigenvalues, $\mathbf{x}^*$ is a saddle point.

Proof. Omitted. □

Intuition: The Hessian has both positive and negative eigenvalues, so the function curves up in some directions and down in others.
If the Hessian is only semi-definite, second-order conditions alone cannot classify the point (example: $f(x_1, x_2) = x_1^2 - x_2^4$ at $(0,0)$).
Since we only used local information, we can at best guarantee local optimality of stationary points. But how about global optimality? Well, for that we need to know something about the global structure of f.

Theorem 6.2.8: Sufficient Condition for Global Optimality

Suppose f is twice differentiable. If $\nabla^2 f(\mathbf{x}) \succeq 0$, $\forall \mathbf{x} \in \mathbb{R}^d$, then every stationary point of f is a global minimizer.

Proof. Let $\mathbf{x}^*$ be a stationary point of f. By Taylor's theorem, $\forall\, \mathbf{x} \in \mathbb{R}^d$ and $t \in (0,1)$ we have,

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \langle \nabla f(\mathbf{x}^*),\, \mathbf{x} - \mathbf{x}^* \rangle + \tfrac{1}{2} \left\langle \mathbf{x} - \mathbf{x}^*,\, \nabla^2 f\big(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)\big)\, (\mathbf{x} - \mathbf{x}^*) \right\rangle$$

$$= f(\mathbf{x}^*) + \tfrac{1}{2} \left\langle \mathbf{x} - \mathbf{x}^*,\, \nabla^2 f\big(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)\big)\, (\mathbf{x} - \mathbf{x}^*) \right\rangle \geq f(\mathbf{x}^*).$$

□

Note: We will later call this sufficient condition "convexity".

Example.

$$\min_{\mathbf{x} \in \mathbb{R}^d} \tfrac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \min_{\mathbf{x} \in \mathbb{R}^d} \tfrac{1}{2} \langle \mathbf{x}, \mathbf{A}^T\mathbf{A}\mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{A}^T\mathbf{b} \rangle + \tfrac{1}{2} \langle \mathbf{b}, \mathbf{b} \rangle.$$

- Stationary points:
$$\nabla f(\mathbf{x}) = \mathbf{A}^T\mathbf{A}\mathbf{x} - \mathbf{A}^T\mathbf{b} = 0 \Rightarrow \mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}.$$
  Any $\mathbf{x}$ such that $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$ is a stationary point.
- Minimizers, maximizers, or saddle points?
  - $\nabla^2 f(\mathbf{x}) = \mathbf{A}^T\mathbf{A} \succeq 0 \Rightarrow$ every stationary point is a global minimizer.
  - If $\mathbf{A}$ has linearly independent columns, then $\nabla^2 f(\mathbf{x}) \succ 0 \Rightarrow \mathbf{x}^* = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$ is strict global minimizer.

6.3 Constrained Optimization

Generic Constrained Optimization

$$\min f(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{x} \in \mathcal{C},$$

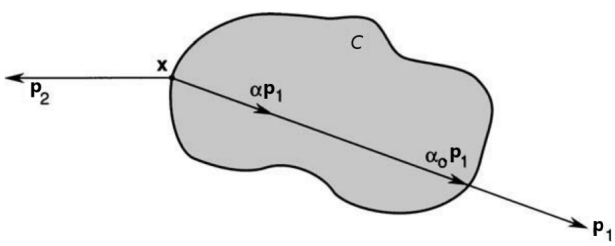
where $f : \mathbb{R}^d \to \mathbb{R}$, and $\mathcal{C} \subseteq \mathbb{R}^d$ is closed.

- If $\mathbf{x}^\star \in \operatorname{int}(\mathcal{C})$ (or $\mathcal{C}$ is open), the optimality conditions are the same as those of unconstrained optimization.
- If $\mathbf{x}^\star \in \partial\mathcal{C}$ (boundary of $\mathcal{C}$), then the constraints restrict feasible directions.

Definition 6.3.1

A vector $\mathbf{p} \in \mathbb{R}^d$ is a feasible direction at $\mathbf{x} \in \mathcal{C}$ if there exists $\bar{\alpha}$ such that $\mathbf{x} + \alpha\mathbf{p} \in \mathcal{C}$, $\forall\, \alpha \in [0, \bar{\alpha}]$.

Intuition: A feasible direction is any direction you can "step into" from your current point without leaving the feasible set. Think of it as a direction that keeps you inside the allowed region, at least for a small step.



When $\mathcal{C}$ is convex, the set of feasible directions at $\mathbf{x}^\star \in \mathcal{C}$ can be written as

$$\mathcal{F}(\mathbf{x}^\star) = \{\mathbf{x} - \mathbf{x}^\star \mid \mathbf{x} \in \mathcal{C}\}.$$

Proof. 1. Any feasible direction comes from a point in $\mathcal{C}$. If $\mathbf{p}$ is feasible, then

$$\mathbf{x}^\star + \alpha\mathbf{p} \in \mathcal{C}, \quad \alpha \in [0, \delta].$$

Taking $\alpha = \min\{1, \delta\}$, and setting $\mathbf{x} = \mathbf{x}^\star + \alpha\mathbf{p}$ gives

$$\mathbf{x} - \mathbf{x}^\star = \alpha\mathbf{p} \in \mathcal{F}(\mathbf{x}^\star).$$

2. Any point in $\mathcal{C}$ gives a feasible direction. If $\mathbf{x} \in \mathcal{C}$, then convexity implies

$$\mathbf{x}^\star + \alpha(\mathbf{x} - \mathbf{x}^\star) \in \mathcal{C}, \quad \alpha \in [0, 1].$$

Thus $\mathbf{p} = \mathbf{x} - \mathbf{x}^\star$ is a feasible direction.

□

Theorem 6.3.2: First-Order Necessary Condition

If $\mathbf{x}^\star \in \mathcal{C}$ is a local minimizer of f over $\mathcal{C}$, then

$$\langle \nabla f(\mathbf{x}^\star), \mathbf{p} \rangle \geq 0, \quad \forall\, \mathbf{p} \in \mathcal{F}(\mathbf{x}^\star),$$

where $\mathcal{F}(\mathbf{x}^\star)$ is the set[a] of feasible directions at $\mathbf{x}^\star$.

[a] It is actually a cone: a set $\mathcal{K} \subseteq \mathbb{R}^d$ is a cone if $\mathbf{p} \in \mathcal{K}$, then $\alpha\mathbf{p} \in \mathcal{K}$, $\forall\, \alpha \geq 0$.

Intuition: At a constrained local minimum, the directional derivative of f along any feasible direction is non-negative. That is, moving a small amount in any allowed direction cannot decrease the function. For unconstrained optimization, or when $\mathbf{x}^\star \in \operatorname{int}(\mathcal{C})$, the set of feasible directions is all of $\mathbb{R}^d$. In this case, the first-order condition

$$\langle \nabla f(\mathbf{x}^\star), \mathbf{p} \rangle \geq 0, \quad \forall\, \mathbf{p} \in \mathbb{R}^d$$

can only hold if

$$\nabla f(\mathbf{x}^\star) = \mathbf{0}.$$

Definition 6.3.3: Stationary Point

Any point $\mathbf{x}^* \in \mathcal{C}$ for which $\langle \nabla f(\mathbf{x}^*), \mathbf{p} \rangle \geq 0$ for all feasible directions $\mathbf{p}$ is called a stationary point.

Stationary points can be defined equivalently as those $\mathbf{x}^\star \in \mathcal{C}$ satisfying

$$\langle \nabla f(\mathbf{x}^\star),\, \mathbf{x} - \mathbf{x}^\star \rangle \geq 0, \quad \forall\, \mathbf{x} \in \mathcal{C}.$$

Theorem 6.3.4: Second-Order Necessary Condition

If $\mathbf{x}^\star \in \mathcal{C}$ is a local minimizer of f over $\mathcal{C}$, then

$$\langle \nabla f(\mathbf{x}^\star),\, \mathbf{p} \rangle \geq 0, \quad \forall\, \mathbf{p} \in \mathcal{F}(\mathbf{x}^\star),$$

and

$$\mathbf{p}^\top \nabla^2 f(\mathbf{x}^\star)\mathbf{p} \geq 0, \quad \forall\, \mathbf{p} \in \mathcal{F}(\mathbf{x}^\star) \text{ with } \langle \nabla f(\mathbf{x}^\star),\, \mathbf{p} \rangle = 0.$$

Intuition: At a constrained local minimum, the slope along any feasible direction is non-negative, so you cannot move downhill. Along directions where the slope is zero, the surface cannot curve downward.

Theorem 6.3.5: Second-Order Sufficient Condition

If $\mathbf{x}^\star \in \mathcal{C}$ is stationary, and

$$\langle \mathbf{p},\, \nabla^2 f(\mathbf{x}^\star)\, \mathbf{p} \rangle > 0, \quad \text{for all } \mathbf{p} \in \mathcal{F}(\mathbf{x}^\star) \setminus \{\mathbf{0}\} \text{ with } \langle \nabla f(\mathbf{x}^\star),\, \mathbf{p} \rangle = 0,$$

then $\mathbf{x}^\star$ is a strict local minimizer.

Interpretation:

- In feasible directions where $\langle \nabla f(\mathbf{x}^\star), \mathbf{p} \rangle > 0$, the first-order term already pushes up the function, so curvature is less relevant.
- But along feasible directions with zero slope, the Hessian must be strictly positive definite so the surface curves upwards.

Theorem 6.3.6: Second-Order Sufficient Conditions: Convex Constraints

Suppose $\mathcal{C}$ is convex and $\mathbf{x}^\star \in \mathcal{C}$ is stationary. If for some $\gamma > 0$, we have

$$\langle \mathbf{x} - \mathbf{x}^\star,\ \nabla^2 f(\mathbf{x}^\star)\,(\mathbf{x} - \mathbf{x}^\star) \rangle \ \geq\ \gamma \, \|\mathbf{x} - \mathbf{x}^\star\|^2, \quad \forall\, \mathbf{x} \in \mathcal{C},$$

then $\mathbf{x}^\star$ is a strict local minimizer.

Intuition: If the feasible set is convex and the Hessian curves upward strongly in every feasible direction, then the surface around the stationary point forms a bowl within the feasible region, guaranteeing a strict local minimum.

6.4 Convexity

We saw that for a twice differentiable function, having $\nabla^2 f(\mathbf{x}) \succeq 0$ was sufficient to ensure global optimality for any stationary point.
These types of functions are examples of a more general class of functions called convex.
Intuition: Convexity means the function has no "valleys within valleys", i.e., the surface always curves upwards or is flat, so every local minimum is automatically a global minimum.

Definition 6.4.1: Convex Function: Geometric Perspective

A function $f : \mathcal{C} \to \mathbb{R}$ over a convex set $\mathcal{C} \subseteq \mathbb{R}^d$ is said to be convex if

$$f(\alpha \mathbf{x} + (1-\alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1-\alpha) f(\mathbf{y}), \quad \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{C},\ \forall\, \alpha \in [0,1].$$

- Geometrically, this inequality means that the line segment between $(\mathbf{x}, f(\mathbf{x}))$ and $(\mathbf{y}, f(\mathbf{y}))$ lies above the graph of f.
- A function f is said to be concave if $-f$ is convex.

Example.

- Affine function $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle + b$ is both convex and concave. Conversely, any function that is convex and concave is affine.
- Any norm $f(\mathbf{x}) = \|\mathbf{x}\|$ is convex.
- Quadratic functions of the form

$$f(\mathbf{x}) = \tfrac{1}{2}\, \mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c,$$

  where $\mathbf{A} \succeq 0$, are convex.

Theorem 6.4.2: First-order condition for convexity

Suppose $f : \mathcal{C} \to \mathbb{R}$ is differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$. Then f is convex if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}),\, \mathbf{y} - \mathbf{x} \rangle, \quad \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{C}.$$

In words: A differentiable function is convex exactly when the tangent line at any point lies below the graph everywhere else. So, at any point $\mathbf{x}$, the tangent line is always an under-estimator of f.

Proof. "$\Rightarrow$": Assume that f is differentiable and convex. So for any $\mathbf{x}, \mathbf{y} \in \mathcal{C}$ and $\alpha \in (0,1)$, we have

$$f(\alpha\mathbf{y} + (1-\alpha)\mathbf{x}) \leq \alpha f(\mathbf{y}) + (1-\alpha) f(\mathbf{x}),$$

which implies

$$\frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Now by the definition of directional derivative we have

$$\langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle = \lim_{\alpha \downarrow 0} \left(\frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} \right) \leq f(\mathbf{y}) - f(\mathbf{x}).$$

"$\Leftarrow$": For any $\mathbf{x}, \mathbf{y} \in \mathcal{C}$, define $\mathbf{w} \triangleq \alpha\mathbf{x} + (1-\alpha)\mathbf{y} \in \mathcal{C}$. By assumption, we have

$$f(\mathbf{x}) \geq f(\mathbf{w}) + \langle \nabla f(\mathbf{w}), \mathbf{x} - \mathbf{w} \rangle,$$

$$f(\mathbf{y}) \geq f(\mathbf{w}) + \langle \nabla f(\mathbf{w}), \mathbf{y} - \mathbf{w} \rangle.$$

Multiplying the first and the second inequalities by α and by $1-\alpha$, respectively, and adding them together we get

$$\alpha f(\mathbf{x}) + (1-\alpha) f(\mathbf{y}) \geq f(\mathbf{w}) + \left\langle \nabla f(\mathbf{w}), \underbrace{\alpha\mathbf{x} + (1-\alpha)\mathbf{y}}_{\mathbf{w}} - \mathbf{w} \right\rangle, = f(\mathbf{w}) = f(\alpha\mathbf{x} + (1-\alpha)\mathbf{y}).$$

□

Theorem 6.4.3: Second-order condition for convexity

Suppose $f : \mathcal{C} \to \mathbb{R}$ is twice differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$. We have

- If $\nabla^2 f(\mathbf{x}) \succeq 0$, $\forall\, \mathbf{x} \in \mathcal{C}$, then f is convex over $\mathcal{C}$.
- If $\mathcal{C}$ is open and f is convex over $\mathcal{C}$, then $\nabla^2 f(\mathbf{x}) \succeq 0$, $\forall\, \mathbf{x} \in \mathcal{C}$.

Intuition: The Hessian tells us about the local curvature of f.

- PSD Hessian $\Rightarrow$ surface never curves downward $\Rightarrow$ Convexity.
- For convex functions over an open set, the curvature is never negative.

Note: The openness of $\mathcal{C}$ was essential in the second-order condition for convexity.

Example.

Consider

$$\mathcal{C} = \{(x, 0) \mid x \in \mathbb{R}\}, \qquad f(x, y) = x^2 - y^2.$$

On $\mathcal{C}$

the function is convex (it is just x^2 along the line $y = 0$), but the Hessian is

$$\nabla^2 f(x, y) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \not\succeq 0.$$

Intuition: Convexity only requires the function to lie below chords connecting points in the set. On a boundary, the Hessian may have negative eigenvalues in directions that cannot be taken (outside the set), so the PSD condition fails.

Theorem 6.4.4: Necessary and Sufficient Conditions for Optimality

Consider the optimization problem

$$\min f(\mathbf{x}) \quad \text{s.t. } \mathbf{x} \in \mathcal{C},$$

where $f : \mathbb{R}^d \to \mathbb{R}$ is convex and differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$.

1. Necessary Condition: Any local optimal point is stationary.
2. Sufficient Condition: Any stationary point is global optimal.

So in particular, for any convex optimization problem, there are no saddle points or local/global maxima. In fact, all stationary points are global minima.

Proof. 1. This is just a consequence of First-Order Necessary Condition in constrained problems; see lecture on Optimality.

2. Suppose f is convex and $\mathbf{x} \in \mathcal{C}$ is stationary. From first-order condition on convexity, we get for any $\mathbf{y} \in \mathcal{C}$

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \geq f(\mathbf{x}),$$

which proves the global optimality of $\mathbf{x} \in \mathcal{C}$.

□

Theorem 6.4.5

If $f \in \mathcal{C}_L^{1,1}(\mathbb{R}^d)$ and convex, then

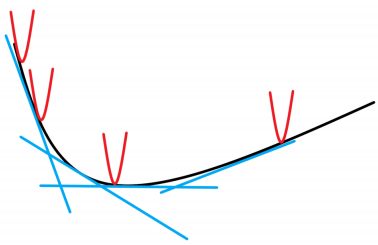
$$0 \leq f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \leq \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|^2.$$

Intuition:

- The left inequality comes from convexity: the linear approximation of f at any point underestimates the function.
- The right inequality comes from the smoothness (L-Lipschitz gradient): the quadratic term bounds how much f can rise above its linear approximation.

Proof. This simply follows from the definition of convex function and also the result on Lipschitz continuity of the gradient.

□



Visual Interpretation: For $f \in \mathcal{C}_L^{1,1}(\mathbb{R}^d)$ and convex, at any point $\mathbf{x}$:

- The tangent line at $\mathbf{x}$ is a global underestimator of f.
- The quadratic with curvature L is a global overestimator of f.
- The function f always lies between these two bounds.

Definition 6.4.6: Strictly Convex Function

A function $f : \mathcal{C} \to \mathbb{R}$ over a convex set $\mathcal{C} \subseteq \mathbb{R}^d$ is said to be strictly convex if $\forall\, \mathbf{x} \neq \mathbf{y} \in \mathcal{C}$, $\forall\, \alpha \in (0, 1)$,

$$f(\alpha\mathbf{x} + (1-\alpha)\mathbf{y}) < \alpha f(\mathbf{x}) + (1-\alpha) f(\mathbf{y}).$$

Intuition: The function curves upward between any two distinct points, so the line connecting two points always lies strictly above the graph.

Theorem 6.4.7: First-order condition for strict convexity

Suppose $f : \mathcal{C} \to \mathbb{R}$ is differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$. Then f is strictly convex if and only if

$$f(\mathbf{y}) > f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle, \quad \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{C},\ \mathbf{x} \neq \mathbf{y}.$$

Theorem 6.4.8: Second-order condition for strict convexity

Suppose $f : \mathcal{C} \to \mathbb{R}$ is twice differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$. If $\nabla^2 f(\mathbf{x}) \succ 0$, $\forall\, \mathbf{x} \in \mathcal{C}$, then f is strictly convex over $\mathcal{C}$.

Note: The converse is not necessarily true, e.g., $f(x) = x^4$.

Theorem 6.4.9

Consider the optimization problem

$$\min f(\mathbf{x}) \quad \text{s.t. } \mathbf{x} \in \mathcal{C},$$

where $f : \mathcal{C} \to \mathbb{R}$ is strictly convex and also $\mathcal{C} \subseteq \mathbb{R}^d$ is convex. Then the minimizer is unique.

Proof. Suppose there are two optimal solutions, $\mathbf{x}, \mathbf{y} \in \mathcal{C}$. By strict convexity, we have

$$f\left(\tfrac{1}{2}\mathbf{x} + \tfrac{1}{2}\mathbf{y}\right) < \tfrac{1}{2} f(\mathbf{x}) + \tfrac{1}{2} f(\mathbf{y}) = \tfrac{1}{2} f(\mathbf{x}) + \tfrac{1}{2} f(\mathbf{x}) = f(\mathbf{x}),$$

which contradicts optimality of $\mathbf{x}$. □

Definition 6.4.10: Strongly Convex Function

A function $f : \mathcal{C} \to \mathbb{R}$ over a convex set $\mathcal{C} \subseteq \mathbb{R}^d$ is said to be strongly convex if $\exists\, \mu > 0$ such that the function $f(\mathbf{x}) - \frac{\mu}{2}\|\mathbf{x}\|^2$ is convex.

Theorem 6.4.11: First-order condition for strong convexity

Suppose $f : \mathcal{C} \to \mathbb{R}$ is differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$. Then f is strongly convex with parameter μ if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \tfrac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2, \quad \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{C}.$$

In words, when f is strongly convex over $\mathbb{R}^d$, we can put a quadratic with positive curvature below f at any point, and f will always lie above this quadratic over all of $\mathbb{R}^d$.

Theorem 6.4.12

Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable and μ-strongly convex. Then,

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{1}{2\mu} \|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\|^2, \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{C}.$$

Theorem 6.4.13: Second-order condition for strong convexity

Suppose $f : \mathcal{C} \to \mathbb{R}$ is twice differentiable on an open set containing the convex set $\mathcal{C} \subseteq \mathbb{R}^d$. We have

- If $\nabla^2 f(\mathbf{x}) \succeq \mu \mathbf{I}$, $\forall \mathbf{x} \in \mathcal{C}$, then f is strongly convex over $\mathcal{C}$ with parameter μ.
- If $\mathcal{C}$ is open and f is strongly convex over $\mathcal{C}$ with parameter μ, then $\nabla^2 f(\mathbf{x}) \succeq \mu \mathbf{I}$, $\forall \mathbf{x} \in \mathcal{C}$.

Example.

Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} \tfrac{1}{2} \langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle + \langle \mathbf{b}, \mathbf{x} \rangle + c,$$

with symmetric $\mathbf{A}$. We have that $\nabla^2 f(\mathbf{x}) = \mathbf{A}$.

- If $\mathbf{A} \succeq 0$, the objective is convex. The optimal value may or may not be bounded (why?). If it is bounded, then there could be many optimal solutions (why?).
- If $\mathbf{A} \succ 0$, the objective is strictly convex (in fact, it is strongly convex with $\mu = \lambda_{\min}(\mathbf{A})$). Hence the solution is unique (what is it?).
- If $\mathbf{A} \not\succeq 0$, the objective is non-convex and always unbounded below (why?).

Example.

Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} \tfrac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2.$$

We have that $\nabla^2 f(\mathbf{x}) = \mathbf{A}^T \mathbf{A}$.

- The objective is always at least convex. The optimal value is always bounded below. But there could be many optimal solutions.
- If $\mathbf{A}$ has linearly independent columns, the objective is strictly convex (in fact, it is strongly convex with $\mu = \sigma^2_{\min}(\mathbf{A})$). Hence the solution is unique.

Chapter 7

Unconstrained Optimization

We now focus unconstrained optimization of smooth functions.
Unconstrained Optimization:

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x})$$

In particular, we will zoom in on several descent methods...

7.1 Descent Method

Like hiking downhill, descent methods follow directions that reduce the function value, step by step, until reaching a (local) minimum (approximately).

Definition 7.1.1: Descent Direction

A direction $\mathbf{p} \in \mathbb{R}^d$ is a descent direction for f at $\mathbf{x}$ if for some $\alpha_0 > 0$ we have

$$f(\mathbf{x} + \alpha\,\mathbf{p}) < f(\mathbf{x}), \quad \forall \alpha \in (0, \alpha_0].$$

In words: There is a line segment from $\mathbf{x}$ along which the function has smaller values than $f(\mathbf{x})$. Geometrically, it is like standing on a hill: any direction that initially goes downhill is a descent direction.
Q: Given a direction $\mathbf{p}$, how do we know it is a descent direction?
A: There is an algebraic test that serves as a sufficient condition for descent direction.

Theorem 7.1.2: Sufficient Condition for Descent Direction

If $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable and $\mathbf{p} \in \mathbb{R}^d$ is such that

$$\langle \nabla f(\mathbf{x}), \mathbf{p} \rangle < 0,$$

then $\mathbf{p}$ is a descent direction for f at $\mathbf{x}$.

Intuition: Here $\langle \nabla f(\mathbf{x}), \mathbf{p} \rangle$ measures the slope of f at $\mathbf{x}$ along direction $\mathbf{p}$, i.e., directional derivative. If this slope is negative, moving a little along $\mathbf{p}$ decreases f, so $\mathbf{p}$ must be a descent direction.

Proof. Omitted. □

So we are guaranteed to get a strict decrease in objective function if we move, by an appropriate amount, along a descent direction.
Hence, for descent methods in unconstrained optimization, there are two main ingredients:

1. Find a search/descent direction $\mathbf{p} \in \mathbb{R}^d$.
2. Find a step size/length $\alpha > 0$ such that $f(\mathbf{x} + \alpha\mathbf{p}) < f(\mathbf{x})$.

These two ingredients give rise to basic iterative descent methods as follows:

Definition 7.1.3: Basic Iterative Descent Method

Starting from an initial guess $\mathbf{x}_0 \in \mathbb{R}^d$, we do

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k, \quad k = 0, 1, \ldots$$

where

1. $\mathbf{p}_k \in \mathbb{R}^d$ is a descent direction at $\mathbf{x}_k$, and
2. $\alpha_k > 0$ is an appropriate step-size at k^{th} iteration.

Think of descent methods as hiking downhill: the direction tells you which way to go, while the step-size tells you how far to go without overshooting or stalling.

7.1.1 Descent Direction

Recall: $\nabla f(\mathbf{x})$ is the direction of maximum rate of increase of f at $\mathbf{x}$. Hence, the simplest descent direction is $\mathbf{p} = -\nabla f(\mathbf{x})$, which is known as the direction of steepest descent.
More generally, consider any $\mathbf{S} \succ 0$, then it is easy to see that $\mathbf{p} = -\mathbf{S}\nabla f(\mathbf{x})$ is a direction of descent. Indeed, we have

$$\langle \nabla f(\mathbf{x}), \mathbf{p} \rangle = -\langle \nabla f(\mathbf{x}), \mathbf{S}\nabla f(\mathbf{x}) \rangle < 0.$$

Example.

Examples include:

- $\mathbf{S} = \mathbf{I}$: Gradient descent
- f strongly convex, $\mathbf{S} = [\nabla^2 f(\mathbf{x})]^{-1}$: Newton Method
- f strongly convex, $\mathbf{S} = [\text{diag}(\nabla^2 f(\mathbf{x}))]^{-1}$: Diagonal Scaling
- $\mathbf{S}$ from Gauss–Newton, L-BFGS, Natural Gradient, etc.

Intuition: Scaling reshapes the space so that the negative gradient points in a better adapted downhill direction.

Definition 7.1.4: Scaled Gradient Descent Methods

Starting from an initial guess $\mathbf{x}_0 \in \mathbb{R}^d$, we do

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{S}_k \nabla f(\mathbf{x}_k), \qquad k = 0, 1, \ldots,$$

where $\mathbf{S}_k \succ 0$, $\forall k$.

(Scaled) gradient direction is always a descent direction unless $\nabla f(\mathbf{x}) = 0$.

- For convex problems, this is a good thing: we have reached a solution.
- For non-convex problems, though, we might be at a saddle point and we cannot rely solely on (scaled) gradient direction.

However, it turns out that we might still be able to find descent directions at/near saddle points.

Example.

For example, suppose we choose $\mathbf{p}$ to be parallel to the eigenvector corresponding to a negative eigenvalue of $\nabla^2 f(\mathbf{x})$. Then assuming sufficient smoothness, e.g., Lipschitz Hessian, we can show that $\mathbf{p}$ is a descent direction.

Intuition: At a saddle point, the gradient vanishes but the curvature of the surface reveals escape di-

rections — moving along a negative curvature allows us to descend.
Recall: If f has L_H-Lipschitz Hessian, then for all $\mathbf{x}, \mathbf{y}$

$$\left| f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle - \tfrac{1}{2} \langle \mathbf{y} - \mathbf{x},\, \nabla^2 f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \rangle \right| \leq \frac{L_H}{6} \|\mathbf{y} - \mathbf{x}\|^3.$$

Now suppose:

- $\nabla f(\mathbf{x}) = 0$ (stationary point),
- $\nabla^2 f(\mathbf{x})\, \mathbf{p} = \lambda \mathbf{p}$ with $\lambda < 0$, $\|\mathbf{p}\| = 1$,
- choose $\mathbf{y} = \mathbf{x} + \alpha \mathbf{p}$ with $\alpha \leq \frac{2|\lambda|}{L_H}$.

Then, we get

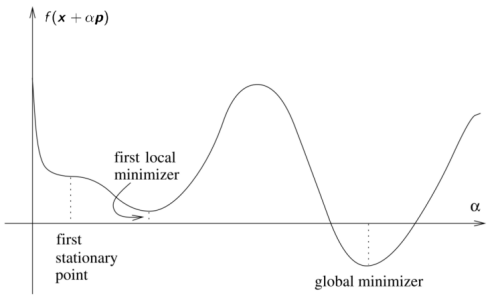
$$f(\mathbf{x} + \alpha \mathbf{p}) \leq f(\mathbf{x}) + \alpha \langle \nabla f(\mathbf{x}), \mathbf{p} \rangle + \frac{\alpha^2}{2} \langle \mathbf{p},\, \nabla^2 f(\mathbf{x})\, \mathbf{p} \rangle + \frac{\alpha^3 L_H}{6} \|\mathbf{p}\|^3 = f(\mathbf{x}) + \frac{\alpha^2 \lambda}{2} + \frac{\alpha^3 L_H}{6}$$

$$\leq f(\mathbf{x}) + \frac{2\lambda^3}{L_H^2} + \frac{8|\lambda|^3}{6L_H^2} = f(\mathbf{x}) - \frac{2|\lambda|^3}{L_H^2} + \frac{4|\lambda|^3}{3L_H^2} = f(\mathbf{x}) - \frac{2|\lambda|^3}{3L_H^2} < f(\mathbf{x}).$$

7.1.2 Step-size

Perhaps, the ideal choice is the **exact line-search**:

$$\alpha_k \in \arg\min_{\alpha \geq 0} f(\mathbf{x}_k + \alpha\, \mathbf{p}_k).$$



However, except for very special cases, computing the exact line search is generally impractical.

Example.

Suppose

$$f(\mathbf{x}) = \tfrac{1}{2} \langle \mathbf{x},\, \mathbf{A}\mathbf{x} \rangle - \langle \mathbf{b},\, \mathbf{x} \rangle, \qquad \mathbf{A} \succeq 0.$$

Given any search direction $\mathbf{p}$, define

$$\phi(\alpha) \triangleq f(\mathbf{x} + \alpha \mathbf{p}) = \tfrac{1}{2} \langle \mathbf{x} + \alpha \mathbf{p},\, \mathbf{A}(\mathbf{x} + \alpha \mathbf{p}) \rangle - \langle \mathbf{b},\, \mathbf{x} + \alpha \mathbf{p} \rangle.$$

The derivative is

$$\phi'(\alpha) = \langle \mathbf{p},\, \mathbf{A}(\mathbf{x} + \alpha \mathbf{p}) \rangle - \langle \mathbf{p},\, \mathbf{b} \rangle.$$

Setting $\phi'(\alpha) = 0$ gives the optimal step:

$$\alpha^\star = \frac{\langle \mathbf{p},\, \mathbf{b} - \mathbf{A}\mathbf{x} \rangle}{\langle \mathbf{p},\, \mathbf{A}\mathbf{p} \rangle} = \frac{\langle \mathbf{p},\, \mathbf{r} \rangle}{\langle \mathbf{p},\, \mathbf{A}\mathbf{p} \rangle}.$$

For gradient descent, $\mathbf{p}_k = -\nabla f(\mathbf{x}_k) = \mathbf{r}_k$, so the update is

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k^\star \mathbf{r}_k, \qquad \alpha_k^\star = \frac{\langle \mathbf{r}_k,\, \mathbf{r}_k \rangle}{\langle \mathbf{r}_k,\, \mathbf{A}\mathbf{r}_k \rangle}.$$

As mentioned, in general, computing the exact line-search is too expensive or impossible.
A naive approach: Since $\mathbf{p}_k$ is a descent direction, one might think it is enough to pick $\alpha > 0$ such that

$$f(\mathbf{x}_k + \alpha\, \mathbf{p}_k) < f(\mathbf{x}_k).$$

But this condition cannot guarantee sufficient descent: it can allow extremely small steps that make almost no progress.
Simply decreasing the function is not enough—we want steps that make meaningful progress.
More practical strategies are based on **inexact line-search**, which aim to:

- Find a step length that gives sufficient reduction in f.
- Avoid the high cost of exact line search.

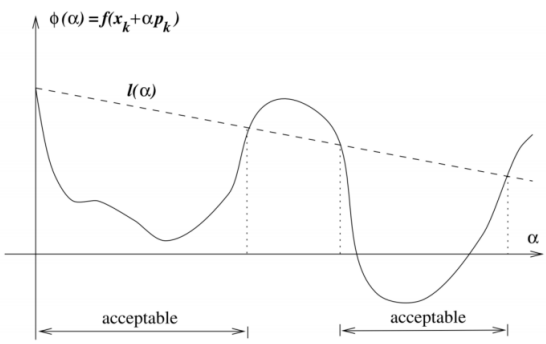
We are willing to trade "perfect" steps for good enough steps that are cheap to compute.

Definition 7.1.5: Armijo Condition

If $\langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle < 0$, we choose $\alpha > 0$ such that

$$\underbrace{f(\mathbf{x}_k + \alpha \mathbf{p}_k)}_{\phi(\alpha)} \leq \underbrace{f(\mathbf{x}_k) + \alpha\beta\langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle}_{\ell(\alpha)}, \qquad \text{for some } \beta \in (0, 1).$$

Intuition: The Armijo condition ensures that the step α reduces f by at least a fraction of what the linear approximation predicts.



Practical Armijo Implementation:

- Typically, we set $\beta = 10^{-4}$.
- Armijo condition alone can still allow tiny steps.
- To avoid this, we often try to find the largest α satisfying Armijo.
- Backtracking: Start from $\alpha_{\max} > 0$ and reduce geometrically

  $$\alpha = \rho^i \alpha_{\max}, \quad i = 0, 1, 2, \ldots$$

  until Armijo is satisfied. Usually $\rho = \frac{1}{2}$, $\alpha_{\max} = 1$ for Newton/quasi-Newton.
- This variant of Armijo is useless at saddle points in non-convex problems.

Lemma 7.1.6: Guarantee of Armijo Step

Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is continuously differentiable and $\langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle < 0$. Then there exists $\bar{\alpha} > 0$ such that the Armijo condition holds for all $\alpha \in (0, \bar{\alpha}]$.

Note: $\bar{\alpha}$ may depend on $\mathbf{x}_k$.

Proof. Omitted. □

Even with backtracking, the step α satisfying the Armijo condition might be far from a (local) minimizer of $\phi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{p}_k)$.
Ideally, we would like $\phi'(\alpha) = 0$, but this is usually impossible to compute in practice.
Intuition: We want a step where the slope along the search direction is not too steep, indicating that we are getting closer to a minimum along that line.
So, we search for α such that $\phi'(\alpha)$ is less negative than $\phi'(0)$ (recall $\phi'(0) = \langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle < 0$).

Definition 7.1.7: Wolfe Conditions

If $\langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle < 0$, we find α such that

$$\text{(i)} \quad f(\mathbf{x}_k + \alpha \mathbf{p}_k) \leq f(\mathbf{x}_k) + \alpha\, \beta_1 \langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle \quad \text{(Armijo Condition)}$$

$$\text{(ii)} \quad \underbrace{\langle \nabla f(\mathbf{x}_k + \alpha \mathbf{p}_k), \mathbf{p}_k \rangle}_{\phi'(\alpha)} \geq \beta_2 \underbrace{\langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle}_{\phi'(0)} \quad \text{(Curvature Condition)}$$

where $0 < \beta_1 < \beta_2 < 1$.

Practical Remarks on Wolfe Conditions:

- Curvature condition rules out very short steps that would satisfy Armijo alone.
- Typical parameters: $\beta_1 = 10^{-4}$, $\beta_2 = 0.9$ (Newton/quasi-Newton), $\beta_2 = 0.1$ (nonlinear conjugate gradient).
- Wolfe steps may not be very close to a minimizer of ϕ. Strong Wolfe conditions improve this:

$$|\langle \nabla f(\mathbf{x}_k + \alpha \mathbf{p}_k),\, \mathbf{p}_k \rangle| \leq \beta_2\, |\langle \nabla f(\mathbf{x}_k),\, \mathbf{p}_k \rangle|\,.$$

- (Strong) Wolfe conditions are not helpful at saddle points in non-convex problems.
- They are especially important in quasi-Newton methods.
- Backtracking alone is insufficient; bisection or interpolation/extrapolation is needed to find α satisfying these conditions.

Lemma 7.1.8: Guarantee of Wolfe Step

Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is continuously differentiable and bounded below, and $\langle \nabla f(\mathbf{x}_k), \mathbf{p}_k \rangle < 0$. Then there exist intervals of α satisfying the (strong) Wolfe conditions.

Note: The interval may depend on the current point $\mathbf{x}_k$.

Proof. Omitted. □

7.2 Gradient Descent

Recall (General Descent Methods):

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$$

where $\mathbf{p}_k$ is a descent method, and α_k is step-size.

Definition 7.2.1: Gradient Descent

$$\mathbf{p}_k = -\nabla f(\mathbf{x}_k)$$

Definition 7.2.2: Algorithm Gradient Descent

1. $\mathbf{x}_0$
2. for $k = 1, 2, \ldots$ until convergence, e.g., $\|\nabla f(\mathbf{x}_k)\| \leq \tau$ do
3. $\mathbf{p}_k = -\nabla f(\mathbf{x}_k)$
4. Compute α_k, e.g., constant step-size, exact linesearch, or Armijo line-search
5. $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$
6. end for

A very typical, and problematic, behavior of the gradient method is the "zig-zagging" effect.

Lemma 7.2.3

Let $\{\mathbf{x}_k\}_{k\geq 0}$ be the sequence generated by gradient descent (GD) with exact line-search for unconstrained optimization of a differentiable function. Then we have

$$\langle \mathbf{x}_{k+2} - \mathbf{x}_{k+1},\, \mathbf{x}_{k+1} - \mathbf{x}_k \rangle = 0.$$

With exact line search, each new step is taken in a direction that is orthogonal to the previous step. This causes the method to "zig-zag" back and forth, especially in narrow valleys.

Proof. Suppose we take two consecutive GD updates using exact line-search. Note that

$$\mathbf{x}_{k+1} - \mathbf{x}_k = -\alpha_k \nabla f(\mathbf{x}_k), \qquad \mathbf{x}_{k+2} - \mathbf{x}_{k+1} = -\alpha_{k+1} \nabla f(\mathbf{x}_{k+1}).$$

Since

$$\alpha_k \in \arg\min_{\alpha \geq 0} \ \phi(\alpha) \triangleq f\big(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)\big),$$

and $\alpha_k \neq 0$, it follows that $\phi'(\alpha_k) = 0$. Hence,

$$-\langle \nabla f(\mathbf{x}_k),\, \nabla f(\mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)) \rangle = -\langle \nabla f(\mathbf{x}_k),\, \nabla f(\mathbf{x}_{k+1}) \rangle = 0.$$

□

We now establish the convergence properties of gradient descent.
We will heavily rely on the assumption that f has Lipschitz-continuous gradient, i.e.,

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L_g \|\mathbf{x} - \mathbf{y}\|,$$

which implies

$$\big| f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}),\, \mathbf{y} - \mathbf{x} \rangle \big| \leq \frac{L_g}{2} \|\mathbf{y} - \mathbf{x}\|^2, \qquad \forall\, \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

Lemma 7.2.4: Sufficient Descent Lemma

Suppose f has L_g-Lipschitz-continuous gradient. For any $\mathbf{x} \in \mathbb{R}^d$ and $\alpha > 0$, set $\mathbf{y} = \mathbf{x} - \alpha \nabla f(\mathbf{x})$. We have

$$f(\mathbf{y}) \leq f(\mathbf{x}) - \alpha \left(1 - \frac{\alpha L_g}{2}\right) \|\nabla f(\mathbf{x})\|^2.$$

Intuition: Each step of gradient GD decreases the function value, provided $\alpha < \frac{2}{L_g}$.

Proof. From L_g-Lipschitz-continuity of gradient, we have

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}),\, \mathbf{y} - \mathbf{x} \rangle + \frac{L_g}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

$$= f(\mathbf{x}) - \alpha\|\nabla f(\mathbf{x})\|^2 + \frac{\alpha^2 L_g}{2}\|\nabla f(\mathbf{x})\|^2 = f(\mathbf{x}) - \left(\alpha - \frac{\alpha^2 L_g}{2}\right)\|\nabla f(\mathbf{x})\|^2.$$

□

Choices for Step-size

- Constant step-size: for any $\alpha \in (0, 2/L_g)$, we get descent since

  $$\alpha\left(1 - \frac{\alpha L_g}{2}\right) > 0.$$

  In particular, the term $\alpha(1 - \alpha L_g/2)$ is maximized at $\alpha = 1/L_g$, and hence

  $$f(y) \leq f(x) - \frac{1}{2L_g}\|\nabla f(x)\|^2.$$

  Choosing $\alpha = 1/L_g$ gives the largest guaranteed decrease in f per step.
  However, in most problems, it is not possible to compute or even reliably estimate L_g.

- Armijo condition: find the largest α such that

  $$f(y) \leq f(x) - \alpha\beta\|\nabla f(x)\|^2.$$

  Suppose $\alpha \leq \frac{2(1-\beta)}{L_g}$. From the descent lemma:

  $$f(y) \leq f(x) - \alpha\left(1 - \frac{\alpha L_g}{2}\right)\|\nabla f(x)\|^2 \leq f(x) - \alpha\beta\|\nabla f(x)\|^2.$$

  Thus, the largest α that satisfies the Armijo condition must be at least

  $$\alpha_{\max} \geq \frac{2(1-\beta)}{L_g},$$

  giving

  $$f(y) \leq f(x) - \frac{2\beta(1-\beta)}{L_g}\|\nabla f(x)\|^2.$$

  Armijo line search adaptively picks the step-size to guarantee sufficient decrease, while possibly allowing larger steps than a conservative constant step-size.

So putting it all together, we have

$$f(\mathbf{y}) \leq f(\mathbf{x}) - M\|\nabla f(\mathbf{x})\|^2,$$

where

$$M \triangleq \begin{cases} \dfrac{1}{2L_g}, & \text{Constant step-size with } \alpha = \dfrac{1}{L_g}, \\ \dfrac{2\beta(1-\beta)}{L_g}, & \text{Armijo line-search.} \end{cases}$$

Theorem 7.2.5: Asymptotic Convergence of GD for $C^{1,1}_{L_g}(\mathbb{R}^d)$

Let $f \in C^{1,1}_{L_g}(\mathbb{R}^d)$, and $\{\mathbf{x}_k\}_{k\geq 0}$ be the sequence generated by GD method for solving

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}),$$

with any of the previous step-size choices. If f is bounded below, then we have

1. The sequence $\{f(\mathbf{x}_k)\}_{k\geq 0}$ is non-increasing. In particular, we have $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$ unless $\nabla f(\mathbf{x}_k) = \mathbf{0}$.
2. $\lim_{k\to\infty} \nabla f(\mathbf{x}_k) = \mathbf{0}$.

Proof. 1. This follows simply from the earlier discussion on the choices of step-size.

2. Since the sequence $\{f(\mathbf{x}_k)\}_{k\geq 0}$ is non-increasing and bounded below, it must converge.[1] Thus, in particular, we have

$$\lim_{k\to\infty} \big(f(\mathbf{x}_k) - f(\mathbf{x}_{k+1})\big) = 0,$$

which, again by the earlier discussion on the choices of step-size, implies

$$0 \leq \lim_{k\to\infty} \|\nabla f(\mathbf{x}_k)\|^2 \leq \frac{1}{M} \lim_{k\to\infty} \big(f(\mathbf{x}_k) - f(\mathbf{x}_{k+1})\big) = 0,$$

and hence $\lim_{k\to\infty} \nabla f(\mathbf{x}_k) = \mathbf{0}$.

□

Theorem 7.2.6: Convergence Rate of GD for $C^{1,1}_{L_g}(\mathbb{R}^d)$

Under the same settings as the previous theorem, we have

$$\min_{i=0,1,\ldots,k} \|\nabla f(\mathbf{x}_i)\| \leq \sqrt{\frac{f(\mathbf{x}_0) - f^\star}{M(k+1)}},$$

where $f^\star \triangleq \min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x})$, and M is defined before.

Intuition: The smallest gradient norm decreases like $O(1/\sqrt{k})$. Gradient descent gets closer to a stationary point at a predictable rate.

Proof. From the previous discussion, we have

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \geq M \|\nabla f(\mathbf{x}_k)\|^2.$$

Now summing this over k, we get

$$f(\mathbf{x}_0) - f(\mathbf{x}_{k+1}) \geq M \sum_{i=0}^{k} \|\nabla f(\mathbf{x}_i)\|^2.$$

Since $f^\star \leq f(\mathbf{x}_{k+1})$, we have $f(\mathbf{x}_0) - f^\star \geq M \sum_{i=0}^{k} \|\nabla f(\mathbf{x}_i)\|^2$. So from

$$\min\{\|\nabla f(\mathbf{x}_0)\|^2, \|\nabla f(\mathbf{x}_1)\|^2, \ldots, \|\nabla f(\mathbf{x}_k)\|^2\} \leq \|\nabla f(\mathbf{x}_i)\|^2, \quad i = 0, \ldots, k,$$

it follows that

$$f(\mathbf{x}_0) - f^\star \geq M(k+1) \min_{i=0,1,\ldots,k} \|\nabla f(\mathbf{x}_i)\|^2.$$

[1] A non-decreasing sequence converges if and only if it is bounded above. A non-increasing sequence is convergent if and only if it is bounded below.

□

In words, the previous theorem states that for functions in $C^{1,1}_{L_g}(\mathbb{R}^d)$, the smallest gradient encountered in first k iterations shrinks like $1/\sqrt{k}$ in the worst case.
It turns out that if the function is convex and in $C^{1,1}_{L_g}(\mathbb{R}^d)$, GD can do better than $1/\sqrt{k}$ in the worst case.

Theorem 7.2.7: Convergence Rate of GD for convex and $C^{1,1}_{L_g}(\mathbb{R}^d)$

Suppose f is convex and in $C^{1,1}_{L_g}(\mathbb{R}^d)$. For GD with $\alpha = 1/L_g$, we have

$$f(\mathbf{x}_k) - f^\star \leq \frac{L_g}{2k}\left(\inf_{\mathbf{x}^\star \in \mathcal{X}^\star} \|\mathbf{x}_0 - \mathbf{x}^\star\|^2\right),$$

where $f^\star \triangleq \min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x})$, and $\mathcal{X}^\star$ is the set of optimal points, i.e.,

$$\mathcal{X}^\star \triangleq \{\mathbf{x}^\star \mid f(\mathbf{x}^\star) = f^\star\}.$$

Intuition: For convex functions, the error in function value decreases like $\mathcal{O}(1/k)$. Convexity allows us to exploit geometry to get faster convergence than the general $\mathcal{O}(1/\sqrt{k})$ bound.

Proof. From descent lemma, we have

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L_g}\|\nabla f(\mathbf{x}_k)\|^2 \leq f(\mathbf{x}^\star) + \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x}^\star\rangle - \frac{1}{2L_g}\|\nabla f(\mathbf{x}_k)\|^2$$

$$= f(\mathbf{x}^\star) + \frac{L_g}{2}\left(\|\mathbf{x}_k - \mathbf{x}^\star\|^2 - \left\|\mathbf{x}_k - \mathbf{x}^\star - \frac{1}{L_g}\nabla f(\mathbf{x}_k)\right\|^2\right) = f(\mathbf{x}^\star) + \frac{L_g}{2}\left(\|\mathbf{x}_k - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{k+1} - \mathbf{x}^\star\|^2\right).$$

So, we have

$$\sum_{i=0}^{k}\left(f(\mathbf{x}_{i+1}) - f(\mathbf{x}^\star)\right) \leq \frac{L_g}{2}\sum_{i=0}^{k}\left(\|\mathbf{x}_i - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{i+1} - \mathbf{x}^\star\|^2\right) = \frac{L_g}{2}\left(\|\mathbf{x}_0 - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{k+1} - \mathbf{x}^\star\|^2\right) \leq \frac{L_g}{2}\|\mathbf{x}_0 - \mathbf{x}^\star\|^2.$$

Since $\{f(\mathbf{x}_k)\}_{k\geq 0}$ is non-increasing, we have

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^\star) \leq \frac{1}{k+1}\sum_{i=0}^{k}\left(f(\mathbf{x}_{i+1}) - f(\mathbf{x}^\star)\right) \leq \frac{L_g}{2(k+1)}\|\mathbf{x}_0 - \mathbf{x}^\star\|^2.$$

This bound holds for any $\mathbf{x}^\star$, so we can take infimum over $\mathcal{X}^\star$, and we get the result. □

In words, the previous theorem states that, for convex functions in $C^{1,1}_{L_g}(\mathbb{R}^d)$, GD with constant step-size of $1/L_g$ converges at the rate of $1/k$ in the worst case.
It turns out that if the function is strongly-convex and in $C^{1,1}_{L_g}(\mathbb{R}^d)$, GD can do better than $1/k$ in the worst case.

Theorem 7.2.8: Convergence Rate of GD for strongly-convex and $C^{1,1}_{L_g}(\mathbb{R}^d)$

Suppose f is μ-strongly convex and in $C^{1,1}_{L_g}(\mathbb{R}^d)$. For GD, we have

$$f(\mathbf{x}_{k+1}) - f^\star \leq (1 - 2\mu M)\left(f(\mathbf{x}_k) - f^\star\right),$$

where M is as defined before.

Note: From our previous discussion on the choices of step-size and the corresponding M values, and noting that for μ-strongly convex and $C^{1,1}_{L_g}(\mathbb{R}^d)$ functions we have $\mu \leq L_g$, it follows that $0 \leq 1 - 2\mu M < 1$.

Intuition: Strong convexity ensures a geometric (linear) decrease in the error. Each iteration reduces the remaining sub-optimality gap by a fixed fraction.

Proof. Omitted. □

Recall: Consider a PD matrix

$$\lambda_{\min} I \preceq \mathbf{A} \preceq \lambda_{\max} I.$$

Recall that the condition number was defined as

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

For μ-strongly convex and $C^{1,1}_{L_g}(\mathbb{R}^d)$ functions, we have

$$\mu I \preceq \nabla^2 f(\mathbf{x}) \preceq L_g I.$$

Thus, we can **define the condition number of** f as

$$\kappa \triangleq \frac{L_g}{\mu}.$$

Intuition: The condition number measures how "stretched" the level sets of f are. Small $\kappa \approx 1$ means nearly spherical level sets, while large $\kappa \gg 1$ means elongated level sets.
For μ-strongly convex and $\mathcal{C}^{1,1}_{L_g}$ functions, GD satisfies

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^\star) \leq \left(1 - \frac{c}{\kappa}\right)(f(\mathbf{x}_k) - f(\mathbf{x}^\star)),$$

where c is a constant depending on the choice of step-size (constant or Armijo line-search).
The smaller the condition number κ, the faster GD converges. As κ grows, convergence slows down because the level sets are more elongated.

Definition 7.2.9: Rates of Convergence

For a sequence $a_k \to 0$ (e.g., $a_k = \|\nabla f(\mathbf{x}_k)\|$, $\|\mathbf{x}_k - \mathbf{x}^\star\|$, or $f(\mathbf{x}_k) - f(\mathbf{x}^\star)$), the convergence rate can be classified as:

- Sub-linear: $a_k \leq \frac{C}{\sqrt{k}},\ \frac{C}{k},\ \frac{C}{k^2},\ C > 0.$
- Linear: $a_{k+1} \leq (1-\eta)a_k,\ \eta \in (0,1).$
- Super-linear: $\lim_{k\to\infty} \frac{a_{k+1}}{a_k} = 0.$
- Quadratic: $a_{k+1} \leq C a_k^2,\ C > 0.$

Intuition: Q-convergence tracks individual steps; R-convergence tracks the overall trend.